Not Dead Yet: Options Trading Floors

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ABSTRACT

Over 20% of U.S. options dollar volume trades on physical floors, almost exclusively in S&P 500 index options. We study the COVID-19 closure of the Cboe floor, when Cboe activated SPX complex auctions, to identify the impact of floor trading. The floor attracts complex trades, particularly the largest. Floor trades cost 36% more than comparable electronic trades during the closure. Smaller trades are more expensive on the floor, while the largest are cheaper. Without the floor, the largest and most complex trades become half as prevalent. The floor facilitates the most difficult trades, but raises the cost of less difficult ones.

Keywords: Floor trading, electronic trading, auction, liquidity, market microstructure
JEL codes: G20, G14, L10
Internet Appendix

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I have nothing to disclose.

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I have received compensation exceeding $10,000 during the past three years for expert witness services provided to a variety of clients. Clients may or may not have an interest in the research in this paper. I have no other potential conflicts to disclose.

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HUMAN INTERMEDIATION HAS LARGELY DISAPPEARED from modern financial markets. In equities, the transition is essentially complete (O'Hara, 2015). Brokers and designated market makers still stand on the New York Stock Exchange (NYSE), but floor trading accounts for roughly 3% of consolidated share volume that is concentrated in the opening and closing auctions. Nevertheless, in U.S. listed options the floor's footprint is an order of magnitude larger, with over 20% of dollar volume executing through open outcry, about three quarters of it in S&P 500 index options (SPX) on the Chicago Board Options Exchange (Cboe).[1] Why has open outcry persisted in this corner of the market when it disappeared elsewhere, and what would be lost if it disappeared?

This paper exploits the COVID-19 closure of the Cboe floor to ask what the floor contributes to execution in modern options markets. The floor attracts complex trades, particularly the largest. Cboe activated an electronic auction for complex SPX orders during the closure. When floor trading resumes on June 15, 2020, and the auction is withdrawn, floor trades cost 36% more than comparable electronic trades executed during the closure. The cost of the S&P 500 ETF (SPY) complex orders that are economically similar and execute away from Cboe does not change. During the closure, the share of complex SPX dollar volume from the largest and most complex trades falls by about half, from 24% to 13%, and recovers to 21% when the floor reopens. The floor facilitates the most difficult trades, but raises the cost of less difficult ones.

These results contribute to the literature on: human intermediation in financial markets, options market microstructure, the design of options execution mechanisms, and the execution of large block trades. A literature on human intermediation in financial markets evaluates whether floor or electronic execution delivers better outcomes (Venkataraman, 2001; Barclay, Hendershott, and McCormick, 2003; Hendershott and Madhavan, 2015), with recent work studying the 2020 NYSE floor closure (Brogaard, Ringgenberg, and Roesch, 2025; Hu and Murphy, 2026).[2] Other papers focus on options market microstruc-

[1] The trading floor has operated since Cboe first opened in 1973 and was the exchange's sole execution mechanism until 1984 when electronic trading for retail orders was first introduced. Cboe trades SPX options exclusively and invested in a new, larger trading floor in Chicago in 2022 (Source: Cboe), as well as partnered with CNBC to broadcast from the Chicago trading floor in 2026 (Source: Cboe).

[2] In equities, the floor's footprint is small enough that disentangling its contribution from confounding variation is difficult. When the NYSE floor suspended operations in March 2020, listed stocks continued to trade simultaneously on more than a dozen venues, while Options Clearing Corporation rules require all listed options to trade on an exchange. Brogaard et al. (2025) find that the suspension widened spreads and

ture and execution costs (e.g., Muravyev and Pearson, 2020; Kaeck, van Kervel, and Seeger, 2022; Hu, Kirilova, Park, and Ryu, 2024). Kaeck et al. (2022) document that effective spreads in SPX options are an order of magnitude larger than price impact, leaving much of the spread unexplained by information asymmetry. Our paper relates to papers on the institutional mechanisms through which options orders execute. Bryzgalova, Pavlova, and Sikorskaya (2023) document an increase in retail options trading via wholesalers, Ernst and Spatt (2026) show how wholesalers internalize retail option order flow through designated market maker assignments and price improvement auctions, Anand and Muravyev (2025) show that auctions facilitating internalization can harm quote competition, and Hendershott, Khan, and Riordan (2026) document how electronic auction mechanisms affect execution quality in options markets. We shed light on the literature that studies how markets execute large block trades. In Seppi (1990), non-anonymous block trading separates liquidity-motivated traders from informed traders. In Grossman (1992), upstairs intermediaries know investors' unexpressed trading interest and use it to match large orders at lower risk. Empirical studies of upstairs equity markets find that traders who can credibly signal that their trades are liquidity-motivated use the upstairs market (Madhavan and Cheng, 1997; Bessembinder and Venkataraman, 2004) and that upstairs brokers access liquidity that the downstairs limit order book does not display (Bessembinder and Venkataraman, 2004).

This literature shares a methodological challenge. Orders are endogenously routed based on characteristics that also determine execution quality, so direct comparison of floor and electronic execution may conflate the mechanism with the selection of orders into it. The literature often addresses this endogeneity using endogenous switching models that impose strong distributional assumptions on the selection equation, making results sensitive to specification error (Madhavan and Cheng, 1997; Conrad, Johnson, and Wahal, 2003; Bessembinder and Venkataraman, 2004; Hendershott and Madhavan, 2015). The COVID-19 closure of the Cboe floor sidesteps this problem. The floor closed from mid-March through mid-June 2020, allowing us to observe how execution costs behave when the mechanism is unavailable.[3] The floor is used almost exclusively for complex trades

increased pricing errors, and Hu and Murphy (2026) find that the floor reduces closing price efficiency for low-spread stocks.

[3]The floor had not previously closed while electronic trading continued. During prior exchange-wide shutdowns, such as the four-day closure following September 11, 2001, all trading venues ceased operations

involving multiple options, called legs, and accommodates about 70% of complex SPX dollar volume; floor-executed complex SPX trades alone account for 16% of OPRA dollar volume across all options. Because SPX trades only on Cboe, displaced orders that would have gone to the floor have no alternative venue. During the closure Cboe activated the Automated Improvement Mechanism (AIM) for SPX, an electronic auction in which a broker exposes a pre-negotiated contra order to competing responses (Hendershott et al., 2026).[4] The changes were reversed at the floor re-opening. The opening therefore restores the floor and removes AIM on the same day. We focus on the opening because it separates the return of the floor from changes in market conditions.[5] The March 2020 closure coincided with the onset of COVID-19, when the VIX rose above 80. Around the June 2020 opening, the VIX averages about 29 in the two weeks before and 33 in the two weeks after, with no jump on June 15. Complex trades on the SPY show no change in effective spreads at the opening (−0.05 cents).[6]

The floor and the electronic mechanisms serve different trades. Floor trades after the opening average 784 contracts and 12.1 legs. AIM attracts the largest electronic trades, but AIM trades in the last two weeks of the closure average only 75 contracts and 3.2 legs, against 36 contracts and 2.6 legs for all electronic trades. A logit model of floor routing after the opening shows that trade size alone yields a pseudo-R^2 of 0.64, and adding the number of legs raises it to 0.70. We therefore pair each floor trade after the opening with its nearest electronic trade during the closure, and separately with its nearest AIM trade, on absolute delta, absolute vega, implied volatility, trade size, time to expiration, and number of legs.

The 1,177 matched pairs cover 36% of floor trades. Matched floor trades execute at an effective spread of 4.45 cents versus 3.15 cents for matched electronic trades. In a regression with the full set of controls and strategy fixed effects, the difference is 1.14

simultaneously.

[4]The permissible leg-ratio limit increased from 3:1 to 25:1. Orders with leg ratios above 25:1 account for 7.50% and 6.99% of dollar volume among the largest and most complex SPX trades during the pre-closure and opening periods, respectively. Therefore, these orders do not account for the decline in the largest and most complex trades documented below during the closure period.

[5]Cboe announced a June 8 opening on May 26 and postponed it to June 15 after civil unrest restricted physical access to the building (Cboe Global Markets, 2020a,b). Around the floor closure, matched floor trades also cost more than their electronic matches (Internet Appendix Section IA.I).

[6]SPY is an ETF that tracks the S&P 500. SPX options are European and cash-settled on the S&P 500 index, while SPY options are American and settle into ETF shares.

cents, 36% of the electronic spread. Matching floor trades only to AIM trades doubles the gap. The 1,026 pairs matched to AIM cover 31% of floor trades, and matched floor trades execute at 4.72 cents versus 2.18 cents for AIM trades. With the full set of controls, the difference is 2.44 cents, so floor trades pay about twice as much as comparable AIM trades. Market maker revenue subtracts from the effective spread the cost of delta-hedging. Market maker revenues on floor trades exceed those on their electronic matches by 1.15 cents. Against AIM, the difference ranges from 2.31 to 2.47 cents.

The floor spread differences change sign with trade size: small floor trades have wider effective spreads than electronic trades of the same size, and the largest have narrower spreads. In a regression setting, floor trades at or below the 25th percentile of the floor size distribution (166 contracts) pay 3.49 cents more than electronic trades of the same size. Above the 90th percentile (1,949 contracts), the relationship reverses, and floor trades pay 4.35 cents less.

The cost comparison, however, covers only the trades that electronic markets accommodate. When the floor closes, the largest and most complex trades become less prevalent. Floor trades without a matched electronic counterpart average 1,031 contracts and 17 legs, against 342 contracts and 3.1 legs for matched floor trades. Replicating a floor trade at the 90th percentile of size (1,950 contracts) electronically requires an average of 65 trades spread over 1.3 minutes and costs about $6,200, against $2,400 to $4,200 for floor trades of similar size.

Additionally, as is standard in the literature, we measure only the costs of realized trades. Without firm quotes for all possible complex trades, we cannot measure the costs of trades that are desired but not executed. Opportunity costs of failing to trade arise when the expected or realized cost of trading exceeds the benefit of the trade (Perold, 1988). For floor trades, we do not observe the cost of a trade when the investor does not bother to submit an order or when the cost negotiated on the floor exceeds the investor's willingness to pay. This biases measured trading costs downward (Hendershott, Li, Livdan, and Schürhoff, 2024). AIM auctions never fail once initiated, but an investor may be unable to find a firm to initiate one for a large trade. This selects which desired trades are executed electronically and again biases measured trading costs downward.

Selection matters least for smaller, less complex trades, so it is unlikely to drive the lower cost of electronic auctions for small trades. In contrast, few large complex

trades execute electronically, and those that do are likely the ones that were cheapest to execute, which tilts the comparison of large trades toward electronic execution. Hence, although selection affects which trades are executed both on the floor and in the electronic mechanisms, it is unlikely to explain why smaller, less risky complex trades are cheaper electronically and larger, riskier complex trades are cheaper on the floor.

Why do the largest and most complex trades become less prevalent without the floor? Electronic auctions and the floor differ in three ways. The first, and the main one, is pre-commitment. AIM cannot start until the initiating broker has paired the order with contra interest that guarantees execution of its full size (Cboe Global Markets, 2026, Rules 5.37 and 5.38). Neither the complex order book nor the floor requires a contra party in advance. The book, however, offers little committed liquidity: market makers are not required to quote complex strategies (Cboe Global Markets, 2026, Rule 5.33, Interpretation .01), they initiate 94.5% of the packages that execute against the book, and the Complex Order Auction exposes an order for about 100 milliseconds and fills what it can. On the floor, a broker represents the order to the crowd without first lining up a contra party. Without the floor, a large complex trade can therefore execute as a single trade only if a firm commits to the entire position before the auction starts.

The second difference is flexibility. Several in-crowd market makers can share an order at a common price, and an unfilled balance can keep working (Cboe Global Markets, 2026, Rules 5.52(b) and 5.85), while AIM often allocates an order in fixed proportions according to an allocation algorithm (Cboe Global Markets, 2026, Rule 5.38). Pooled across trade sizes and conditional on execution, 26.8% of AIM fills cover the whole order leg, against 14.3% of floor fills. AIM fills cluster at the 50% and 40% shares of the leg that its allocation rule assigns to the initiating order. Floor fills also cluster at half the leg, consistent with a similar participation entitlement for the designated or lead market maker on the floor (Cboe Global Markets, 2019, Rule 5.85(a)), but they cluster less at 40% and otherwise come in many sizes. The crowd sizes its own pieces. Small floor trades more often go whole to a single counterparty: at or below the 25th percentile of size, 46.8% of floor fills cover the whole leg, against 29.9% of AIM fills. Large floor trades are more often divided among several market makers: above the 90th percentile, 1.3% of floor fills cover the whole leg, against 3.9% of AIM fills. Each market maker in the crowd takes only the portion of the trade that fits its inventory and risk limits, so it holds less unhedged risk until it can hedge

and can offer a better price (Grossman and Miller, 1988). Consistent with this risk-sharing channel, floor trades above the 90th percentile of size pay less than electronic trades of the same size.

The third difference is screening. Identities are known in the crowd, and a trader can forgo anonymity through a floor broker (Grossman, 1992). Floor brokers also know latent traders whose interest the electronic book does not display, and they can share the initiator's motive and remaining size with the crowd, which draws competing liquidity and divides the position among suppliers (Harris, 2003). Non-anonymity lets a liquidity-motivated trader credibly commit not to trade further, which separates it from informed traders (Seppi, 1990). In Benveniste, Marcus, and Wilhelm (1992), repeated dealing lets the specialist discipline brokers who bring informed orders. On the Paris Bourse, upstairs block trades carry less information than block trades in the electronic limit order book, despite being larger (Bessembinder and Venkataraman, 2004). Futures price impact suggests a similar role for the floor in larger SPX trades. Across all trades, each $10,000 of signed delta per index point moves futures prices by 0.43 basis points within 10 seconds for electronic trades, and by 0.54 and 0.68 basis points less for floor trades at 10 and 15 seconds, so floor trades have no positive futures price impact. At or below the 25th percentile of size, neither floor nor electronic trades have detectable futures price impact, so price impact does not explain the higher floor cost of small trades. Above the 90th percentile, floor trades move futures prices by 0.99 to 1.09 basis points per $10,000 of signed delta less than electronic trades with the same signed delta, at horizons of 10 and 15 seconds. The lower impact of the largest floor trades is consistent with screening, although uninformed traders may also select into the floor for their largest trades, the difference is not significantly larger than for mid-sized trades, and hedging pressure and the timing of the return window can also differ between venues.

These findings speak to market design. The floor accounts for 70% of complex SPX dollar volume but only 1.6% of orders. A small number of large, complex positions generate most of its activity. For the largest trades, the floor costs less than electronic execution and has lower futures price impact. The largest multi-leg positions have no electronic match at all. Electronic execution is cheaper for smaller, easier trades. Having both mechanisms simultaneously available could lower aggregate transaction costs, provided orders are routed to the mechanism that suits them. Why Cboe does not offer an electronic auction

with the flexibility of the floor, and whether the floor's economics survive when its smaller trades execute electronically, are questions for future research.

I. Data and Institutional Details

Cboe Options Exchange ("C1") operates a hybrid market in which electronic order handling and open outcry floor trading coexist (Cboe Global Markets, 2019, 2026). On the electronic side, individual option series trade in a limit order book (LOB), the Simple Book, where resting orders execute against incoming marketable orders in price-time priority. Multi-leg orders have a parallel facility, the complex limit order book (CLOB), where an order for two or more legs at a guaranteed net price can rest and execute against other complex orders, or against the individual series books when their combined liquidity permits. Cboe also operates the Complex Order Auction (COA), which exposes a marketable complex order to competing responses for 100 milliseconds in most classes and then fills what it can against the responses and resting interest. Any unfilled balance rests in the CLOB.[7]

C1 also has a physical trading floor in Chicago. Floor brokers receive orders from clients, such as asset managers, hedge funds, and the proprietary desks of clearing members, and represent them in the trading crowd, where registered market makers respond with competing bids and offers in open outcry. Orders reach the floor through PAR (Public Automated Routing System), the workstation that connects the electronic system with floor execution.

A. The Concentration of Floor Trading in SPX

Floor trading in U.S. listed options is concentrated in one product. Table I reports average daily volume across all option classes reported through OPRA, by order type and execution venue. In Panel A, complex orders account for about half of OPRA dollar volume (50.67%) and 28% of contract volume, and floor execution accounts for 22% of dollar volume and 8.5% of contract volume. A floor contract trades at about three times the dollar value of

[7] Under Cboe rules, a marketable complex order that does not opt out of the COA process initiates an auction upon entry if its net limit price is at or within the synthetic best bid and offer derived from the individual leg books. The COA is available for all classes during our sample period.

an electronic one, mostly because floor volume concentrates in SPX.

Panel B turns to SPX. The class accounts for $3.96 billion in average daily dollar volume, about 26% of OPRA dollar volume, and for 4% of contract volume. Complex orders make up 86% of SPX dollar volume ($3.39 of $3.96 billion). Floor-executed complex SPX orders account for $2.38 billion per day, about 70% of SPX complex dollar volume and 16% of OPRA dollar volume. Floor single-leg SPX volume is 0.36% of OPRA dollar volume. SPX accounts for about three quarters of all floor dollar volume.[8]

TABLE I HERE

Because SPX is a proprietary Cboe product, all SPX volume, floor and electronic, trades on one exchange under the same rule book and clearing arrangements. Differences between floor and electronic SPX trades therefore do not reflect a choice of listing venue, clearing, or regulation.

B. Complex Options Strategies

A complex options order is a single instruction to execute two or more option legs simultaneously at a guaranteed net price. The net price is the sum of the leg prices weighted by the leg ratios (a call bought at $3.00 and a put sold at $1.00 in a 1:1 ratio trade at a net debit of $2.00, wherever the individual legs trade within that constraint). Simultaneous execution eliminates execution risk, the possibility that one leg fills while another does not, which would leave the trader with an unintended position. For strategies whose legs offset each other's exposures, the package hedges only if every leg executes.

SPX complex strategies include vertical and calendar spreads, straddles and strangles, butterflies, condors, and ratio spreads, which Table IA.IX of the Internet Appendix defines. A Combo pairs a long (short) call with a short (long) put at the same strike and expiration and replicates a long (short) S&P 500 futures position. Packages built from Combos, sometimes with additional legs or other Combos, are among the largest packages.

C. The COVID-19 Cboe Trading Floor Closure

On March 16, 2020, Cboe closed the C1 trading floor in response to the COVID-19 pandemic, effective immediately and until further notice (Cboe Global Markets, 2020c,d).

[8] These figures cover the pre-closure and opening periods defined in the notes to Table I.

All routing to the floor was disabled, and orders marked for floor execution were rejected. Floor brokers could no longer use the PAR workstations or operate in the trading crowd.

To keep complex SPX orders trading during the closure, Cboe made two temporary changes. First, complex SPX orders with leg ratios up to 25:1, including strategies previously restricted to floor execution, became eligible for electronic execution in the CLOB and the COA. Second, Cboe activated the Automated Improvement Mechanism (AIM) for SPX complex orders during regular trading hours, a function it had not previously allowed for that class. An AIM auction can start only after the initiating member pairs the agency order with contra-side interest that guarantees execution of the full order at a stop price (Cboe Global Markets, 2026, Rules 5.37 and 5.38). Other participants can then improve on that price and receive allocations.[9] SAM, the solicitation auction mechanism, remained inactive for proprietary classes including SPX throughout the closure.

Floor trading resumed on June 15, 2020, and all temporary rule changes were reversed that day. The 3:1 ratio limit on electronic eligibility was reinstated, AIM for SPX during regular trading hours was deactivated, and the floor again became the required channel for SPX complex orders with leg ratios above 3:1. The closure and the opening provide the identifying variation. The opening restores the floor and removes AIM and the expanded electronic eligibility on the same day.

D. Data

We combine trade records from the Options Price Reporting Authority (OPRA) and from the Cboe Options trade-by-trade execution file with option and futures quotes from LSEG Tick History. OPRA reports every U.S. listed options trade in real time with its price, contract size, millisecond timestamp, exchange identifier, and a condition code that records whether the trade executed on the floor or electronically and whether it was part of a complex package. We use OPRA for total dollar volume across all U.S. options exchanges and for the SPY placebo samples, because SPY options trade on many exchanges.[10]

The Cboe data is the Enhanced U.S. Options Trade-by-Trade Execution Detail file from

[9] Cboe's regulatory notice specifies that both changes applied only during the electronic-only period and would be reverted when open outcry trading resumed (Cboe Global Markets, 2020d).

[10] Because OPRA neither links legs nor identifies initiators, we group SPY legs into packages with the time-window rule of Li (2022) and sign them with the quote rule.

Cboe DataShop. It reports every execution on C1 in 2020 for all underlyings, electronic and open outcry, single-leg and complex, and therefore contains every SPX execution. Each execution appears as two records, one for the buyer and one for the seller, linked by a common execution identifier. Each record reports the participant's side and capacity, which Cboe classifies as customer, professional customer, broker-dealer, firm, or market maker.[11] A trade-type field identifies the mechanism of each execution and, for some executions, which record belongs to the initiating order.

OPRA reports complex trades leg by leg, without an identifier linking the legs of a package. The Cboe file links them. For electronic trades, a complex execution identifier links all legs of a complex execution. For open outcry trades, the file records the time at which the trade was agreed in the crowd, and we define a floor package as all records sharing an agreement timestamp on a given day.[12] Because the file has no order identifier, an electronic order that executes against several counterparties can appear under several complex execution identifiers, and we combine complex executions with the same legs and sides, leg ratio, initiator capacity, mechanism, and microsecond timestamp into one electronic order. The Cboe identifiers link the legs of each SPX package directly, and we do not need a time-window rule (such as the 3-millisecond rule) of Li (2022), which would split many floor orders.[13] We classify each package into a standard strategy type (vertical spread, butterfly, condor, Combo, Combo paired with additional legs, and others) with the leg-configuration rules of Li (2022). About 95% of complex packages are classified. The remaining 4.9%, labeled non-standard, are used only for volume shares, such as the share of SPX volume that is complex, and in none of the spread or other analyses. Sections IA.IV and IA.III of the Internet Appendix detail the procedure.[14]

The Cboe data also identify the initiating side of most trades, which OPRA does not. For electronic trades, the trade type marks the initiating side, such as the order that removes liquidity from the CLOB, starts a COA, or is the agency order in AIM. On the floor,

[11] Cboe records the capacity of each order (Cboe Global Markets, 2026, Rule 1.1): customer (C), a person that is not a broker-dealer; professional customer (U), a non-broker-dealer that places more than 390 listed options orders per day on average; firm (F), the proprietary account of a clearing member; broker-dealer (B); and market maker (M).

[12] Cboe confirms that this agreement timestamp is common to all legs and fills of the same floor order.

[13] About two thirds of multi-leg SPX floor orders in 2020 span more than 3 milliseconds between their first and last reported leg.

[14] The Internet Appendix may be found in the online version of this article.

an order represented in the crowd trades against the in-crowd market makers who respond to it. We use the trade type when it separates the record of the order represented in the crowd from the response. Otherwise, the side opposite a single market maker initiates or, failing that, the initiator of other fills on the same leg of the same order. The fills that remain unsigned are crosses of two orders represented in the crowd, mostly between two market makers and in some cases between two institutional investors. A package spread requires a direction on every leg, and we exclude the floor orders that contain unsigned fills (16% of floor orders in the two weeks before the closure and 22% in the two weeks after the opening). When a market maker initiates a trade, we measure execution costs from the side of its counterparty, the end user (Section II.C).

LSEG Tick History (formerly Thomson Reuters Tick History) provides intraday quotes for SPX options, including the PM-settled SPXW series. We measure option spreads and implied volatilities against the last quote on each leg before the package price is fixed, which is the agreement timestamp for floor trades and the execution timestamp for electronic trades.[15] Quote midpoints drift in the direction of the trade before execution on every venue, by 4.81 cents for floor packages and by 1.61 to 4.26 cents for electronic packages between 15 minutes and 1 minute before execution. A benchmark taken before the price is fixed would raise floor spreads most. Measuring spreads when the price is fixed does not inflate floor costs relative to electronic ones (Internet Appendix Section IA.V).

All variables are constructed at the package level. Trade size is the total number of contracts across all legs. Dollar volume is the sum over legs of the leg price times the leg's contracts, multiplied by the SPX contract multiplier of $100. The contract ratio is the ratio of the largest to the smallest leg size in contracts. Absolute net delta, vega, and gamma are contract-weighted averages across legs of option sensitivities evaluated when the package price is fixed. Implied volatility comes from the quote midpoint on each leg. Days to expiration is the contract-weighted average of the remaining calendar days across legs. Following Li (2022), quoted and effective spreads are computed at the package level against a synthetic midpoint, the contract-weighted average of leg midpoints with each leg signed by the initiator's observed side on that leg. The hedge cost is the cost of crossing half the E-mini quoted spread on the package's absolute delta, and market maker revenue is the effective spread minus the hedge cost. Spreads are expressed in cents per unit of the

[15] The Cboe file also reports a national best bid and offer for each record.

underlying and winsorized at the 0.5% level in each tail.

Futures data are microsecond-stamped trades and quotes for E-mini S&P 500 futures (ES) on the CME Globex platform, also from LSEG Tick History. We use the front-month contract and roll to the next contract on the first day of the expiration month.[16] Each package is matched to the prevailing futures quote and the most recent futures trade at the time its price is fixed, the timestamp used for option quotes.

The sample runs from January 2 to October 21, 2020. It contains a pre-closure period in which floor and electronic execution are available, the floor closure from March 16 to June 14, 2020, and the opening period. The sample begins in January 2020 to avoid contamination from the rule change effective on October 21, 2019, and it ends in October 2020 to exclude the November 16, 2020 rule amendment. We exclude trades executed outside regular trading hours. Section II describes who trades on each venue.

II. Trading Activity, Floor Routing, and Trade Initiators

The closure changes the mechanisms available to complex SPX orders. Before the closure and after the opening, an order can execute on the floor or electronically, and orders with leg ratios above 3:1 must execute on the floor. During the closure, an order can execute only electronically, and AIM is available. This section describes which trades execute on the floor and who initiates them. We first describe trading activity and execution costs in the pre-closure, closure, and opening periods. We then estimate which order characteristics predict floor execution, and find that size and the number of legs provide most of the fit. Finally, we explain how the identity of the initiator determines the side from which we measure execution costs.

A. Descriptive Statistics

Complex trades account for most SPX dollar volume, and the floor executes most of it when it is open. Figure 1 plots daily SPX option dollar volume as a share of total OPRA dollar volume. Panel A covers the full sample from January to October 2020. Complex SPX trades account for about 20 to 30% of OPRA dollar volume in January, with a dip to about 12% in early February, and single-leg SPX trades for roughly 3 to 5%. Both series

[16] E-mini futures expire quarterly on the third Friday of March, June, September, and December.

rise from late February. Complex volume peaks above 50% of OPRA dollar volume in the first weeks of the closure and then reverts toward pre-closure levels. Panel B covers the four weeks around the opening. Neither series shows a discontinuity at June 15.

FIGURE 1 HERE

Figure 2 splits complex SPX volume by venue. Before the closure, the floor executes the majority of complex SPX dollar volume (Panel A). During the closure, floor volume drops to zero and electronic venues execute the flow. At the opening, floor trading resumes and electronic volume, now without AIM, falls back toward pre-closure levels. Panel B shows that the change at June 15 is immediate. Floor volume jumps from zero to about 20% of OPRA dollar volume, and electronic volume falls from about 29% to about 7%, a drop significant at the 1% level. The share of complex SPX dollar volume from the largest and most complex trades also falls during the closure and recovers after the opening (Section IV.C).

FIGURE 2 HERE

Table II reports daily means of trading activity and execution costs for complex SPX trades (Panel A) and complex SPY trades (Panel B) in the pre-closure, closure, and opening periods. Complex trades account for 86 to 88% of SPX dollar volume in every period. Before the closure, complex SPX trades average 53 contracts at a net price of $0.01 per unit of the underlying, because offsetting legs lower the net price of a package. Average trade size falls from 53 to 40 to 34 contracts across the three periods.

TABLE II HERE

Effective spreads are measured from the side of the initiator, or from the side of its counterparty when a market maker initiates the trade (Section II.C). Equally-weighted effective spreads on complex SPX trades are 1.93, 2.34, and 1.82 cents in the three periods, and value-weighted effective spreads are 5.03, 6.51, and 3.31 cents. Spreads rise during the closure and fall at the opening, and market maker revenues follow the same path under each weighting. Hedge costs range from 0.34 to 0.39 cents equally-weighted, about a seventh to a fifth of the average complex effective spread.

Panel B reports complex SPY trades, none of which can route to the Cboe floor (Section III.E). Their equally-weighted effective spreads follow the same path as SPX spreads (1.09, 1.34, and 0.77 cents), and the decline from the closure period to the opening period is significant at the 1% level. The regressions in Section III control for these market-wide changes with the VIX and the daily SPY complex spread. We do not study single-leg SPX trades, of which only a small part trades on the floor. A single-leg trade executes against the quotes of its own series, and those quotes are also the benchmark for its effective spread. When a market maker takes an investor's limit order that sets the best quote, the benchmark already includes that order, so the effective spread does not measure the cost the investor pays. The benchmark of a complex trade is the synthetic quote built from the quotes of its legs, which the complex trade usually does not trade against. Floor volume comes from a small share of complex orders.

B. Floor Routing

Average costs mix trades of very different size and complexity, and the mix changes with the mechanisms available. To see which orders route to the floor, we estimate a logit model of floor execution on complex SPX orders executed during the opening period (June 15 to October 21, 2020). In this period, as before the closure, orders can execute on the floor or electronically, and AIM is no longer available. The floor trades in our matched sample also come from this period. Electronic executions are grouped into orders before estimation.

$$\Pr(\text{Floor}_i = 1) = \Lambda(\alpha + \beta' X_i + \mu_s) \tag{1}$$

The dependent variable equals one if order i executes on the trading floor and zero if it executes electronically. The vector X_i includes log trade size, log number of legs, absolute delta, days to expiration, vega, gamma, implied volatility, and an SPXW indicator for PM-settled contracts. μ_s denotes strategy fixed effects and $\Lambda(\cdot)$ the logistic function. If the floor serves the trades that are hardest to execute electronically, the probability of floor execution rises with trade size and the number of legs, holding the option characteristics of the package fixed.

TABLE III HERE

Table III reports logit coefficients in Panel A and average marginal effects in Panel B. Floor execution occurs for 1.6% of the orders, because most complex orders are small packages that execute electronically. Trade size and the number of legs provide most of the model's fit. Size alone yields a pseudo-R^2 of 0.64. Together with the number of legs it reaches 0.70, against 0.80 with all controls and strategy fixed effects. Size and legs are significant at the 1% level in every specification, including those with strategy fixed effects. They predict floor execution within strategy types as well as across them. Absolute delta is not statistically significant.

In the full specification, a doubling of trade size is associated with a 0.51 percentage point higher probability of floor execution, and a doubling of the number of legs with a 0.93 percentage point higher probability, against an unconditional probability of 1.6%. Floor execution is also more likely for higher-vega packages and less likely for longer maturities, higher implied volatility, and PM-settled contracts. Floor execution is associated mainly with how large and how complex the trade is. Many floor trades have no electronic counterpart, and Section IV.C estimates the cost of assembling a trade of their size from electronic orders.

C. *Trade Initiators*

Floor trades are large and complex, and who initiates them matters for how execution costs are measured. Internet Appendix Table IA.XV reports the capacity of the initiating party of complex SPX packages on each venue.

Effective spreads are conventionally measured from the side of the trade initiator, on the assumption that the initiator demands liquidity and pays the spread. In the CLOB, market makers rarely post resting orders. They monitor the complex limit orders that brokers post for their customers and execute against those priced to their advantage (Li, Musto, and Pearson, 2023). The initiator is then a market maker, and the investor who posted the resting order pays the spread. Our focus is the cost paid by end users, the traders other than market makers, such as retail and institutional investors, as in Gârleanu, Pedersen, and Poteshman (2005). We therefore measure execution costs from the side of the end user: the initiator, or its counterparty when a market maker initiates the trade.[17]

[17]Measured from the side of the initiator, the effective spread of a trade that a market maker initiates is a gain to the market maker rather than a cost to the investor on the other side. On the floor, where market

Market maker revenue follows the same convention. All execution costs in the paper, including those in Table II, are end-user measures.

The behavior of market makers in the CLOB is consistent with little standing liquidity for complex strategies. Complex strategies far outnumber individual series, and Cboe market makers are not required to quote in the CLOB (Cboe Global Markets, 2026, Rule 5.33, Interpretation .01). A large complex order therefore cannot expect resting interest of its size in the CLOB. It can go to the floor, where a broker represents it to the crowd, or to an electronic auction during the floor closure. Section IV.D compares these mechanisms.

The logit describes the segment of complex SPX order flow that concentrates on the floor. It consists of large, multi-leg packages initiated by retail and institutional investors and filled by the trading crowd. Because most electronic trades are small and simple, only the smaller and simpler floor trades have close electronic counterparts. Section III compares matched floor and electronic trades around the opening.

III. Execution Quality

Size and the number of legs predict which complex orders route to the floor (Section II.B). This section compares the execution costs of floor and electronic trades of similar size, complexity, and option characteristics. Trades are not randomly assigned to venues, and a direct comparison of floor and electronic execution would confound venue effects with differences in order flow (Section II). During the closure, every complex SPX order executes electronically, including orders that would otherwise route to the floor. We pair floor trades after the opening with observably similar electronic trades from the closure through one-to-one nearest-neighbor matching. We first describe the sample and the matched pairs. We then compare daily mean costs and estimate the cost difference trade by trade for all matched pairs. Finally, SPY complex trades that do not route to the Cboe floor provide a placebo for a market-wide change in execution costs at the opening.

A. Sample

We focus on the opening of the trading floor in June 2020. The closure in March 2020 coincided with the onset of the COVID-19 pandemic, a period of extreme volatility, volume

makers initiate 1.2% of packages, the two conventions give similar average costs.

surges, and shifting market participation that could confound any comparison through channels unrelated to the venue. The opening occurs during a period with less extreme market conditions.[18]

FIGURE 3 HERE

Figure 3 shows the volatility environment. The VIX rises from about 15 to above 80 at the onset of the closure and declines through the spring. In June it moves between about 24 and 41, well below the March peak but above its January level (Panel A). In the matching window, the VIX averages about 29 before the opening and 33 after it (Panel B). It jumps on June 11, two trading days before the opening, and shows no discontinuity at the opening date. Every specification controls for the VIX.

B. Matching

For each floor trade executed in the two weeks after the opening (June 15 to June 26, 2020), we find its nearest electronic counterpart from the two weeks before the opening (June 1 to June 14, 2020). Matching is one-to-one and without replacement. The six matching covariates are absolute delta, absolute vega, implied volatility, trade size in contracts, days to expiration, and number of legs. We use a caliper of 0.25 standard deviations, following Rosenbaum and Rubin (1985), applied to each matching variable, which keeps matched pairs close on every dimension.[19] Electronic executions are grouped into orders before matching.

TABLE IV HERE

Before matching, floor trades are much larger and more complex than electronic trades (Table IV, Panel A). Floor trades average 784 contracts and 12.1 legs, against 36 contracts and 2.6 legs electronically. They also have longer maturities (84 versus 22 days to expiration) and higher absolute vega (1.20 versus 0.20), with similar absolute delta (0.02 versus 0.03). After matching, the differences disappear. Trade size is 342

[18] Internet Appendix Section IA.I repeats the matched analysis around the closure. Matched floor trades also cost more than their electronic matches, by 6.33 cents with all controls, amid much higher volatility.

[19] Our results are robust to calipers of 0.15, 0.20, 0.30, and 0.35 standard deviations (not tabulated). Internet Appendix Section IA.II repeats the match with the electronic sample restricted to AIM trades.

versus 339 contracts, the number of legs is 3.08 versus 3.07, and days to expiration is 25.17 versus 25.16. The 1,177 matched pairs, a match rate of 35.8%, cover the overlap of the two distributions, and the cost differences we estimate apply to this shared support. Unmatched floor trades are larger and more complex still, with 1,031 contracts, 17.2 legs, 117 days to expiration, and absolute vega of 1.64. Section IV.C studies them.

Panel B reports execution quality measures, which are not matching variables. Hedge costs are 0.28 cents on each side, because matching on absolute delta equalizes them. Matched floor trades pay an average effective spread of 4.45 cents, against 3.15 cents for matched electronic trades. The difference of 1.30 cents per unit of underlying is about 40% of the electronic mean. Market maker revenue, the effective spread net of the hedge cost, shows the same gap (4.17 versus 2.87 cents). With equal hedge costs, the full spread difference accrues to intermediaries as revenue. Unmatched floor trades pay 5.46 cents in effective spread and 5.25 cents in market maker revenue. Measuring spreads when the package price is fixed does not inflate this gap. Midpoints drift in the direction of the trade on every venue before execution, most for floor packages (Section I and Internet Appendix Section IA.V).

C. Univariate Analysis

The cost of matched SPX trades rises at the opening. Figure 4 plots daily mean effective spreads, market maker revenues, and hedge costs in the two weeks before and after the opening. The SPX series follows the matched sample, with matched electronic trades before the opening and matched floor trades after it. The SPY series covers matched SPY complex trades, a matched version of the placebo group of Section III.E. The mean daily matched SPX effective spread rises from 2.88 cents before the opening to 4.30 cents after it (Panel A). The increase of 1.42 cents is significant at the 5% level. SPY complex effective spreads average about 1 cent both before and after the opening. A market-wide change in execution conditions would have moved the SPY series as well. Because the SPX orders are matched, the shift is unlikely to merely reflect differences in size, legs, or maturity.

FIGURE 4 HERE

Panels B and C decompose the effective spread into market maker revenue and hedge cost. Market maker revenues on matched SPX trades rise from 2.63 to 4.03 cents. The

increase of 1.40 cents accounts for almost all of the increase in the effective spread. Hedge costs of matched SPX trades change by an insignificant 0.02 cents between the two periods. SPY complex market maker revenues stay at about 0.9 cents and hedge costs at 0.07 cents.

D. Regression Analysis

The daily averages do not control for market conditions or for the characteristics of each trade. We estimate the cost difference trade by trade on the matched sample:

$$Y_i = \alpha + \beta \,\text{Opening}_i + \gamma' X_i + \mu_s + \varepsilon_i \tag{2}$$

The dependent variable Y_i is the effective spread or the market maker revenue of trade i, in cents per unit of underlying. Opening_i equals one for floor trades executed after the opening and zero for matched electronic trades during the closure. X_i contains the VIX, the average daily effective spread of SPY complex trades, and trade-level controls (quoted spread, absolute delta, days to expiration, vega, gamma, implied volatility, log trade size, and a PM-settled indicator). The market maker revenue regressions omit absolute delta, which determines the hedge cost. μ_s denotes strategy fixed effects. The electronic matches execute during the closure, when AIM is available for SPX complex orders. The coefficient β compares the floor with the electronic mechanisms available during the closure, including AIM.[20] Standard errors are clustered by date (20 days). If the floor executes comparable trades at no higher cost than the electronic mechanisms of the closure, β is zero or negative. A positive β means that end users pay more on the floor for trades that these mechanisms also execute.

TABLE V HERE

Matched floor trades execute at wider spreads than their electronic counterparts (Table V). With the VIX as the only control, the Opening coefficient is 0.76 cents and is not statistically significant. Part of the raw difference of 1.30 cents loads on the VIX, which is higher after the opening (Figure 3). Controlling also for the daily SPY complex spread

[20] AIM trades account for 24.13% of the electronic matches. Internet Appendix Section IA.II matches post-opening floor trades directly to AIM trades from the closure. With the full set of controls, the Opening coefficient for effective spreads is 2.44 cents, significant at the 1% level.

raises the estimate to about 0.98 cents, significant at the 5% level. With the full set of controls, the estimate is 1.14 cents, significant at the 1% level, or 36% of the 3.15-cent mean effective spread of matched electronic trades. Market maker revenues mirror this result. The full-specification estimate is 1.15 cents, significant at the 1% level and 40% of the 2.87-cent mean market maker revenue of matched electronic trades. Intermediaries earn more per unit of underlying on floor trades than on comparable electronic trades.

E. Placebo Tests

The matched comparison could also pick up a market-wide change in option execution costs around June 15. SPY complex trades carry the same equity exposure as SPX. SPY options are multi-listed, and we keep electronic executions on the fourteen exchanges other than Cboe Options and C2. No trade in the sample routes to the Cboe floor.[21] Table VI estimates Equation (2) on these trades, without the SPY complex spread and the PM-settled indicator among the controls. No SPY complex trade changes venue at the opening. We estimate the specification without matching on all 428,627 electronic SPY complex trades, with Opening_i equal to one for trades executed after the opening.

TABLE VI HERE

The Opening coefficient for SPY complex effective spreads is −0.05 cents with the full set of controls and is not statistically significant. Market maker revenues give the same estimate. Matched SPX complex trades cost 1.14 cents more. The SPX complex cost difference does not appear to simply reflect a market-wide shift in execution costs around the opening date.

Matched floor trades after the opening cost 1.14 cents more than matched electronic trades during the closure, and SPY complex spreads do not change at the opening. The matched comparison covers 35.8% of floor trades. These smaller and simpler trades cost more on the floor than their electronic matches. Section IV compares floor and electronic trades across the size distribution and studies the larger and more complex floor trades.

[21] We exclude Cboe Options and C2 because the return of floor activity may alter the composition of electronic order flow on these venues. NYSE American and Nasdaq PHLX operate trading floors, which account for a residual share of SPY activity and re-opened before June 15, 2020. The other twelve exchanges are NYSE Arca, Nasdaq ISE, Nasdaq GEMX, Nasdaq MRX, Nasdaq BX Options, Nasdaq NOM, BOX, MIAX, MIAX Pearl, MIAX Emerald, Cboe BZX Options, and Cboe EDGX Options.

IV. Differences in Floor and Electronic Trading and Costs

The matched comparison in Section III holds the characteristics of each trade fixed and covers 35.8% of floor trades. The matched difference averages over trades of very different size. Size and the number of legs predict floor routing (Section II.B), and the floor trades outside the match are larger and have more legs. This section studies all complex trades around the opening and builds up the comparison by trade size. We first show that total execution costs fall at the opening, that the decline is concentrated among the largest trades, and that the cost difference between floor and electronic trades changes sign with trade size. We then test whether floor trades carry information about the index. Next, we study the floor trades that have no electronic match and estimate what it costs to assemble a large floor trade from electronic orders. Section IV.D compares the electronic auctions with the floor in pre-commitment, flexibility, and screening, with evidence from the size of fills and from futures price impact. Unless noted otherwise, the analyses use the full sample, which contains every complex SPX floor trade in the two weeks after the opening (June 15–26, 2020) and every electronic complex SPX order in the last two weeks of the closure (June 1–12, 2020), 3,392 floor trades and 205,725 electronic orders. Electronic executions are grouped into orders, and costs are end-user measures (Section II.C). We sort trades into five bands of the floor trade-size distribution. The bands are at or below the 25th percentile (166 contracts or fewer), between the 25th and 50th percentiles (167 to 380 contracts), between the 50th and 75th (381 to 839), between the 75th and 90th (840 to 1,948), and above the 90th percentile (1,949 contracts or more).

A. Execution Costs by Trade Size

The daily mean effective spread of matched complex SPX trades rises from 2.88 to 4.30 cents at the opening (Section III.C). Total dollar effective spreads move in the opposite direction. In Panel A of Figure 5, the total dollar effective spread paid on complex SPX trades falls from about $2.5 million per day during the closure to about $1.6 million per day after the opening. In Panel B, complex SPX dollar volume falls from about $3.1 billion to $2.2 billion per day. The two declines are significant at the 5% level. The decline in total spreads is concentrated among the largest trades. For trades above the 90th percentile of floor trade size (Large, > P90, in Panel A), total effective spreads fall from about $0.7

million to \$0.15 million per day, significant at the 1% level. For all other trades (Other, ≤ P90), they fall from about \$1.8 million to \$1.45 million per day, a change that is not statistically significant.

FIGURE 5 HERE

The two measures answer different questions. The daily mean follows matched trades whose characteristics are held fixed. The dollar total sums spreads over all complex trades and contracts, whose composition changes at the opening, and the floor-electronic difference changes sign with trade size.

We compare floor and electronic trades within size bands with the following regression:

$$Y_i = \alpha + \sum_k \beta_k \, \text{Opening}_i \times \text{Size}_{k,i} + \sum_k \delta_k \, \text{Size}_{k,i} + \gamma' X_i + \mu_s + \varepsilon_i \tag{3}$$

Y_i is the effective spread or market maker revenue of trade i in cents. Opening_i equals one for floor trades after the opening and zero for electronic trades during the closure. $\text{Size}_{k,i}$ equals one if trade i falls in size band k. The coefficient β_k measures the floor-electronic difference within band k. The controls X_i are those of Equation (2). As in Table V, the specifications start from the VIX alone, then add the daily SPY complex effective spread, and then add the full set of trade-level controls with strategy fixed effects μ_s. Standard errors are clustered by date.

TABLE VII HERE

Table VII shows the reversal at the two ends of the size distribution. At or below the 25th percentile, the floor-electronic difference in effective spreads is 5.63 cents with the VIX as the only control and 5.86 cents with the SPY complex spread. With the full set of controls, it falls to 3.49 cents and remains significant at the 1% level. Above the 90th percentile, the difference is between −4.00 and −4.35 cents and significant at the 1% level in every specification. Between the 25th and 50th percentiles, the difference of 1.73 to 1.93 cents with market-level controls falls to 1.10 cents and loses significance with trade-level controls. The two bands between the 50th and 90th percentiles show no significant difference. With the full set of controls, market maker revenues give close estimates, 3.49 cents in the smallest band and −4.30 cents in the largest.

The matched difference of 1.14 cents in Section III comes from trades in the lower half of the floor size distribution, where floor trades pay more with market-level controls. Matched floor trades average 342 contracts, and about 5% of them exceed 1,000 contracts (Figure 6). Harris (2003) predicts that options markets with small transaction sizes will automate and that floors will serve mainly large traders. The cost differences line up with that prediction. We next examine whether futures price impact accounts for the higher floor cost of comparable trades.

B. Futures Price Impact

Matched floor trades and small floor trades cost more than comparable electronic trades (Sections III and IV.A). A higher floor spread on these trades could compensate market makers for adverse selection. If floor trades carry more information about the index, futures prices should move further in the direction of floor trades than in the direction of electronic trades with the same delta. We regress short-horizon E-mini S&P 500 futures returns on the signed delta of each complex trade:

$$r^f_{t+h} = \alpha + \beta_1 \,\text{Signed Delta}_i + \beta_2 \,\text{Floor}_i \times \text{Signed Delta}_i + \gamma' X_i + \mu_t + \mu_s + \varepsilon_i \tag{4}$$

r^f_{t+h} is the log E-mini futures return, in basis points, from 100 milliseconds before the trade to h after it, for h equal to 10 and 15 seconds. Signed Delta is the trade's net delta exposure in units of \$10,000 per one-point SPX move, signed by the side of the end user. Floor_i equals one for floor trades. X_i includes days to expiration, vega, gamma, implied volatility, trade size, a PM-settled indicator, and the number of legs. μ_t and μ_s are date and strategy fixed effects. Floor and electronic trades occur on different dates, and the date fixed effects absorb the Floor main effect. The coefficient β_1 measures the futures price impact of electronic trades, and $\beta_1 + \beta_2$ the impact of floor trades. Standard errors are clustered by date. The regressions use the full sample, 204,969 trades with complete controls.

Because Floor is collinear with the date, β_2 also absorbs any change in how futures respond to option order flow between early and late June. Short-horizon futures returns also mix information with the pressure of delta hedging. Market makers who take on delta hedge part of it in futures, and the hedge moves the futures price in the direction of the

trade whether or not the trade is informed.

TABLE VIII HERE

Table VIII reports the estimates at 10 and 15 seconds. Across all trades, electronic trades move futures prices in the direction of their delta. Each $10,000 of signed delta per SPX point is associated with a futures return of 0.43 basis points within 10 seconds and 0.42 basis points within 15 seconds. The floor interaction is −0.54 and −0.68 basis points, significant at the 5% and 1% levels. The implied impact of floor trades, $\beta_1 + \beta_2$, is −0.11 basis points at 10 seconds and −0.26 at 15 seconds. Floor trades with the same signed delta as electronic trades have no positive futures price impact.

Futures price impact does not account for the higher floor cost of comparable trades. Floor trades cost more than their electronic matches and have no positive futures price impact. This adds to the evidence in Kaeck et al. (2022) that much of the SPX spread is not explained by information asymmetry. For floor trades, the return window starts 100 milliseconds before the price is agreed, and the broker represents the order to the crowd before that time. Futures moves during representation, including hedges placed by market makers in the crowd, fall outside the window. The clustering by date relies on 20 trading days. Floor trades are also much larger than electronic trades, and price impact can differ with trade size. Section IV.D estimates Equation (4) separately for small and large trades and relates the estimates to screening on the floor.

We next examine the floor trades that the matched comparison leaves out.

C. Floor Trades without an Electronic Match

Many large floor trades have no electronic counterpart. The matching in Section III.B pairs 1,177 of 3,284 floor trades with an electronic trade from the closure.[22] The 2,107 floor trades without a match average 1,031 contracts, 17.2 legs, 117 days to expiration, and an absolute vega of 1.64 (Table IV). Matched floor trades average 342 contracts, 3.1 legs, 25 days to expiration, and an absolute vega of 0.41. Average absolute delta is 0.02 in each group. Unmatched floor trades pay an average effective spread of 5.46 cents, and their market maker revenue is 5.25 cents. No electronic trade from the closure is close enough to provide a cost comparison.

[22] The full sample of this section includes 3,392 floor trades.

FIGURE 6 HERE

Figure 6 compares the trade size of matched and unmatched floor trades. In Panel A, about 28% of unmatched floor trades exceed 1,000 contracts, against about 5% of matched floor trades. The unmatched trades occupy the right tail of the floor size distribution, where electronic trades are scarce. Panel B repeats the comparison for floor trades matched to AIM trades (Internet Appendix Table IA.IV). Matched floor trades cluster at round sizes of about 100 and 200 contracts, while about 27% of unmatched floor trades exceed 1,000 contracts, against about 4% of matched floor trades. Unmatched floor trades average 1,006 contracts and 15.9 legs, against 295 contracts and 3.8 legs for floor trades matched to AIM trades. They are the large, multi-leg packages for which the two weeks of AIM trades before the opening contain no counterpart.

FIGURE 7 HERE

Trades of this kind become less prevalent when the floor closes. Figure 7 plots the share of complex SPX dollar volume from trades whose size and number of legs each exceed the median of floor trades without an electronic match. In Panel A, the weekly share averages about 24% before the closure, about 13% during the closure, and about 21% after the opening. Its lowest weekly value, about 8.5%, falls in the week of June 1, after the May 26 announcement of the opening and in the week of the June 3 announcement of its delay. This low is consistent with traders deferring large trades until the floor returned. Lower demand for large, complex positions during the pandemic may account for part of the decline during the closure. In Panel B, the daily share rises from about 11.6% in the two weeks before the opening to about 19.6% in the two weeks after it. The difference of 8.04 percentage points is significant at the 1% level, and the VIX was higher after the opening than in early June (Figure 3). The recovery coincides with the return of the floor.

When a large complex trade cannot execute as one electronic package, the trader can split it into smaller electronic trades. We estimate the cost of doing so for a floor trade at the 90th percentile of size, 1,950 contracts.[23] If electronic orders supply large size as cheaply as the floor, the replication should cost about as much as floor trades of similar

[23]The target rounds the 90th percentile of floor trade size, 1,948.8 contracts, which is also the lower cutoff of the large trades in Section IV.D.

size. From the electronic orders of June 1 to 12, 2020, we chain consecutive orders within each trading day until their cumulative size reaches 1,950 contracts and pro-rate the last order in each chain. The dollar effective spread of each order is its effective spread in cents times its number of contracts, because the cents-to-dollars conversion and the SPX multiplier of 100 cancel. A chain of trades also accumulates delta, and the hedging of that delta moves futures prices against the trader. We compute this slippage as

$$\text{Slippage} = \frac{\hat{\beta}_1 \cdot D^2 \cdot S}{10^8} \tag{5}$$

where $\hat{\beta}_1$ is the electronic price impact coefficient for all trades without strategy fixed effects in Table VIII, at 10 and 15 seconds, D is the absolute delta of the target in dollars per SPX point, and $S = 3{,}100$ is the average SPX level in June 2020. With linear price impact (Kyle, 1985), the futures price move scales with D. Equation (5) charges this move on the full position, an upper bound for a position built in steps, which pays about half of it. We set $D = 2{,}549$ by scaling the mean absolute net delta of opening-period floor trades, \$1,100 per SPX point at a mean size of 841 contracts, to 1,950 contracts. The full-sample coefficients are an upper range for chains of small orders, whose impact is not distinguishable from zero (Section IV.D).[24] Electronic total cost is the sum of the spread and the smaller or larger slippage across the two horizons. Floor cost is the mean dollar effective spread of floor trades in bands of ±1%, ±2.5%, ±5%, and ±10% of percentile rank around the 90th percentile.

TABLE IX HERE

Table IX reports the replication. Assembling 1,950 contracts electronically takes 65.3 orders on average (median 64) over 1.3 minutes (median 1.0). The cumulative electronic spread is \$6,081, or 3.12 cents per unit of underlying. Slippage adds \$85 to \$87, for an electronic total cost of \$6,166 to \$6,168. Floor trades of similar size cost \$2,439 in the ±1% band, \$3,357 in the ±2.5% band, and \$4,246 in the ±5% band. The floor advantage is \$1,920 to \$3,729, or 31% to 60% of the electronic total cost, and almost all of it comes

[24] With the corresponding coefficients for electronic trades above the 90th percentile (Section IV.D), 0.26 and 0.27 basis points, slippage is about \$52 to \$54 and total electronic cost \$6,133 to \$6,135.

from spreads. The ±10% band spans the 80th to the 100th percentile, has a mean size of 2,668 contracts, and costs $3,745 on the floor.[25]

The replication reproduces the size of a large floor trade and leaves out its composition. The chains combine consecutive orders of any strategy, and a trader who needs a specific multi-leg package (unmatched floor trades average 17.2 legs) cannot assemble it from unrelated small packages. Legging into the package one piece at a time exposes the trader to execution risk on the legs that remain. A large floor trade costs less than the same size assembled from electronic orders, before any cost of reproducing the legs of a specific package.

All execution costs in this paper are measured on trades that execute. Without firm quotes for every possible complex trade, the cost of trades that investors want but do not execute is unobserved. A trade fails when its expected cost exceeds its benefit. On the floor, we do not observe orders that investors decide not to submit, or orders whose negotiated cost exceeds the investor's willingness to pay. An AIM auction never fails once it starts, but an investor may not find a member willing to guarantee the full size, and a COA fills only what responses and resting interest allow. In each mechanism, selection of which desired trades execute biases measured costs downward (Hendershott et al., 2024). The bias is smallest for small, simple trades, which execute readily in each mechanism. A different selection applies to small floor trades. After the opening, they are 0.5% of small complex trades (853 of 180,860) and are routed to the floor while electronic execution is available. The gap at or below the 25th percentile falls from 5.86 to 3.49 cents with trade-level controls and strategy fixed effects, and characteristics outside the controls can account for part of the remainder. For the largest complex trades, selection removes many of them from the electronic sample during the closure, and the measured electronic cost of those that execute understates the cost of the trades that do not. Deferral works in the opposite direction. If traders postponed large trades until the floor returned, the large trades that execute electronically in early June come from traders who could not wait, which can raise their measured cost relative to the floor. We next examine how the rules of the electronic mechanisms and of the floor treat large, complex trades.

[25] Internet Appendix Table IA.VIII builds the chains from AIM trades only. The AIM chain requires 36.2 trades and costs $6,166 in effective spreads.

D. *Electronic Constraints and Risk Sharing on the Floor*

Large, many-leg packages become much less prevalent without the floor (Figure 7), and assembling one from electronic orders costs more than executing a trade of similar size on the floor (Section IV.C). Electronic auctions and the floor differ in three ways that matter for such packages. They differ in whether liquidity must be committed before the auction starts, in how an order can be divided among liquidity suppliers, and in how much the parties know about each other. We call these differences pre-commitment, flexibility, and screening. The first is institutional, and we bring evidence on the other two.

The first difference is pre-commitment. Market makers have no quoting obligation on complex orders (Cboe Global Markets, 2026, Rule 5.33, Interpretation .01), and they initiate 94.5% of CLOB packages (Section II.C). The Complex Order Auction exposes an order for about 100 milliseconds and fills what it can against the responses and resting interest. Any unfilled balance rests in the CLOB. AIM, activated for SPX during the closure (Cboe Global Markets, 2020d), can start only after the initiating member pairs the agency order with interest that guarantees execution of the full size (Cboe Global Markets, 2026, Rules 5.37 and 5.38). Other participants can then improve the price. AIM tests whether a pre-paired trade can be price-improved.[26] On the floor, a broker represents the order to the crowd without first lining up a contra party, and the crowd can share a large, many-leg package without any member guaranteeing its full size in advance.

The second difference is flexibility. On the floor, several in-crowd market makers can share an order at a common price, and an unfilled balance can keep working (Cboe Global Markets, 2026, Rules 5.52(b) and 5.85). In AIM, the allocation follows a fixed algorithm. After priority customer orders, the initiating order receives up to 50% of the agency order when one other participant responds at the final price and 40% when two or more respond (Cboe Global Markets, 2019, Rule 5.38(e)). On the floor, after Priority Customer orders in the book, a DPM or LMM quoting the best price can receive 50%, 40%, or 30% of the order,

[26] The closure filing states that large customer orders on the floor are primarily filled by appointed market makers solicited as contras, and that the temporary AIM rule change was needed because the standing prohibition on soliciting appointed market makers in single-leg AIM would otherwise leave such orders without electronic liquidity (Securities and Exchange Commission, 2020, at 15824–25). Complex AIM already allowed appointed market makers to be solicited as contras (Cboe Global Markets, 2020d). Internet Appendix Section IA.II reports strategy composition in the CLOB, COA, and AIM separately and matches floor trades to AIM trades.

depending on the number of other in-crowd quotes, if Cboe applies this participation entitlement to the class. The remaining contracts go to in-crowd bids and offers in the sequence in which they were made, split equally when the sequence cannot be determined, and contracts that one market maker declines go to the others (Cboe Global Markets, 2019, Rule 5.85(a)). Figure 8 plots the size of each fill as a share of the order leg it fills.[27]

FIGURE 8 HERE

Pooled across trade sizes, floor fills are smaller relative to their order legs, and they come in many sizes. Across all electronic mechanisms, 52.2% of fills cover the whole order leg, against 14.3% of floor fills (Panel A). In AIM, 26.8% of fills cover the whole leg (Panel B). AIM fills also cluster at half of the leg and at 40% of it, about 13% and 7% of AIM fills against about 11% and 4% of floor fills. These are the shares that the AIM allocation rule assigns to the initiating order, and the floor fills at half the leg are consistent with the DPM or LMM entitlement. Floor fills spread over intermediate shares, in particular below 30% of the leg. The average fill covers 37.5% of its order leg on the floor, against 48.1% in AIM and 66.9% across all electronic mechanisms. Floor trades are larger than AIM trades, so these pooled shares partly reflect trade size. Within size bands, the average floor fill covers a larger share of its leg than the average AIM fill below the 75th percentile and about the same share above it (not tabulated). The evidence on flexibility is the shape of the distribution and how it changes with trade size. Beyond the entitlement, the crowd divides an order into pieces of many sizes, while AIM fills concentrate at the shares its algorithm fixes. Several market makers can then share a large position, each holding a smaller part until it can hedge (Grossman and Miller, 1988). Table X reports how often a fill covers the whole order leg in AIM and on the floor, by trade size.

TABLE X HERE

The crowd sizes its own pieces. Conditional on execution, small floor trades more often go to a single counterparty: 46.8% of floor fills cover the whole leg at or below the 25th percentile, against 29.9% of AIM fills, and 27.2% against 16.3% between the 25th and 50th percentiles. Large floor trades are more often divided among several market makers:

[27] We observe only executed quantities and not the orders that fail to execute, so the fill evidence in this section is conditional on execution.

3.4% of floor fills cover the whole leg between the 75th and 90th percentiles, against 7.0% of AIM fills, and 1.3% against 3.9% above the 90th percentile.[28] Consistent with this risk sharing, floor trades above the 90th percentile pay less than electronic trades of the same size (Section IV.A). The steeper gradient on the floor fits the division of labor in block markets: a single dealer can fill a small block and trade out of it, while the largest blocks are assembled from many liquidity suppliers because dealers are unwilling to hold very large positions (Harris, 2003). One way a small order is taken whole is crossing. After exposing an order to the crowd, the broker's firm can cross it against its own principal interest (Cboe Global Markets, 2026, Rule 5.87).

The third difference is screening. Identities are known on the floor. A trader can forgo anonymity through a floor broker, who builds a reputation for the trader on the floor (Grossman, 1992), and floor brokers know latent traders whose interest the electronic book does not display (Harris, 2003). In Grossman (1992), intermediaries know and use investors' unexpressed demand to match trades and reduce their own risk. A floor broker can credibly share the initiator's motive and remaining size with the crowd. This draws competing liquidity and divides the position among several suppliers. Sharing this information also carries a leakage cost (Harris, 2003), so the broker decides how much of an order to disclose.

Non-anonymity also lets liquidity suppliers screen the orders they fill, as in the literature on block trading. In Seppi (1990), a liquidity-motivated block trader can credibly commit not to trade further, which separates it from informed traders. In Benveniste et al. (1992), repeated, non-anonymous dealing between brokers and the specialist lets the specialist partly identify informed order flow and discipline the brokers who bring it through future penalties. In the upstairs market, liquidity suppliers rely on block brokers to determine whether an initiator is well informed and whether it has revealed the full size of its order (Harris, 2003). On the Paris Bourse, upstairs block trades carry less information than block trades in the electronic limit order book, despite being larger (Bessembinder and Venkataraman, 2004). On the NYSE, whose downstairs market is itself a floor, upstairs trades carry only slightly less information (Madhavan and Cheng, 1997), and Bessem-

[28] In linear probability regressions of a whole-leg indicator on the interactions of the floor indicator with the size bands, with the full set of controls and strategy fixed effects, the floor-minus-AIM differences are significant at the 1% level in the two smallest bands and at the 5% and 10% levels in the two largest bands (not tabulated).

binder and Venkataraman (2004) argue that the NYSE floor could replicate some benefits of upstairs trading, because floor brokers can work orders without fully revealing them and specialists and floor brokers can sometimes deduce who initiates a trade. If the Cboe floor screens orders in this way, floor trades should move futures prices less than electronic trades with the same signed delta. Across all trades, they do (Section IV.B). Screening should matter most for large trades, which a liquidity supplier must hold until it can trade out of them (Harris, 2003). Table XI estimates Equation (4) separately for small trades, at or below the 25th percentile of the floor trade-size distribution (166 contracts or fewer), in Panel A, and for large trades, above the 90th percentile (more than 1,948.8 contracts), in Panel B.

TABLE XI HERE

Among the 672 large trades in Panel B, electronic trades move futures prices by 0.26 and 0.27 basis points per $10,000 of signed delta at 10 and 15 seconds. The floor interaction is −0.99 and −1.03 basis points, significant at the 5% and 10% levels, and with strategy fixed effects it is −1.09 at both horizons, significant at the 5% level. The point estimates of $\beta_1 + \beta_2$ range from −0.73 to −0.85 basis points. Among the 194,870 small trades in Panel A, which are almost all electronic, neither the electronic coefficient nor the floor interaction is statistically significant. Small trades on either venue have no detectable price impact.

The evidence suggests that the floor screens larger trades, although it is not conclusive. At or below the 25th percentile, floor trades pay 3.49 cents more than electronic trades with the full set of controls (Section IV.A), and neither venue's trades have a detectable futures price impact, so price impact does not explain the higher floor cost of small trades. Above the 90th percentile, floor trades move futures prices less than electronic trades with the same signed delta. This is consistent with the separation of liquidity-motivated traders in Seppi (1990) and the discipline of brokers in Benveniste et al. (1992), and it parallels the lower information content of upstairs block trades (Bessembinder and Venkataraman, 2004). Several caveats apply. In Seppi (1990), separation requires the block to be small relative to what an informed trader could trade elsewhere, so the model does not predict that the largest trades are the least informed. The difference between floor and electronic impact among the largest trades is not significantly larger than among trades between the

25th and 90th percentiles.[29] The same pattern fits self-selection of uninformed traders into the floor for their largest trades. Hedging pressure, the timing of the return window, and any change in how futures respond to option order flow between early and late June, discussed in Section IV.B, can also differ between floor and electronic trades, and the large-trade regressions have 672 observations.

Floor and electronic execution serve different trades. In the regression setting, small complex trades cost 3.49 cents less on the electronic mechanisms than on the floor, and the largest trades cost 4.35 cents less on the floor, without positive futures price impact. Total dollar effective spreads on complex SPX trades fall at the opening, mostly among the largest trades. When the floor closes, the largest and most complex trades become less prevalent, and those that execute electronically cost more per contract than on the floor. Assembling a 1,950-contract trade from electronic orders takes about 65 orders and costs 1.5 to 2.5 times the floor cost of trades of similar size. The logit of Section II.B points to the same trades, with size and the number of legs as the main predictors of floor routing.

V. Conclusion

The transition from human-intermediated to electronic trading is one of the defining structural shifts in modern financial markets. In equities, the process is largely complete. In options it is not. The largest complex trades in S&P 500 index options continue to execute predominantly through open outcry, and the floor handles about 70% of complex SPX dollar volume. Whether the floor's persistence reflects a genuine intermediation advantage for the largest and most complex trades, or merely the absence of a competitive electronic alternative, is a question the prior literature has not addressed.

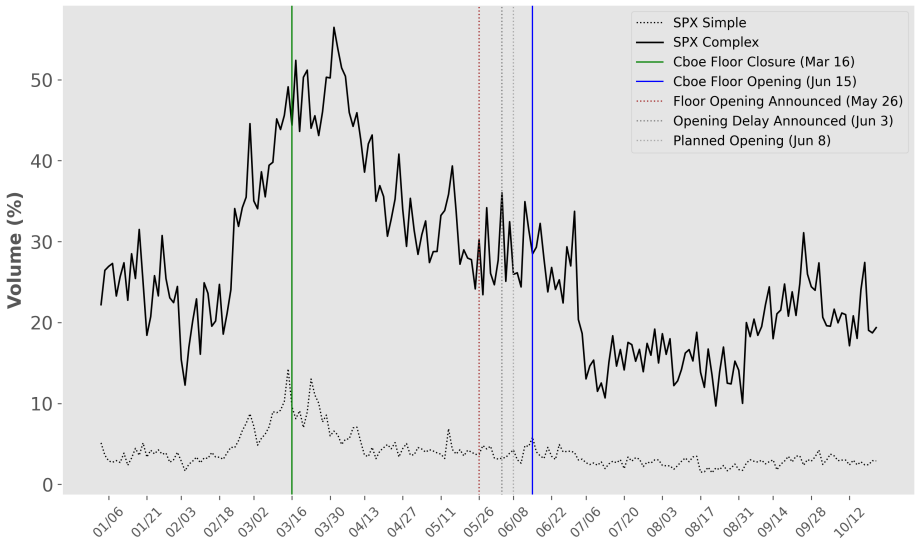
We address it by treating the June 15, 2020, opening of the Cboe trading floor as a shock to the availability of open outcry intermediation in SPX, options on the S&P 500 index that only trade on Cboe. We match each floor trade after the opening to its nearest electronic counterpart during the closure on option characteristics, trade size, time to expiration, and number of legs. Across all matched trades, floor trades pay effective spreads more

[29] In a pooled regression with indicators for the two size groups, their interactions with Floor and Signed Delta, and strategy fixed effects, the triple interaction of Floor, Signed Delta, and the large-trade indicator is −0.44 basis points at 10 seconds and −0.31 at 15 seconds, and neither is statistically significant (not tabulated).

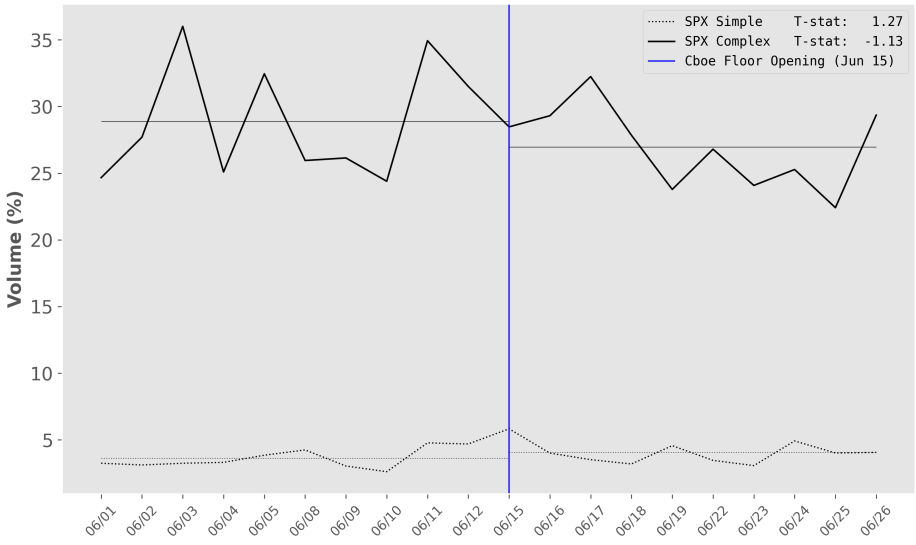
than a third higher than their electronic counterparts, with a comparable difference in market maker revenues. Because the match holds option characteristics and trade size fixed, the difference reflects how each mechanism prices intermediation for observably equivalent positions. The electronic matches come from the closure, when Cboe ran the Automated Improvement Mechanism (AIM) for SPX. SPY complex trades, which track the same index on other venues, show no comparable shift. Within size groups, the average combines opposite effects. With the full set of controls, small floor trades pay 3.49 cents more than electronic trades of the same size, without higher futures price impact. The largest floor trades pay 4.35 cents less and have lower futures price impact.

The average cost advantage of electronic execution applies only to the trades electronic markets accommodate. The largest and most complex trades become much less prevalent during the closure, when they can only execute electronically. This is consistent with electronic liquidity providers being unwilling to absorb large, complex positions, although lower demand during COVID may also matter. Floor trading appears to share risk more efficiently. Assembling a 1,950-contract trade from electronic orders takes about 65 orders and costs 1.5 to 2.5 times the floor cost of trades of similar size. Market makers need not quote complex strategies, the Complex Order Auction fills what it can within 100 milliseconds, and AIM cannot start until the broker finds counterparty interest for the entire order. Preserving floor access while extending AIM to complex SPX orders could lower aggregate transaction costs without eliminating the execution service that large traders rely on. Whether the floor would remain viable if its smaller trades migrated to a coexisting electronic auction, and why Cboe does not offer an electronic auction with the flexibility of the floor, are open questions. The trading floor is not dead yet. It facilitates the most difficult trades, the largest and most complex, which Cboe's electronic mechanisms execute at higher cost or not at all.

Figures

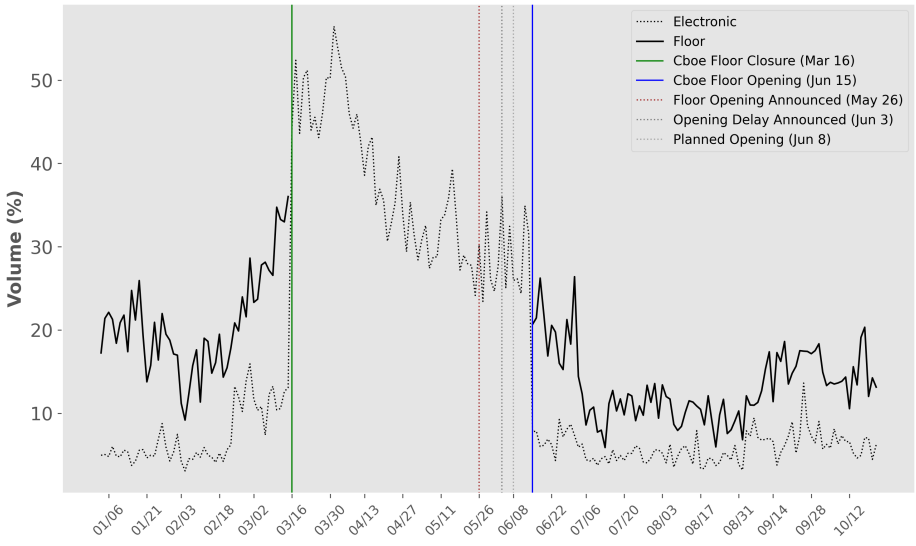


Panel A. Around the Closure and Opening

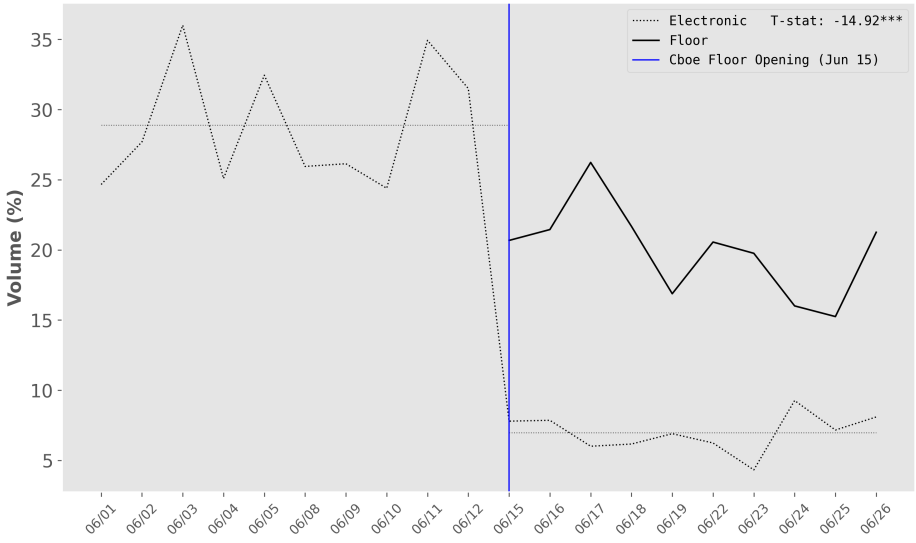


Panel B. Around the Opening (Jun 1 – Jun 26)

Figure 1. SPX Option Volume as Percentage of Total OPRA Volume: Complex versus Simple
This figure plots daily SPX option dollar volume as a percentage of total OPRA volume. Panel A covers the full sample from January to October 2020 and separates complex trades (packages of two or more legs) from single-leg trades. Panel B zooms in on the four weeks surrounding the floor opening on June 15, 2020. The solid green line marks the Cboe floor closure on March 16, 2020, and the solid blue line marks the floor opening on June 15, 2020. The dotted vertical lines mark the announcement of the planned opening on May 26, the announcement of the delay on June 3, and the originally planned opening date of June 8. In Panel B, horizontal lines show closure and opening period means for each series, and the legend reports Welch t-statistics for the difference in means between the two sub-periods. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.



Panel A. Around the Closure and Opening

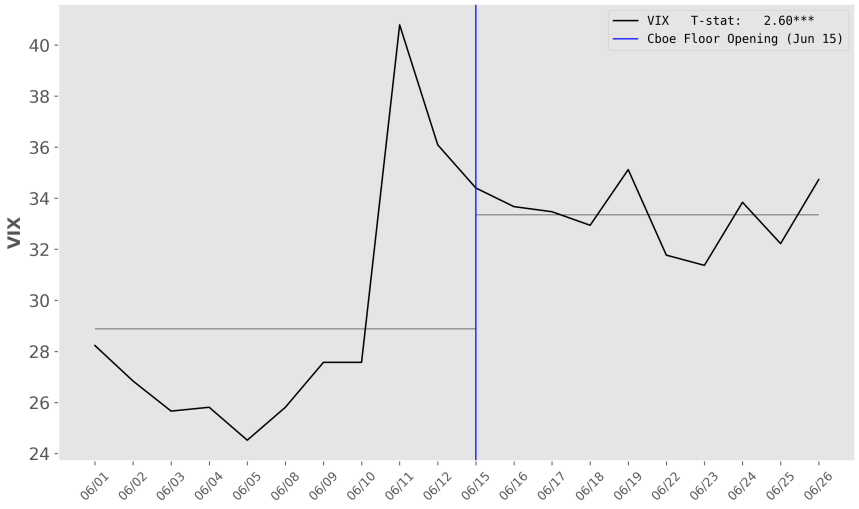


Panel B. Around the Opening (Jun 1 – Jun 26)

Figure 2. SPX Option Volume as Percentage of Total OPRA Volume: Floor versus Electronic
This figure plots daily floor and electronic complex SPX option dollar volume as a percentage of total OPRA volume. Panel A covers the full sample from January to October 2020. Panel B narrows the window to the four weeks surrounding the floor opening on June 15, 2020. The solid green line marks the Cboe floor closure on March 16, 2020, and the solid blue line marks the floor opening on June 15, 2020. The dotted vertical lines mark the announcement of the planned opening on May 26, the announcement of the delay on June 3, and the originally planned opening date of June 8. In Panel B, horizontal lines show the closure and opening period means of electronic volume, and the legend reports the Welch t-statistic for their difference. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}\ p < 0.01$.

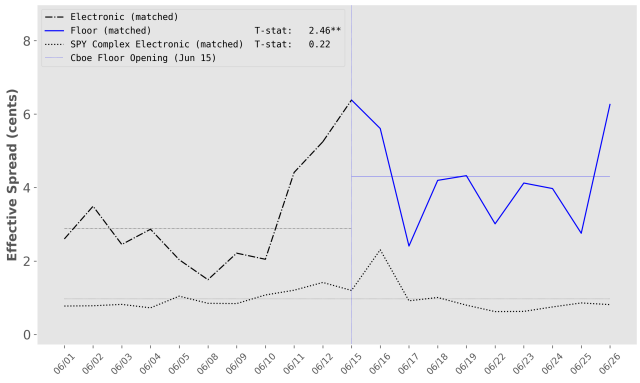


Panel A. Around the Closure and Opening

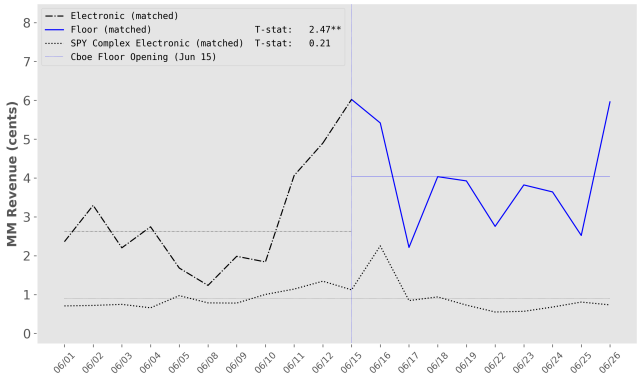


Panel B. Around the Opening (Jun 1 – Jun 26)

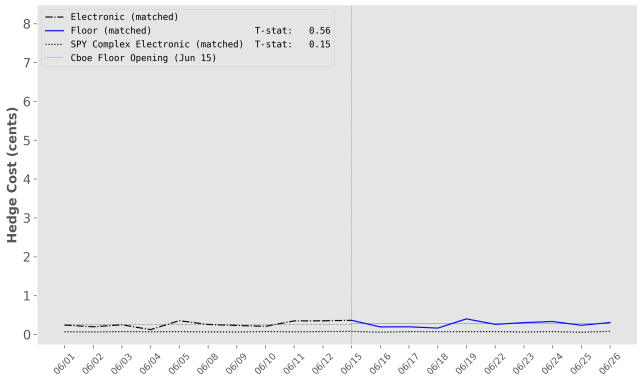
Figure 3. VIX Around the Closure and Opening
This figure plots the daily closing level of the Cboe Volatility Index (VIX). Panel A covers the full sample from January to October 2020. The solid green line marks the Cboe floor closure on March 16, and the solid blue line marks the floor opening on June 15. The dotted vertical lines mark the announcement of the planned opening on May 26, the announcement of the delay on June 3, and the originally planned opening date of June 8. Panel B narrows the window to the two weeks before and after the opening (June 1 to June 26, 2020). In Panel B, horizontal lines show closure and opening period means, and the legend reports Welch t-statistics for the difference in means between the two sub-periods. $^{*}p < 0.10$; $^{**}p < 0.05$; *** $p < 0.01$.



Panel A. Effective Spreads

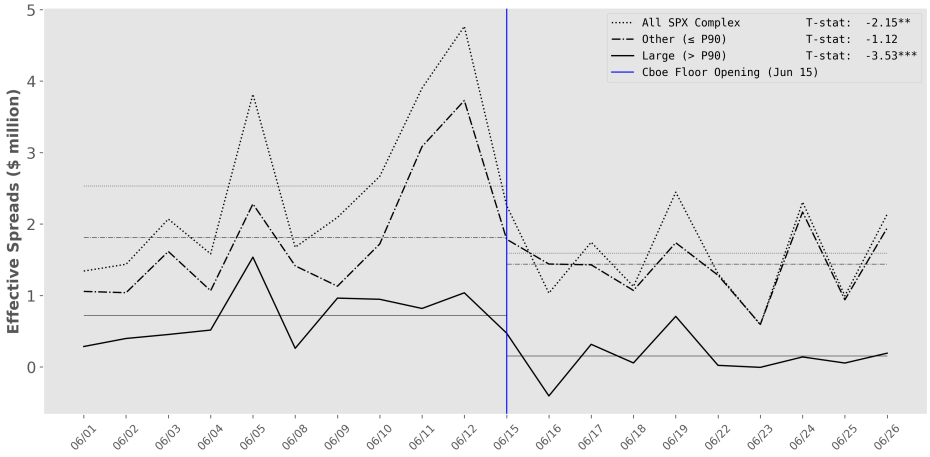


Panel B. Market Maker Revenues

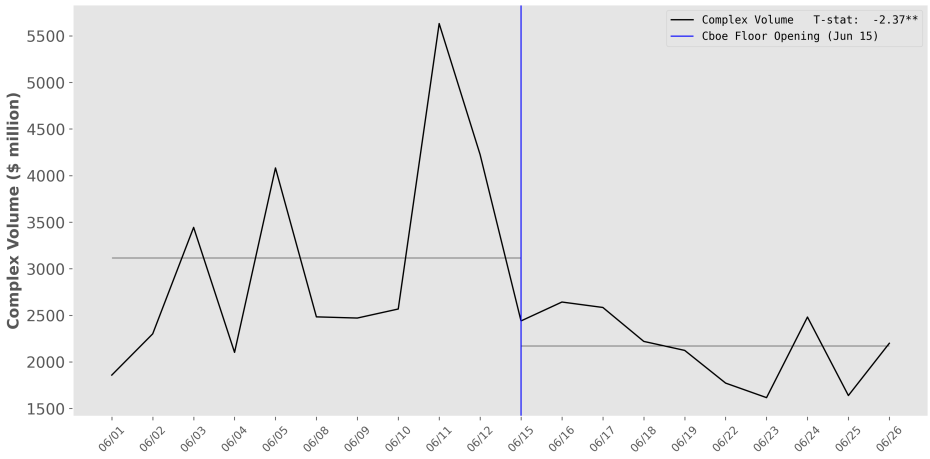


Panel C. Hedge Costs

Figure 4. Effective Spread, Market Maker Revenue, and Hedge Cost
This figure plots daily mean effective spreads (Panel A), market maker revenue (Panel B), and hedge costs (Panel C) in cents for matched SPX complex trades, with matched electronic trades before the opening (dash-dotted line) and matched floor trades after it (solid line), and for matched SPY complex trades (dotted line), which serve as a placebo. All panels cover the two weeks before and after the Cboe floor opening (June 1 to June 26, 2020). The dotted blue line marks the opening on June 15, 2020. Horizontal lines show closure and opening period means for each series. The legend in each panel reports Welch t-statistics for the difference between the opening and closure means of the SPX and SPY series. $^{*}p < 0.10$; $^{**}p < 0.05$; *** $p < 0.01$.

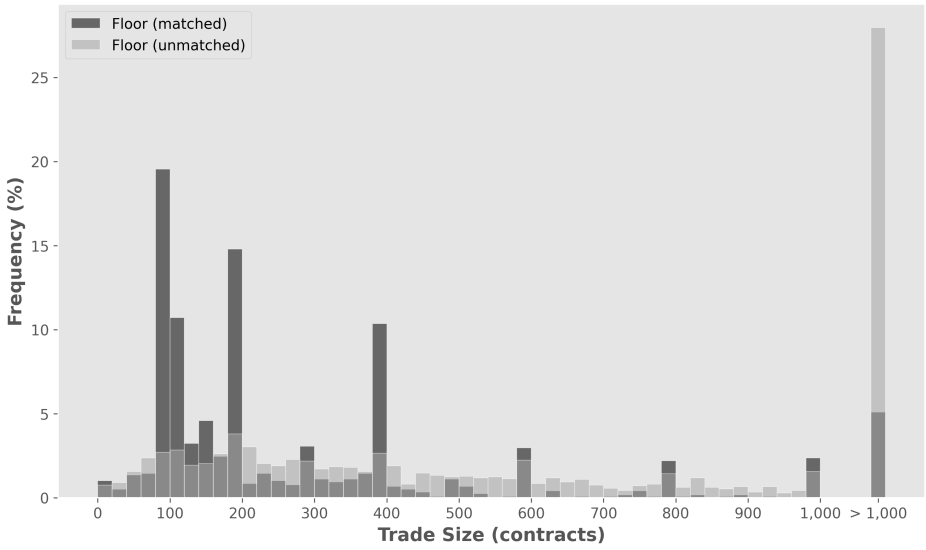


Panel A. Total Effective Spread Paid

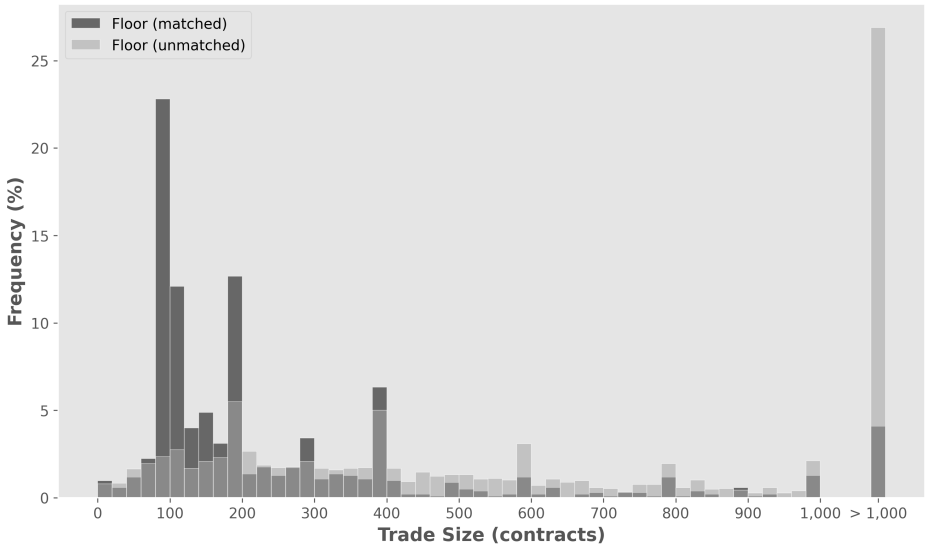


Panel B. Total Complex SPX Volume

Figure 5. Trading Activity Around the Opening
This figure plots daily aggregates for complex SPX trades during the two weeks before and after the floor opening (June 1 to June 26, 2020). Panel A reports the total dollar effective spread paid across all complex SPX trades in millions. The dotted line covers all complex SPX trades, the solid line trades above the 90th percentile of the floor trade-size distribution (> P90), and the dash-dotted line all other trades (≤ P90). Panel B reports total complex SPX dollar volume in millions. The solid blue vertical line marks the floor opening on June 15, 2020. The horizontal grey lines denote closure and opening period averages, and each panel reports the Welch t-statistic for the difference in means. *$p < 0.10$; **$p < 0.05$; *** $p < 0.01$.

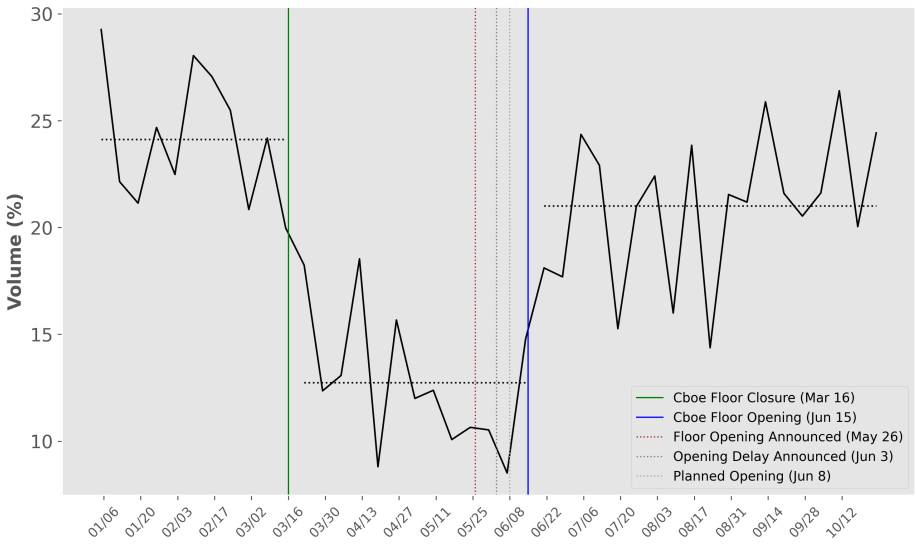


Panel A. Floor Trades Matched to Electronic Trades

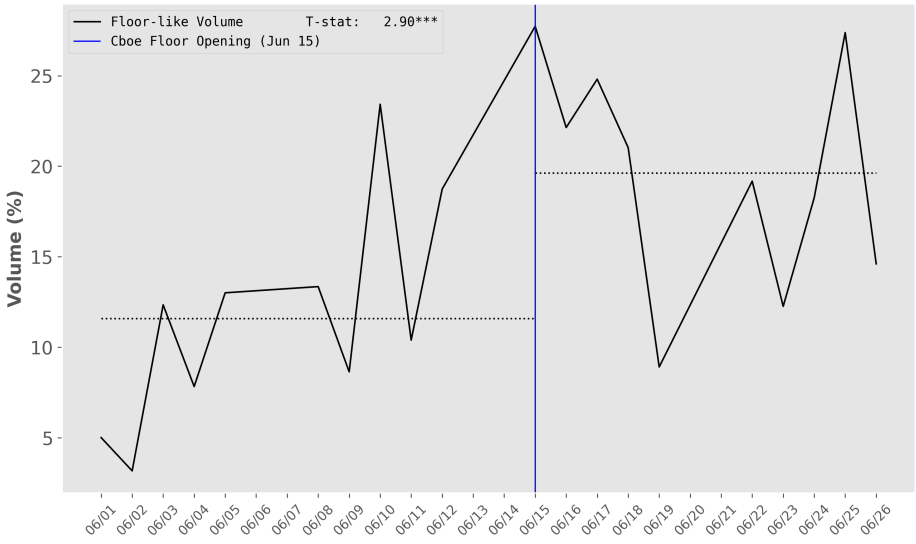


Panel B. Floor Trades Matched to AIM Trades

Figure 6. Trade Size of Matched and Unmatched Floor Trades
This figure plots the distribution of trade size (in contracts) for floor trades executed in the two weeks after the opening (June 15–26, 2020), with and without a match. Panel A uses the matched analysis of Table IV, with 1,177 matched and 2,107 unmatched floor trades. Panel B uses the floor-versus-AIM matched analysis of Internet Appendix Table IA.IV, with 1,026 matched and 2,258 unmatched floor trades. The last bin collects trades above 1,000 contracts. The vertical axis measures the percentage of trades in each bin within each group.



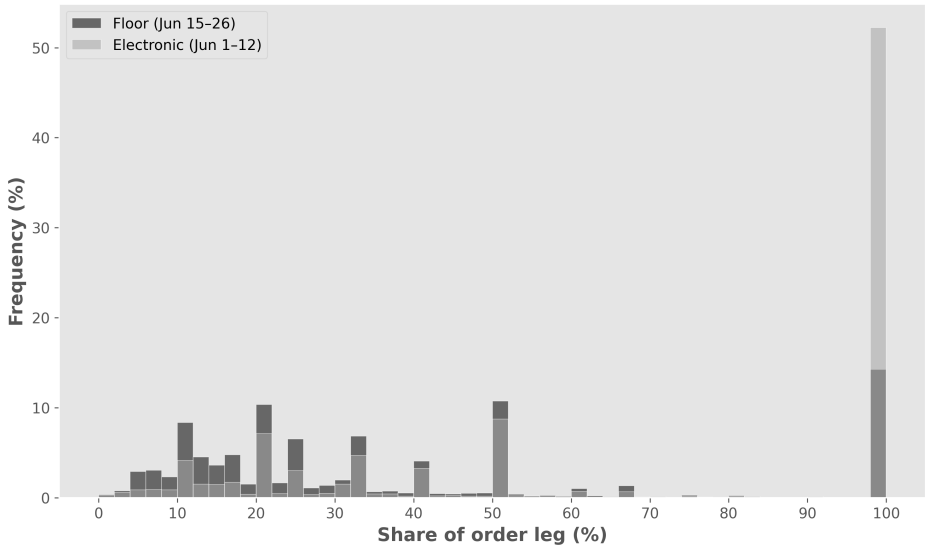
Panel A. Around the Closure and Opening (Weekly)



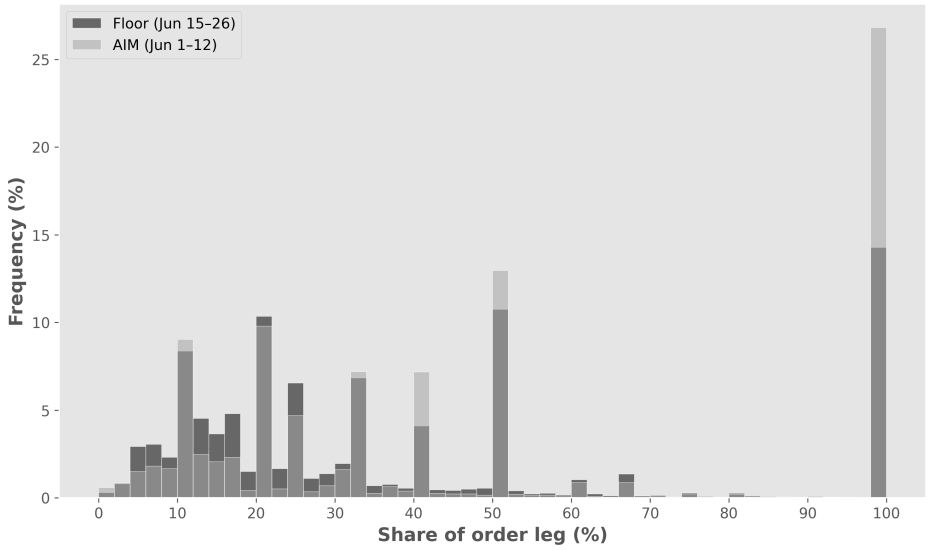
Panel B. Around the Opening (Daily, June 1–26)

Figure 7. Share of Trades Exceeding the Median Unmatched Floor Trade (Trade Size, Number of Legs)

This figure plots the share of SPX complex dollar volume from trades whose trade size and number of legs each exceed the median of floor trades without an electronic match. Panel A reports weekly shares from January to October 2020. The solid green line marks the Cboe floor closure on March 16, and the solid blue line marks the floor opening on June 15. The dotted vertical lines mark the announcement of the planned opening on May 26, the announcement of the delay on June 3, and the originally planned opening date of June 8. The horizontal dotted lines show the pre-closure, closure, and opening period means. Panel B reports daily shares in the two weeks before and after the opening (June 1 to June 26, 2020). The horizontal dotted lines show the closure and opening period means, and the legend reports the Welch t-statistic for their difference. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}\ p < 0.01$.



Panel A. Floor versus All Electronic Fills



Panel B. Floor versus AIM Fills

Figure 8. Fill Size Relative to the Order Leg
This figure plots the distribution of fills by their size as a share of the order leg they fill, in percent of fills. A fill is a single execution in one leg of a package. Each panel compares floor fills in the two weeks after the opening (June 15–26, 2020) with electronic fills during the last two weeks of the closure (June 1–12, 2020). Panel A uses all electronic fills and Panel B AIM fills. Electronic executions are grouped into orders. Bins are two percentage points wide, and the bin at 100% contains fills that cover the whole order leg.

Tables

Table I. Option Trading

This table reports average daily option trading volume in dollar terms and contract terms across all U.S. listed option classes reported through the Options Price Reporting Authority (OPRA). The sample covers the pre-closure (January 2 to March 13, 2020) and opening (June 16 to October 21, 2020) periods, excluding the floor closure period. Dollar volume is reported in billions and contract volume in hundreds of thousands. The % OPRA columns express each category as a share of total OPRA volume. Panel A reports volume for all option classes, decomposed by order type (complex versus single-leg) and execution venue (floor versus electronic). Panel B restricts attention to the S&P 500 index (SPX) option class and further decomposes SPX volume by order type and floor participation.

Panel A: All Option Classes				
	$ Volume		Contract Volume	
	$ billions	% OPRA	100,000	% OPRA
Complex	7.39	50.67	74.53	28.22
Simple	6.90	49.33	189.85	71.78
Floor	3.25	22.00	22.37	8.54
Electronic	11.04	78.00	242.00	91.46
Panel B: SPX Option Class				
	$ Volume		Contract Volume	
	$ billions	% OPRA	100,000	% OPRA
SPX	3.96	25.83	10.87	4.12
Complex	3.39	22.32	8.07	3.06
Simple	0.57	3.51	2.80	1.06
SPX Electronic	1.51	9.52	6.55	2.47
Complex	1.01	6.36	4.07	1.53
Simple	0.50	3.16	2.48	0.94
SPX Floor	2.45	16.31	4.33	1.65
Complex	2.38	15.95	4.01	1.53
Simple	0.07	0.36	0.32	0.12

Table II. Descriptive Statistics

This table reports daily means of trading activity and execution cost measures for complex (multi-leg) SPX options (Panel A) and complex SPY options (Panel B) across the pre-closure (January 2 to March 13, 2020), closure (March 16 to June 14, 2020), and opening (June 15 to October 21, 2020) periods. In Panel A, floor and electronic trades are pooled within the pre-closure and opening periods; the closure period contains electronic trades only by construction. Electronic fills are grouped into orders. Panel B covers electronic SPY complex trades executed on exchanges other than Cboe Options and C2; SPY legs are grouped into packages with the time-window rule of Li (2022) and signed with the quote rule. Volume is the share of SPX (Panel A) or SPY (Panel B) dollar volume executed in complex trades. Trade Size is measured in contracts and millions of dollars. Net Price is the execution price in dollars per unit of the underlying. Effective Spread is the trade direction times the difference between the execution price and the quote midpoint, in cents, measured from the side of the initiator, or, in Panel A, from the side of the counterparty when a market maker initiates the trade (the end-user effective spread of Section II.C). In Panel A, option quotes are from LSEG Tick History, taken at the agreement time for floor trades and at the execution time for electronic trades. Hedge Cost is defined as the absolute delta exposure of the trade multiplied by half the quoted spread in E-mini S&P 500 futures prevailing at the time of the option trade, measured in cents. MM Revenue is the effective spread minus the hedge cost. All three execution cost measures are reported as equally-weighted (EW) and value-weighted (VW) daily means. The last two columns report differences in means between adjacent periods. Statistical significance is assessed using a two-sample Welch t-test with standard errors clustered by date. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}\ p < 0.01$.

Panel A: SPX Complex Trades

Variables	Units	Pre-Closure	Closure	Opening	Closure – Pre-Closure	Opening – Closure
Volume	% of SPX Volume	86.43	87.86	87.49	1.43**	−0.37
Trade Size	Contracts	52.98	39.92	33.58	−13.06***	−6.34***
	$ Million	0.21	0.35	0.15	0.14***	−0.20***
Net Price	$	0.01	0.09	0.07	0.08	−0.02
Effective Spread (EW)	Cents	1.93	2.34	1.82	0.41**	−0.52***
Effective Spread (VW)	Cents	5.03	6.51	3.31	1.48*	−3.20***
Hedge Cost (EW)	Cents	0.39	0.34	0.37	−0.05***	0.03**
Hedge Cost (VW)	Cents	0.29	0.26	0.27	−0.03*	0.01
MM Revenue (EW)	Cents	1.52	1.98	1.43	0.46**	−0.55***
MM Revenue (VW)	Cents	5.05	7.33	3.56	2.28**	−3.77***

Panel B: SPY Complex Trades

Variables	Units	Pre-Closure	Closure	Opening	Closure – Pre-Closure	Opening – Closure
Volume	% of SPY Volume	27.02	29.73	26.88	2.71*	−2.85**
Trade Size	Contracts	37.77	29.13	28.63	−8.64***	−0.50
	$ Million	0.01	0.02	0.01	0.01***	−0.01***
Net Price	$	0.80	1.33	0.63	0.53***	−0.70***
Effective Spread (EW)	Cents	1.09	1.34	0.77	0.25	−0.57***
Effective Spread (VW)	Cents	2.30	2.97	1.93	0.67**	−1.04***
Hedge Cost (EW)	Cents	0.08	0.08	0.07	0.00***	−0.01
Hedge Cost (VW)	Cents	0.12	0.16	0.13	0.04***	−0.03***
MM Revenue (EW)	Cents	1.01	1.27	0.69	0.26*	−0.58***
MM Revenue (VW)	Cents	2.18	2.81	1.80	0.63**	−1.01***

Table III. Determinants of Floor Trading After the Opening

This table reports logistic regressions where the dependent variable equals one if a complex SPX option trade executes on the trading floor and zero if it executes electronically. Electronic executions are grouped into orders before estimation. The sample covers the opening period (June 15 to October 21, 2020) and contains up to 1,698,094 observations, of which 27,839 floor trades; the number of observations in each specification is reported at the bottom of each panel. The estimated specification follows Equation (1):

$$\Pr(\text{Floor}_i = 1) = \Lambda(\alpha + \beta' X_i + \mu_s)$$

where X_i includes log trade size, log number of legs, days to expiration, vega, gamma, implied volatility, absolute delta, and an SPXW indicator for PM-settled contracts. μ_s denotes strategy fixed effects, and $\Lambda(\cdot)$ is the logistic function. Panel A reports logit coefficients with z-statistics in parentheses, computed from standard errors clustered by date. Panel B reports average marginal effects with z-statistics in parentheses, computed as the mean change in the predicted probability of floor execution from a one-unit increase in each variable evaluated at observed values. For variables in natural logs (trade size and legs), a one-unit increase corresponds to a factor of $e \approx 2.72$; a doubling corresponds to an increase of $\ln 2 \approx 0.69$ units. The unconditional floor probability is 1.64%. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}\ p < 0.01$.

Panel A: Estimated Coefficients						
Log Trade Size	2.01***	1.82***	1.80***	1.84***	1.84***	1.83***
	(152.81)	(132.73)	(115.65)	(122.71)	(120.99)	(111.37)
Log Legs		2.13***	1.60***		3.42***	3.34***
		(51.25)	(38.12)		(8.52)	(8.05)
Days-to-Expiration			−0.06***			−0.09***
			(−3.81)			(−7.05)
Vega			0.34***			0.25***
			(30.46)			(20.19)
Gamma			0.03***			−0.00
			(10.04)			(−0.21)
IV			−1.05***			−0.66***
			(−5.69)			(−3.98)
|Delta|			−0.18			−0.01
			(−0.65)			(−0.02)
SPXW			−1.73***			−0.71***
			(−25.71)			(−11.94)
Intercept	−12.11***	−13.65***	−11.70***	−8.03***	−12.75***	−12.49***
	(−175.02)	(−140.38)	(−82.38)	(−9.86)	(−12.98)	(−12.75)
Strategy FE	No	No	No	Yes	Yes	Yes
Pseudo R-squared	0.64	0.70	0.75	0.79	0.79	0.80
Observations	1,698,094	1,698,094	1,678,134	1,698,060	1,698,060	1,678,104

continued on next page

Table III continued

Panel B: Average Marginal Effects						
Log Trade Size	0.0159***	0.0115***	0.0095***	0.0077***	0.0076***	0.0073***
	(95.72)	(79.24)	(53.33)	(60.33)	(59.48)	(51.36)
Log Legs		0.0135***	0.0084***		0.0142***	0.0134***
		(59.64)	(46.33)		(8.63)	(8.10)
Days-to-Expiration			−0.0003***			−0.0004***
			(−3.71)			(−6.96)
Vega			0.0018***			0.0010***
			(28.99)			(18.97)
Gamma			0.0001***			−0.0000
			(10.72)			(−0.21)
IV			−0.0055***			−0.0026***
			(−5.81)			(−4.03)
|Delta|			−0.0009			−0.0000
			(−0.65)			(−0.02)
SPXW			−0.0128***			−0.0033***
			(−17.30)			(−10.40)
Strategy FE	No	No	No	Yes	Yes	Yes
Observations	1,698,094	1,698,094	1,678,134	1,698,060	1,698,060	1,678,104

Table IV. Matched Sample Summary Statistics

This table compares pre-matching and post-matching means for floor and electronic complex SPX trades around the opening of the Cboe trading floor. For each floor trade executed in the two weeks after the opening (June 15–26, 2020), we identify the nearest electronic trade from the two weeks before the opening (June 1–14, 2020) using one-to-one nearest-neighbor matching without replacement. We match on six covariates: absolute delta (|Delta|), absolute vega (|Vega|), implied volatility, trade size in contracts, days to expiration, and number of legs, imposing a per-variable caliper of 0.25 standard deviations. Panel A reports the matching covariates before and after matching. Panel B reports execution quality measures that are not included in the matching algorithm: the effective spread, hedge cost, and market maker revenue, all measured in cents. When a market maker initiates the trade, the effective spread and market maker revenue are measured from the side of its counterparty, so that both are end-user measures (Section II.C), the cost borne by the non-market-maker side. Within each panel, the left columns report pre-matching means and differences, and the right columns report post-matching means and differences. The sample contains 2,354 trades (1,177 matched pairs) with a match rate of 35.8%.

Panel A: Options Trade Characteristics

	Before Matching			After Matching			
	Floor	Electronic	Difference	Floor	Electronic	Difference	Floor Unmatched
|Delta|	0.02	0.03	−0.01	0.02	0.02	0.00	0.02
|Vega|	1.20	0.20	1.00	0.41	0.40	0.00	1.64
Implied Vol.	0.35	0.37	−0.02	0.39	0.38	0.00	0.34
Trade Size	783.62	36.17	747.44	341.51	339.41	2.10	1030.58
Days-to-Exp.	84.18	21.77	62.41	25.17	25.16	0.01	117.15
Legs	12.14	2.58	9.57	3.08	3.07	0.00	17.21

Panel B: Execution Quality

	Before Matching			After Matching			
	Floor	Electronic	Difference	Floor	Electronic	Difference	Floor Unmatched
Effective Spread	5.10	1.97	3.13	4.45	3.15	1.30	5.46
Hedge Cost	0.24	0.36	−0.13	0.28	0.28	0.00	0.21
MM Revenue	4.86	1.60	3.26	4.17	2.87	1.30	5.25

Table V. Execution Quality and Floor Opening

This table reports OLS regressions of effective spread and market maker revenue on matched samples of complex SPX trades. Each post-opening floor trade (June 15–26, 2020) is matched 1:1 without replacement to the nearest pre-opening electronic trade (June 1–12, 2020) on |delta|, |vega|, implied volatility, trade size, days to expiration, and number of legs. Electronic executions are grouped into orders before matching. The estimated specification follows Equation (2):

$$Y_i = \alpha + \beta\, Opening_i + \gamma' X_i + \mu_s + \varepsilon_i$$

where Y_i is measured in cents. $Opening_i$ equals one for trades executed after the opening and zero otherwise. X_i includes the VIX volatility index in all specifications. The specifications successively add the average daily effective spread for SPY complex options and then the trade-level controls (quoted spread, absolute delta, days to expiration, vega, gamma, implied volatility, log trade size, and a PM-settled indicator) with strategy fixed effects. μ_s denotes strategy fixed effects. Absolute delta is omitted from the market maker revenue regressions because it mechanically determines the hedging cost component of the dependent variable. When a market maker initiates the trade, the effective spread and market maker revenue are measured from the side of its counterparty. The sample contains 1,177 matched pairs. t-statistics in parentheses are computed from standard errors clustered by date. Intercept and some controls are omitted for brevity. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}\ p < 0.01$.

	Effective Spread			Market Maker Revenue		
	(1)	(2)	(3)	(1)	(2)	(3)
Opening	0.76	0.98**	1.14***	0.78	1.00***	1.15***
	(1.44)	(2.43)	(3.01)	(1.50)	(2.60)	(2.96)
VIX	0.20***	0.16***	0.10**	0.19***	0.15***	0.12***
	(4.24)	(4.44)	(2.53)	(4.15)	(4.26)	(2.78)
SPY Complex ES		0.96***	0.88***		0.97***	0.93***
		(2.94)	(3.13)		(3.15)	(3.29)
|Delta|			51.82***			
			(3.83)			
Other Controls	No	No	Yes	No	No	Yes
Strategy FE	No	No	Yes	No	No	Yes
R-squared	0.01	0.01	0.11	0.01	0.01	0.09
Observations	2,354	2,354	2,354	2,354	2,354	2,354

Table VI. Execution Costs of Unaffected Trades

This table reports OLS regressions of effective spread and market maker revenue on electronic SPY complex trades, which are unaffected by the Cboe floor opening. Floor trades and trades executed on Cboe Options and C2 are excluded. The estimated specification follows Equation (2):

$$Y_i = \alpha + \beta\, Opening_i + \gamma' X_i + \mu_s + \varepsilon_i$$

where Y_i is measured in cents. $Opening_i$ equals one for trades executed after the opening and zero otherwise. X_i includes trade-level controls: quoted spread, absolute delta, days to expiration, vega, gamma, implied volatility, log trade size, and VIX. μ_s denotes strategy fixed effects. Absolute delta is omitted from the market maker revenue regressions because it mechanically determines the hedging cost component of the dependent variable. t-statistics in parentheses are computed from standard errors clustered by date. Intercept and some controls are omitted for brevity. $^*p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

	Effective Spread		Market Maker Revenue	
	(1)	(2)	(1)	(2)
Opening	−0.17	−0.05	−0.17	−0.05
	(−0.59)	(−0.79)	(−0.61)	(−0.79)
VIX	0.05**	0.02***	0.05**	0.02***
	(2.25)	(3.67)	(2.28)	(3.71)
|Delta|		1.32***		
		(5.71)		
Other Controls	No	Yes	No	Yes
Strategy FE	No	Yes	No	Yes
R-squared	0.01	0.46	0.01	0.46
Observations	428,627	428,627	428,627	428,627

Table VII. Execution Quality and Floor Opening by Trade Size

This table reports OLS regressions of effective spread and market maker revenue on complex SPX trades by trade-size band, without matching. The sample compares floor trades executed in the two weeks after the opening (June 15–26, 2020) with electronic trades executed during the last two weeks of the floor closure (June 1–12, 2020). Electronic executions are grouped into orders. Effective spreads and market maker revenue are computed from LSEG quotes. The estimated specification follows Equation (3), which extends Equation (2) by interacting the Opening indicator with trade-size bands:

$$Y_i = \alpha + \sum_k \beta_k \, Opening_i \times Size_{k,i} + \sum_k \delta_k \, Size_{k,i} + \gamma' X_i + \mu_s + \varepsilon_i$$

where Y_i is measured in cents. $Opening_i$ equals one for floor trades executed after the opening and zero for electronic trades executed during the closure. $Size_{k,i}$ equals one if the trade falls in trade-size band k: at or below the 25th percentile (P25), between the 25th and 50th, 50th and 75th, and 75th and 90th percentiles, or above the 90th percentile (P90) of the floor trade-size distribution. X_i includes the VIX volatility index in all specifications. The specifications successively add the average daily effective spread for SPY complex options and then the trade-level controls (quoted spread, absolute delta, days to expiration, vega, gamma, implied volatility, log trade size, and a PM-settled indicator) with strategy fixed effects. μ_s denotes strategy fixed effects. Absolute delta is omitted from the market maker revenue regressions because it mechanically determines the hedging cost component of the dependent variable. When a market maker initiates the trade, the effective spread and market maker revenue are measured from the side of its counterparty. t-statistics in parentheses are computed from standard errors clustered by date. Intercept, size-band indicators, and some controls are omitted for brevity. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

	Effective Spread			Market Maker Revenue		
	(1)	(2)	(3)	(1)	(2)	(3)
Opening × ≤ P25	5.63***	5.86***	3.49***	5.53***	5.72***	3.49***
	(11.65)	(11.44)	(4.66)	(11.11)	(11.23)	(4.79)
Opening × P25–P50	1.73**	1.93**	1.10	1.89**	2.05**	1.14
	(2.15)	(2.43)	(1.30)	(2.29)	(2.54)	(1.35)
Opening × P50–P75	0.41	0.56	0.66	0.83	0.95*	0.66
	(0.79)	(1.08)	(1.35)	(1.57)	(1.82)	(1.36)
Opening × P75–P90	−1.04	−0.88	−0.48	−0.84	−0.71	−0.47
	(−0.94)	(−0.70)	(−0.38)	(−0.73)	(−0.56)	(−0.38)
Opening × > P90	−4.17***	−4.00***	−4.35***	−4.66***	−4.53***	−4.30***
	(−4.24)	(−3.80)	(−4.16)	(−4.54)	(−4.10)	(−4.16)
VIX	0.07**	0.03**	0.02	0.07***	0.04**	0.02
	(2.58)	(2.03)	(1.42)	(2.76)	(2.17)	(1.30)
SPY Complex ES		0.85***	0.78***		0.73***	0.74***
		(4.65)	(3.25)		(3.59)	(2.96)
Other Controls	No	No	Yes	No	No	Yes
Strategy FE	No	No	Yes	No	No	Yes
R-squared	0.01	0.01	0.04	0.01	0.01	0.03
Observations	209,117	209,117	204,969	204,969	204,969	204,969

Table VIII. Price Impact of Options Trading on Futures Prices

This table reports OLS regressions of E-mini S&P 500 futures returns on the signed delta exposure of complex SPX trades. The dependent variable is the log futures return, in basis points, measured 10 seconds or 15 seconds after the options trade, relative to 100 milliseconds before it. Signed Delta is the trade's net delta exposure, normalized to $10,000 per one-point SPX move, and measured from the side of the end user: from the side of the initiator, or from the side of its counterparty when a market maker initiates the trade. The sample compares floor trades executed in the two weeks after the opening (June 15–26, 2020) with electronic trades executed during the last two weeks of the floor closure (June 1–12, 2020), and includes all trades with complete controls. Electronic executions are grouped into orders. Floor is an indicator that equals one if the trade executes on the Cboe trading floor and zero otherwise. For each horizon, the table reports the specification without and with strategy fixed effects. The estimated specification follows Equation (4):

$$r^{f}_{t+h} = \alpha + \beta_1 \, Signed\ Delta_i + \beta_2 \, Floor_i \times Signed\ Delta_i + \gamma' X_i + \mu_t + \mu_s + \varepsilon_i$$

where r^{f}_{t+h} is the E-mini futures return at horizon $h \in \{10 \text{ seconds}, 15 \text{ seconds}\}$, μ_t denotes date fixed effects, and μ_s denotes strategy fixed effects. Because floor and electronic trades occur on different dates, the Floor indicator is absorbed by the date fixed effects. X_i includes trade-level controls in all specifications: days to expiration, vega, gamma, implied volatility, trade size, a PM-settled indicator, and number of legs. t-statistics in parentheses are computed from standard errors clustered by date. Intercepts and controls are omitted for brevity. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

	10 Seconds		15 Seconds	
	(1)	(2)	(3)	(4)
Signed Delta ($10K)	0.43***	0.44***	0.42***	0.43***
	(3.02)	(3.11)	(3.94)	(3.99)
Floor × Signed Delta	−0.54**	−0.57**	−0.68***	−0.71***
	(−2.03)	(−2.15)	(−2.59)	(−2.68)
Day FE	Yes	Yes	Yes	Yes
Strategy FE	No	Yes	No	Yes
R-squared	0.00	0.00	0.00	0.00
Observations	204,969	204,969	204,969	204,969

Table IX. Replication Costs of Floor Trades

This table reports the cost of replicating large floor trades through electronic execution. The target trade size is 1,950 contracts, the 90th percentile (P_{90}) of the trade size distribution of floor trades in the two weeks after the opening (June 15–26, 2020). Floor costs are computed as the mean dollar effective spread (ES) of floor trades whose trade size rank falls within the band's percentile range (±1%, ±2.5%, ±5%, ±10%) relative to the opening distribution. Effective spreads are measured from the side of the initiator, or from the side of the counterparty when a market maker initiates the trade. Electronic replication targets are constructed by chaining consecutive electronic trades within each trading day until cumulative trade size reaches the target, pro-rated to deliver exact target amounts, with execution costs estimated from the two weeks before the opening (June 1–12, 2020). Electronic fills are grouped into orders. Slippage is computed as Slippage = $\beta_1 \cdot D^2 \cdot S/10^8$, where β_1 is the electronic price-impact coefficient on Signed Delta at the 10-second (10s) and 15-second (15s) horizons from the all-trade specifications without strategy fixed effects in Table VIII (in basis points per $10,000 of signed delta), D is the absolute delta exposure of the target in $/SPX point, and S = 3,100 is the June 2020 average SPX level. D scales the mean absolute net delta of opening-period floor trades ($1,100 per SPX point at a mean trade size of 841 contracts) to the target trade size, which gives D = 2,549. Electronic total cost adds to the electronic spread the smallest (low estimate) and the largest (high estimate) slippage across horizons. Floor advantage is the electronic total cost minus the floor total cost, in dollars and as a percentage of the electronic total cost.

Panel A: Electronic Replication Mechanics (Jun 1–12, 2020)					
Trades Required	(mean)	65.3			
	(median)	64.0			
Elapsed Minutes	(mean)	1.3			
	(median)	1.0			
Trade Size	(target)	1,950			
Panel B: Electronic Execution Costs from Cumulative Futures Price Impact ($)					
Slippage	(10s)	87			
	(15s)	85			
Panel C: Total Execution Costs ($)					
Electronic Spread		6,081			
Electronic Total Cost	(low estimate)	6,166			
	(high estimate)	6,168			
		±1%	±2.5%	±5%	±10%
Floor Total Cost		2,439	3,357	4,246	3,745
Floor Advantage	($)	3,727–3,729	2,809–2,811	1,920–1,922	2,421–2,423
	(%)	60.4–60.5	45.6	31.1–31.2	39.3
Panel D: Floor sample composition					
Band Ranges		±1%	±2.5%	±5%	±10%
Percentile Ranges		P89–P91	P87.5–P92.5	P85–P95	P80–P100
Observations	(each side)	34	85	170	339
Mean Trade Size	(in band)	1,925	1,903	1,969	2,668

Table X. Fills That Cover the Whole Order Leg by Trade Size

This table reports the percentage of fills that cover the whole order leg, for AIM trades executed during the last two weeks of the floor closure (June 1–12, 2020) and floor trades executed in the two weeks after the opening (June 15–26, 2020), for all trades and by trade-size band. A fill is a single execution in one leg of a package, and it covers the whole order leg when its size equals the size of that leg of the order. Size bands are defined by percentiles of the floor trade-size distribution, in contracts. Electronic executions are grouped into orders. The sample contains 175,051 AIM fills and 28,083 floor fills.

Size (Contracts)	AIM (Jun 1–12)	Floor (Jun 15–26)	Floor minus AIM
All Trades	26.8	14.3	−12.5
≤ P25 (≤ 166)	29.9	46.8	16.9
P25–P50 (167–380)	16.3	27.2	10.9
P50–P75 (381–839)	12.4	13.6	1.2
P75–P90 (840–1,948)	7.0	3.4	−3.6
> P90 (≥ 1,949)	3.9	1.3	−2.7

Table XI. Price Impact of Options Trading on Futures Prices by Trade Size

This table reports OLS regressions of E-mini S&P 500 futures returns on the signed delta exposure of complex SPX trades by trade size. The dependent variable is the log futures return, in basis points, measured 10 seconds or 15 seconds after the options trade, relative to 100 milliseconds before it. Signed Delta is the trade's net delta exposure, normalized to $10,000 per one-point SPX move, and measured from the side of the end user: from the side of the initiator, or from the side of its counterparty when a market maker initiates the trade. The sample compares floor trades executed in the two weeks after the opening (June 15–26, 2020) with electronic trades executed during the last two weeks of the floor closure (June 1–12, 2020). Electronic executions are grouped into orders. Floor is an indicator that equals one if the trade executes on the Cboe trading floor and zero otherwise. Panel A restricts the sample to trades at or below the 25th percentile of floor trade size (166 contracts or fewer), and Panel B to trades above the 90th percentile (more than 1,948.8 contracts). Within each panel and horizon, the table reports the specification without and with strategy fixed effects. The estimated specification follows Equation (4):

$$r^f_{t+h} = \alpha + \beta_1 \, Signed\ Delta_i + \beta_2 \, Floor_i \times Signed\ Delta_i + \gamma' X_i + \mu_t + \mu_s + \varepsilon_i$$

where r^f_{t+h} is the E-mini futures return at horizon $h \in \{10 \text{ seconds}, 15 \text{ seconds}\}$, μ_t denotes date fixed effects, and μ_s denotes strategy fixed effects. Because floor and electronic trades occur on different dates, the Floor indicator is absorbed by the date fixed effects. X_i includes trade-level controls in all specifications: days to expiration, vega, gamma, implied volatility, trade size, a PM-settled indicator, and number of legs. t-statistics in parentheses are computed from standard errors clustered by date. Intercepts and controls are omitted for brevity. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Panel A: Small Trades (≤ P25)				
	10 Seconds		15 Seconds	
	(1)	(2)	(3)	(4)
Signed Delta ($10K)	−0.22	−0.08	−0.37	−0.20
	(−0.29)	(−0.11)	(−1.00)	(−0.52)
Floor × Signed Delta	−1.17	−1.04	−1.96	−1.84
	(−0.64)	(−0.59)	(−1.04)	(−0.97)
Day FE	Yes	Yes	Yes	Yes
Strategy FE	No	Yes	No	Yes
R-squared	0.00	0.00	0.00	0.00
Observations	194,870	194,870	194,870	194,870
Panel B: Large Trades (> P90)				
	10 Seconds		15 Seconds	
	(1)	(2)	(3)	(4)
Signed Delta ($10K)	0.26***	0.31**	0.27**	0.24
	(4.78)	(2.20)	(2.38)	(1.63)
Floor × Signed Delta	−0.99**	−1.09**	−1.03*	−1.09**
	(−2.34)	(−2.54)	(−1.83)	(−1.98)
Day FE	Yes	Yes	Yes	Yes
Strategy FE	No	Yes	No	Yes
R-squared	0.05	0.15	0.04	0.14
Observations	672	672	672	672

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Internet Appendix

Not Dead Yet: Options Trading Floors

— not for publication —

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ABSTRACT

This Internet Appendix accompanies "Not Dead Yet: Options Trading Floors." Section IA.I repeats the matched analysis around the March 2020 floor closure. Matched floor trades pay 6.33 cents more than their electronic matches with the full set of controls. Section IA.II splits electronic execution into the complex limit order book (CLOB), the Complex Order Auction (COA), and the Automated Improvement Mechanism (AIM), matches floor trades to AIM trades around the opening and the closure, and computes the cost of replicating large floor trades with AIM trades. Section IA.III defines the complex strategies and reports their volume shares and characteristics, and Section IA.IV describes how packages are reconstructed, classified, and signed from the Cboe trade-by-trade data. Section IA.V shows that quote midpoints drift in the direction of the trade before execution in every venue. Section IA.VI reports the capacity of the initiating party of complex SPX packages on each venue. Section IA.VII documents the institutional details of floor trading and hybrid execution on Cboe.

IA.I: Floor Closure Results

The main analysis in Section III focuses on the opening of the Cboe trading floor in June 2020 because the closure in March 2020 coincided with the onset of the COVID-19 pandemic, a period of extreme volatility and shifts in market participation that could confound any comparison between floor and electronic execution. This appendix repeats the matched analysis around the closure. For each floor trade executed in the two weeks before the closure (March 2 to March 13, 2020), we identify the nearest electronic trade from the two weeks after it (March 16 to March 27, 2020), using the matching algorithm, covariates, and caliper of Section III.B. Electronic executions are grouped into orders, and effective spreads and market maker revenues are measured from the side of the counterparty when a market maker initiates the trade. Of 5,941 floor trades, 1,447 find an electronic match, a match rate of 24.4%.

Table IA.I reports means before and after matching. Before matching, floor trades average 1,246 contracts and 13.0 legs, against 53 contracts and 2.7 legs for electronic trades, and they have longer maturities (83 against 42 days to expiration) and higher absolute vega (1.56 against 0.26). After matching, the two groups are balanced, with 693 and 689 contracts, 3.8 legs each, and 45 and 47 days to expiration. The 4,494 unmatched floor trades average 1,423 contracts and 15.9 legs. Matched floor trades pay an effective spread of 13.83 cents, against 9.73 cents for their electronic matches. The difference of 4.09 cents is 42% of the electronic mean. Market maker revenues show the same gap (13.51 against 9.40 cents), and hedge costs are balanced at about 0.3 cents. Spreads around the closure are about three times their level around the opening, when matched electronic trades pay 3.15 cents (Table IV), which reflects the volatility of March 2020.

TABLE IA.I HERE

Table IA.II reports the matched regressions. The Closure indicator equals one for electronic trades during the closure and zero for floor trades before it, so a negative coefficient means that floor execution costs more. With the VIX as the only control, matched floor trades pay 8.64 cents more than their electronic matches. The gap is larger than the raw difference because the VIX is much higher during the closure window, and each VIX point raises effective spreads by 0.26 cents. Controlling also for the SPY complex spread leaves the estimate at 8.35 cents. With the full set of trade-level controls and strategy fixed effects, the difference is 6.33 cents, 65% of the mean electronic spread, and the VIX coefficient is no longer significant. Market maker revenues give nearly identical estimates, from 8.65 to 6.55 cents. Every estimate is significant at the 1% level.

TABLE IA.II HERE

The closure results have the same sign as the opening results. In both transitions, matched floor trades cost more than electronic trades of similar size and complexity, 1.14 cents (36%) around the opening and 6.33 cents (65%) around the closure with the full set of controls. As in the main analysis, the electronic matches execute during the closure, when AIM is available for SPX complex orders. The larger gap in March coincides with extreme volatility, and we do not interpret the difference in magnitudes. The consistent sign across two transitions three months apart supports the interpretation that the cost difference reflects the presence or absence of floor intermediation rather than transient pandemic conditions.

Table IA.I. Matched Sample Summary Statistics around Floor Closure

This table compares pre-matching and post-matching means for floor and electronic complex SPX trades around the closure of the Cboe trading floor. For each floor trade executed in the two weeks before the closure (March 2–13, 2020), we identify the nearest electronic trade from the two weeks after the closure (March 16–27, 2020) using one-to-one nearest-neighbor matching without replacement. We match on six covariates: absolute delta (|Delta|), absolute vega (|Vega|), implied volatility, trade size in contracts, days to expiration, and number of legs, imposing a per-variable caliper of 0.25 standard deviations. Within each panel, the left columns report pre-matching means and differences, and the right columns report post-matching means and differences. Panel A reports the matching covariates before and after matching. Panel B reports execution quality measures that are not included in the matching algorithm: the effective spread, hedge cost, and market maker revenue, all measured in cents per unit of underlying. Electronic executions are grouped into orders before matching. When a market maker initiates the trade, the effective spread and market maker revenue are measured from the side of its counterparty. The sample contains 2,894 trades (1,447 matched pairs out of 5,941 floor trades) with a match rate of 24.4%.

Panel A: Options Trade Characteristics							
	Before Matching			After Matching			
	Floor	Electronic	Difference	Floor	Electronic	Difference	Floor Unmatched
|Delta|	0.02	0.02	0.00	0.02	0.02	0.00	0.02
|Vega|	1.56	0.26	1.29	0.59	0.58	0.01	1.87
Implied Vol.	0.52	1.14	−0.62	0.67	0.70	−0.03	0.47
Trade Size	1245.83	52.53	1193.30	693.28	689.37	3.91	1423.10
Days-to-Exp.	82.61	42.49	40.12	45.04	47.10	−2.06	94.67
Legs	12.96	2.69	10.27	3.82	3.82	0.00	15.89
Panel B: Execution Quality							
	Before Matching			After Matching			
	Floor	Electronic	Difference	Floor	Electronic	Difference	Floor Unmatched
Effective Spread	16.07	4.23	11.84	13.83	9.73	4.09	16.79
Hedge Cost	0.30	0.33	−0.02	0.31	0.33	−0.02	0.30
MM Revenue	15.77	3.90	11.87	13.51	9.40	4.11	16.49

Table IA.II. Execution Quality and Floor Closure

This table reports OLS regressions of the effective spread and market maker revenue on matched samples of complex SPX trades. Each pre-closure floor trade (March 2–13, 2020) is matched 1:1 without replacement to the nearest closure-period electronic trade (March 16–27, 2020) on |delta|, |vega|, implied volatility, trade size, days to expiration, and number of legs. The estimated specification follows Equation (2):

$$Y_i = \alpha + \beta_1 \, Closure_i + \gamma' X_i + \mu_s + \varepsilon_i$$

where Y_i is measured in cents per unit of underlying. $Closure_i$ equals one for electronic trades executed during the closure and zero for floor trades executed before the closure. Column (1) controls for the VIX. Column (2) adds the average daily effective spread for SPY complex options. Column (3) adds the trade-level controls (quoted spread, absolute delta, days to expiration, vega, gamma, implied volatility, log trade size, and a PM-settled indicator) and strategy fixed effects. μ_s denotes strategy fixed effects. Absolute delta is omitted from the market maker revenue regressions because it mechanically determines the hedging cost component of the dependent variable. Electronic executions are grouped into orders before matching. When a market maker initiates the trade, the effective spread and market maker revenue are measured from the side of its counterparty. The matched sample contains 2,894 trades (1,447 matched pairs), and the regressions use the 2,886 trades with complete controls. t-statistics in parentheses are computed from standard errors clustered by date. Intercept and some controls are omitted for brevity. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

	Effective Spread			Market Maker Revenue		
	(1)	(2)	(3)	(1)	(2)	(3)
Closure	−8.64***	−8.35***	−6.33***	−8.65***	−8.36***	−6.55***
	(−5.92)	(−5.71)	(−5.84)	(−6.02)	(−5.84)	(−5.64)
VIX	0.26***	0.23***	0.08	0.26***	0.23***	0.09
	(6.98)	(3.28)	(1.37)	(7.12)	(3.38)	(1.58)
SPY Complex ES		0.49	0.99		0.50	0.83
		(0.45)	(1.04)		(0.47)	(0.85)
|Delta|			68.92***			
			(3.50)			
Other Controls	No	No	Yes	No	No	Yes
Strategy FE	No	No	Yes	No	No	Yes
R-squared	0.02	0.02	0.13	0.02	0.02	0.12

IA.II: Decomposing Electronic Execution

The main text pools all electronic complex SPX trades into a single comparison group. Electronic execution on Cboe, however, operates through three distinct channels. In the complex limit order book (CLOB), an incoming complex order executes against resting complex orders or legs into the individual series books. The Complex Order Auction (COA) exposes an incoming marketable complex order to competing electronic responses before filling it at the best available price. The Automated Improvement Mechanism (AIM) is an electronic auction in which a Trading Permit Holder submits a pre-paired Agency Order against an Initiating Order, and the market has an opportunity to improve the price. AIM is structurally closer to floor execution because both involve a broker bringing a client's agency order to competing market makers. This appendix documents how strategy composition differs across electronic channels and compares floor execution directly with AIM.

The decomposition also clarifies the identification. AIM for SPX complex orders was activated during the closure and withdrawn at the opening, so the Opening coefficient in the main matched-sample regression reflects both the return of floor intermediation and the removal of the electronic auction. The SPY complex placebo in the main text addresses this in part, as SPY complex trades were unaffected by the AIM rule change and show no change in execution costs at the opening (-0.05 cents, $t = -0.79$, with the full set of controls and strategy fixed effects). Over the same transition, matched floor SPX complex trades cost 1.14 cents more than their electronic matches ($t = 3.01$, Table V). Matching floor trades directly to AIM trades measures the cost difference between floor execution and the electronic mechanism that was withdrawn at the opening and that most closely resembles floor execution. The comparison also tests whether the cost difference varies with the structure of the electronic counterpart, asking whether floor intermediation differs from electronic execution in general or only from a specific auction design.

A. Strategy Composition across Electronic Channels

Table IA.III reports the share of complex SPX dollar volume by strategy group within the CLOB, COA, and AIM. The three channels serve different segments of the complex market. Most CLOB and COA volume is in packages without a Combo. Before the closure, Vertical Spreads account for 29.58% of CLOB volume and 32.79% of COA volume, followed by Butterflies and Condors (22.29% and 25.20%) and Calendars and Diagonals (16.93% and 17.19%). Combo-based packages (Combos, Index Combos, Combo Rolls, Box Spreads, and Multiple Combos) account for only 2.43% of CLOB volume and 1.08% of COA volume before the closure. During the closure, when all complex flow executes electronically, the combo-based share of CLOB volume rises to 31.94%, led by Index Combos (13.48%) and Box Spreads (9.58%), consistent with some displaced floor flow entering the book.

In COA, the combo-based share rises only to 4.93%. In AIM, by contrast, combo-based packages account for 78.54% of closure-period dollar volume, and Index Combos alone account for 37.01%. This composition is close to that of the floor, where combo-based packages account for 76.71% of pre-closure volume (Table IA.X). After the opening, the combo-based share of CLOB volume falls back to 11.65%. The main text compares floor trades from June 15 to June 26 with electronic trades from June 1 to June 14, the last two weeks of the closure. If most of the matched electronic trades were AIM trades, the Opening coefficient would largely reflect the difference between floor and AIM execution. The matched comparison with AIM below measures this difference directly.

TABLE IA.III HERE

B. Matched-Sample Analysis against AIM

We redo the matched-sample analysis of Section III with a new match that restricts the electronic sample to AIM trades. We begin with the opening event and then report the closure event, which Section IA.I analyzes for the pooled electronic sample.

B.1. Opening Event

The opening event is the primary identification strategy in the main text and the setting where the AIM confound is most relevant. The Opening coefficient in the pooled specification captures the combined effect of floor intermediation and AIM withdrawal. Matching post-opening floor trades against AIM trades from the closure measures the cost difference between floor execution and the withdrawn mechanism directly.

Table IA.IV reports pre- and post-matching trade characteristics for the opening event. Each post-opening floor trade is paired with its nearest AIM counterpart during the closure, yielding 2,052 observations (1,026 matched pairs) with a match rate of 31.2%. Before matching, floor trades are roughly ten times larger than AIM trades (784 versus 75 contracts) and have more legs (12.1 versus 3.2). After matching, the six covariates are closely aligned, with a remaining gap in trade size of 25 contracts (295 versus 270). The floor effective spread is 4.72 cents against 2.18 cents for AIM, a difference of 2.54 cents. Hedge costs are balanced at 0.22 cents.

TABLE IA.IV HERE

Table IA.V reports the matched-sample regressions. As in Table V, the controls range from the VIX alone to the SPY complex effective spread and the full set of trade-level controls with strategy fixed effects. The Opening coefficient for effective spreads is 2.31 cents ($t = 4.09$) with VIX as the only control and 2.44 cents ($t = 6.13$) with the full set of controls and strategy fixed effects. Relative to the AIM post-matching mean of 2.18

cents, floor trades pay more than twice the effective spread of comparable AIM trades. The estimate is about twice the pooled estimate of 1.14 cents in the main text. Matched AIM trades pay 2.18 cents, against 3.15 cents for matched electronic trades in the main text (Table IV). Market maker revenue estimates track the effective spread results closely, with Opening coefficients between 2.31 and 2.47 cents, all significant at the 1% level.

TABLE IA.V HERE

B.2. Closure Event

Table IA.VI reports pre- and post-matching trade characteristics for the closure event. Each pre-closure floor trade (March 2–13, 2020) is paired with its nearest AIM counterpart during the closure (March 16–27, 2020). Of 5,941 floor trades, 1,374 find an AIM match, a match rate of 23.1%, for 2,748 observations. Before matching, floor trades average 1,246 contracts and 13.0 legs, against 200 contracts and 4.0 legs for AIM trades, and they have longer maturities (83 against 42 days to expiration) and higher absolute vega (1.56 against 0.46). AIM trades during the closure are larger and have more legs than the average electronic trade of the same two weeks, which has 53 contracts and 2.7 legs (Table IA.I). After matching, floor trades average 662 contracts and AIM trades 635, with 4.9 and 4.8 legs and 57 and 59 days to expiration. The 4,567 unmatched floor trades average 1,421 contracts and 15.4 legs. Matched floor trades pay an effective spread of 14.67 cents, against 6.52 cents for their AIM matches. The difference of 8.15 cents compares with 4.09 cents against all electronic trades (Table IA.I). Market maker revenues show the same gap (14.43 against 6.28 cents), and hedge costs are balanced at about 0.25 cents.

TABLE IA.VI HERE

Table IA.VII reports the matched-sample regressions for the closure event. Closure equals one for AIM trades executed during the closure and zero for matched floor trades executed before it. A negative coefficient therefore indicates higher costs for floor trades. The Closure coefficient for effective spreads is −12.71 cents ($t = -7.33$) with the VIX as the only control and −9.56 cents ($t = -7.10$) with the full set of controls and strategy fixed effects. The VIX coefficient falls from 0.30 cents per point with the VIX as the only control to 0.12 cents with the full set of controls, where it is significant at the 10% level. Market maker revenue estimates range from −9.71 to −12.72 cents. Every Closure estimate is significant at the 1% level. Against all electronic trades, the full-specification estimate is −6.33 cents (Table IA.II). In absolute value, the closure estimate of 9.56 cents is about four times the opening estimate of 2.44 cents, and matched AIM spreads in March are about three times their level in June (6.52 against 2.18 cents). Relative to the mean spread of matched AIM trades, floor trades pay 112% more around the opening and 147% more around the closure.

TABLE IA.VII HERE

C. Electronic Replication Costs

We repeat the replication cost analysis of Table IX using only AIM trades to construct the electronic chains. AIM is the electronic mechanism closest to the floor and the one withdrawn at the opening. Table IA.VIII uses the target of the main text, 1,950 contracts, the 90th percentile of the trade size distribution of floor trades in the two weeks after the opening. We chain consecutive AIM trades within each trading day from June 1 to 12, 2020, until cumulative trade size reaches the target, and pro-rate the last trade in each chain. Slippage is $\beta_1 \cdot D^2 \cdot S/10^8$, where β_1 is the electronic price-impact coefficient for all trades at the 10-second and 15-second horizons from Table VIII, D is the absolute delta exposure of the target, computed as in Table IX, and $S = 3{,}100$ is the June 2020 average SPX level. Slippage depends only on the target and is the same as in the main text.

Assembling the target through AIM requires on average 36.2 trades (median 30.0) over 4.1 minutes (median 3.3) (Panel A). AIM trades are larger than the average electronic order, so the chain needs fewer trades than the 65.3 of the pooled replication, and it takes longer because AIM trades arrive less often. The electronic spread component is \$6,166, slightly above the pooled \$6,081. Adding cumulative futures slippage of \$85 to \$87 across the two horizons (Panel B) yields total electronic costs between \$6,251 and \$6,253 (Panel C). Floor trades of similar size cost \$2,439 in the ±1% band, \$3,357 in the ±2.5% band, and \$4,246 in the ±5% band, for a floor advantage of \$2,005 to \$3,814, or 32 to 61% of the electronic total cost. In the ±10% band, which spans the 80th to the 100th percentile with a mean size of 2,668 contracts, the floor costs \$3,745. Replicating a large floor trade with AIM trades costs about as much as replicating it with all electronic orders, and the floor advantage is close to the 31 to 60% of the pooled replication.

TABLE IA.VIII HERE

Table IA.III. Complex Strategy Composition in CLOB, COA, and AIM

This table reports the percentage of total SPX complex dollar volume attributable to each strategy group in the complex limit order book (CLOB), the Complex Order Auction (COA), and the Automated Improvement Mechanism (AIM). Strategies are classified following Li (2022) based on the configuration of option legs within each package and grouped by package structure; Table IA.IX defines the strategies in each group. The first three columns report strategy shares within CLOB volume and the next three columns report strategy shares within COA volume during the pre-closure (January 2 to March 13, 2020), closure (March 16 to June 14, 2020), and opening (June 15 to October 21, 2020) periods. The final column reports strategy shares within AIM volume during the closure period. Each column sums to 100%. All values are percentages.

Strategy	CLOB			COA			AIM
	Pre	Closure	Opening	Pre	Closure	Opening	Closure
Combo	0.39	0.55	3.65	0.14	0.67	2.04	0.01
Index Combo	0.31	13.48	0.11	0.21	1.56	0.16	37.01
Combo Roll	0.83	7.37	1.37	0.20	1.62	0.44	14.99
Box Spread	0.90	9.58	6.52	0.53	1.08	7.26	23.54
Multiple Combos	0.00	0.96	0.00	0.00	0.00	0.00	2.99
Ratio Spreads	6.69	6.98	4.98	8.00	8.01	3.84	2.49
Straddles and Strangles	8.28	7.19	8.22	2.31	2.41	3.19	2.37
Calendars and Diagonals	16.93	15.93	26.19	17.19	28.35	25.04	4.14
Vertical Spreads	29.58	20.29	24.68	32.79	30.72	28.75	5.85
Vertical Rolls	12.68	8.09	6.20	12.58	12.60	9.15	1.59
Butterflies and Condors	22.29	9.11	17.67	25.20	12.47	19.59	4.87
Other Strategies	1.14	0.48	0.39	0.86	0.50	0.55	0.10

Table IA.IV. Matched Sample Summary Statistics around Opening: Floor versus AIM
This table compares pre-matching and post-matching means for floor complex SPX trades and complex SPX trades executed in the Automated Improvement Mechanism (AIM) around the opening of the Cboe trading floor. For each floor trade executed in the two weeks after the opening (June 15–26, 2020), we identify the nearest AIM trade from the two weeks before the opening (June 1–14, 2020) using one-to-one nearest-neighbor matching without replacement. We match on six covariates: absolute delta (|Delta|), absolute vega (|Vega|), implied volatility, trade size in contracts, days to expiration, and number of legs, imposing a per-variable caliper of 0.25 standard deviations. Panel A reports the matching covariates before and after matching. Panel B reports execution quality measures that are not included in the matching algorithm: the effective spread, hedge cost, and market maker revenue, all measured in cents. When a market maker initiates the trade, the effective spread and market maker revenue are measured from the side of its counterparty, so that both are end-user measures (Section II.C), the cost borne by the non-market-maker side. Within each panel, the left columns report pre-matching means and differences, and the right columns report post-matching means and differences. The sample contains 2,052 trades (1,026 matched pairs) with a match rate of 31.2%.

Panel A: Options Trade Characteristics

	Before Matching			After Matching			
	Floor	AIM	Difference	Floor	AIM	Difference	Floor Unmatched
|Delta|	0.02	0.02	0.00	0.02	0.02	0.00	0.02
|Vega|	1.20	0.24	0.96	0.56	0.56	0.00	1.49
Implied Vol.	0.35	0.34	0.01	0.38	0.37	0.00	0.35
Trade Size	783.62	75.17	708.44	295.11	269.79	25.32	1005.59
Days-to-Exp.	84.18	22.84	61.34	33.13	33.12	0.01	107.38
Legs	12.14	3.22	8.93	3.80	3.78	0.01	15.93

Panel B: Execution Quality

	Before Matching			After Matching			
	Floor	AIM	Difference	Floor	AIM	Difference	Floor Unmatched
Effective Spread	5.10	1.85	3.25	4.72	2.18	2.54	5.27
Hedge Cost	0.24	0.30	−0.06	0.22	0.22	0.01	0.25
MM Revenue	4.86	1.56	3.31	4.50	1.97	2.53	5.03

Table IA.V. Execution Quality and Floor Opening: Floor versus AIM

This table reports OLS regressions of effective spread and market maker revenue on matched samples of floor and AIM complex SPX trades. Each post-opening floor trade (June 15–26, 2020) is matched 1:1 without replacement to the nearest pre-opening trade executed in the Automated Improvement Mechanism (AIM) (June 1–14, 2020) on |delta|, |vega|, implied volatility, trade size, days to expiration, and number of legs. The estimated specification follows Equation (2):

$$Y_i = \alpha + \beta\, Opening_i + \gamma' X_i + \mu_s + \varepsilon_i$$

where Y_i is measured in cents. $Opening_i$ equals one for floor trades executed after the opening and zero for matched AIM trades executed during the closure. X_i includes trade-level controls in all specifications: quoted spread, absolute delta, days to expiration, vega, gamma, implied volatility, log trade size, a PM-settled indicator, the average daily effective spread for SPY complex options, and the VIX volatility index. μ_s denotes strategy fixed effects. Absolute delta is omitted from the market maker revenue regressions because it mechanically determines the hedging cost component of the dependent variable. When a market maker initiates the trade, the effective spread and market maker revenue are measured from the side of its counterparty. The sample contains 2,052 trades (1,026 matched pairs). t-statistics in parentheses are computed from standard errors clustered by date. Intercept and some controls are omitted for brevity. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}\ p < 0.01$.

	Effective Spread			Market Maker Revenue		
	(1)	(2)	(3)	(1)	(2)	(3)
Opening	2.31***	2.45***	2.44***	2.31***	2.45***	2.47***
	(4.09)	(4.93)	(6.13)	(4.26)	(5.19)	(6.07)
VIX	0.10**	0.06*	−0.00	0.10**	0.07*	0.00
	(2.15)	(1.73)	(−0.03)	(2.30)	(1.89)	(0.12)
SPY Complex ES		0.78*	0.53		0.77**	0.54
		(1.94)	(1.21)		(2.06)	(1.20)
|Delta|			28.28**			
			(1.97)			
Other Controls	No	No	Yes	No	No	Yes
Strategy FE	No	No	Yes	No	No	Yes
R-squared	0.01	0.02	0.05	0.01	0.02	0.04

Table IA.VI. Matched Sample Summary Statistics around Closure: Floor versus AIM

This table compares pre-matching and post-matching means for floor complex SPX trades and complex SPX trades executed in the Automated Improvement Mechanism (AIM) around the closure of the Cboe trading floor. For each floor trade executed in the two weeks before the closure (March 2–13, 2020), we identify the nearest AIM trade from the two weeks after the closure (March 16–27, 2020) using one-to-one nearest-neighbor matching without replacement. We match on six covariates: absolute delta (|Delta|), absolute vega (|Vega|), implied volatility, trade size in contracts, days to expiration, and number of legs, imposing a per-variable caliper of 0.25 standard deviations. Panel A reports the matching covariates before and after matching. Panel B reports execution quality measures that are not included in the matching algorithm: the effective spread, hedge cost, and market maker revenue, all measured in cents per unit of underlying. AIM executions are grouped into orders before matching. When a market maker initiates the trade, the effective spread and market maker revenue are measured from the side of its counterparty, so that both are end-user measures (Section II.C), the cost borne by the non-market-maker side. Within each panel, the left columns report pre-matching means and differences, the right columns report post-matching means and differences, and the last column reports means for floor trades without a match. The sample contains 2,748 trades (1,374 matched pairs out of 5,941 floor trades) with a match rate of 23.1%.

Panel A: Options Trade Characteristics

	Before Matching			After Matching			
	Floor	AIM	Difference	Floor	AIM	Difference	Floor Unmatched
|Delta|	0.02	0.02	0.01	0.02	0.02	0.00	0.02
|Vega|	1.56	0.46	1.10	0.79	0.77	0.02	1.79
Implied Vol.	0.52	0.99	−0.47	0.64	0.66	−0.02	0.48
Trade Size	1245.83	200.38	1045.45	661.85	634.98	26.87	1420.70
Days-to-Exp.	82.61	41.67	40.95	57.45	59.35	−1.90	90.15
Legs	12.96	4.00	8.96	4.89	4.84	0.05	15.38

Panel B: Execution Quality

	Before Matching			After Matching			
	Floor	AIM	Difference	Floor	AIM	Difference	Floor Unmatched
Effective Spread	16.07	5.21	10.86	14.67	6.52	8.15	16.49
Hedge Cost	0.30	0.26	0.04	0.24	0.25	−0.01	0.32
MM Revenue	15.77	4.95	10.82	14.43	6.28	8.16	16.17

Table IA.VII. Execution Quality and Floor Closure: Floor versus AIM

This table reports OLS regressions of effective spread and market maker revenue on matched samples of floor and AIM complex SPX trades. Each pre-closure floor trade (March 2–13, 2020) is matched 1:1 without replacement to the nearest post-closure trade executed in the Automated Improvement Mechanism (AIM) (March 16–27, 2020) on |delta|, |vega|, implied volatility, trade size, days to expiration, and number of legs, imposing a per-variable caliper of 0.25 standard deviations. The estimated specification follows Equation (2):

$$Y_i = \alpha + \beta\, Closure_i + \gamma' X_i + \mu_s + \varepsilon_i$$

where Y_i is measured in cents per unit of underlying. $Closure_i$ equals one for AIM trades executed during the closure and zero for matched floor trades executed before the closure. A negative coefficient therefore indicates higher costs for floor trades. Column (1) controls for the VIX. Column (2) adds the average daily effective spread for SPY complex options. Column (3) adds the trade-level controls (quoted spread, absolute delta, days to expiration, vega, gamma, implied volatility, log trade size, and a PM-settled indicator) and strategy fixed effects. μ_s denotes strategy fixed effects. Absolute delta is omitted from the market maker revenue regressions because it mechanically determines the hedging cost component of the dependent variable. AIM executions are grouped into orders before matching. When a market maker initiates the trade, the effective spread and market maker revenue are measured from the side of its counterparty. The matched sample contains 2,748 trades (1,374 matched pairs out of 5,941 floor trades, a match rate of 23.1%), and the regressions use the 2,738 trades with complete controls. t-statistics in parentheses are computed from standard errors clustered by date. Intercept and some controls are omitted for brevity. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

	Effective Spread			Market Maker Revenue		
	(1)	(2)	(3)	(1)	(2)	(3)
Closure	−12.71***	−12.40***	−9.56***	−12.72***	−12.40***	−9.71***
	(−7.33)	(−8.30)	(−7.10)	(−7.37)	(−8.35)	(−6.95)
VIX	0.30***	0.26***	0.12*	0.30***	0.26***	0.13*
	(6.88)	(3.78)	(1.70)	(6.97)	(3.79)	(1.86)
SPY Complex ES		0.58	1.30		0.61	1.20
		(0.54)	(1.28)		(0.57)	(1.17)
|Delta|			73.37***			
			(2.70)			
Other Controls	No	No	Yes	No	No	Yes
Strategy FE	No	No	Yes	No	No	Yes
R-squared	0.04	0.04	0.11	0.04	0.04	0.10

Table IA.VIII. AIM Replication Costs of Floor Trades

This table reports the cost of replicating large floor trades through AIM execution. The target trade size is 1,950 contracts, the 90th percentile (P_{90}) of the trade size distribution of floor trades in the two weeks after the opening (June 15–26, 2020). Floor costs are computed as the mean dollar effective spread (ES) of floor trades whose trade size rank falls within the band's percentile range (±1%, ±2.5%, ±5%, ±10%) relative to the opening distribution. Effective spreads are measured from the side of the initiator, or from the side of the counterparty when a market maker initiates the trade. Electronic replication targets are constructed by chaining consecutive AIM trades within each trading day until cumulative trade size reaches the target, prorated to deliver exact target amounts, with execution costs estimated from the two weeks before the opening (June 1–12, 2020). Electronic fills are grouped into orders. Slippage is computed as Slippage = $\beta_1 \cdot D^2 \cdot S/10^8$, where β_1 is the electronic price-impact coefficient on Signed Delta at the 10-second (10s) and 15-second (15s) horizons from the all-trade specifications without strategy fixed effects in Table VIII (in basis points per $10,000 of signed delta), D is the absolute delta exposure of the target in $/SPX point, and S = 3,100 is the June 2020 average SPX level. D scales the mean absolute net delta of opening-period floor trades to the target trade size, which gives D = 2,549. Electronic total cost adds to the electronic spread the smallest (low estimate) and the largest (high estimate) slippage across horizons. Floor advantage is the electronic total cost minus the floor total cost, in dollars and as a percentage of the electronic total cost.

Panel A: AIM Replication Mechanics (Jun 1–12, 2020)					
Trades Required	(mean)	36.2			
	(median)	30.0			
Elapsed Minutes	(mean)	4.1			
	(median)	3.3			
Trade Size	(target)	1,950			
Panel B: Electronic Execution Costs from Cumulative Futures Price Impact ($)					
Slippage	(10s)	87			
	(15s)	85			
Panel C: Total Execution Costs ($)					
Electronic Spread		6,166			
Electronic Total Cost	(low estimate)	6,251			
	(high estimate)	6,253			
		±1%	±2.5%	±5%	±10%
Floor Total Cost		2,439	3,357	4,246	3,745
Floor Advantage	($)	3,812–3,814	2,894–2,896	2,005–2,007	2,506–2,508
	(%)	61.0	46.3	32.1	40.1
Panel D: Floor sample composition					
Band Ranges		±1%	±2.5%	±5%	±10%
Percentile Ranges		P89–P91	P87.5–P92.5	P85–P95	P80–P100
Observations	(each side)	34	85	170	339
Mean Trade Size	(in band)	1,925	1,903	1,969	2,668

IA.III: Complex Strategy Composition

This appendix describes the strategy composition of complex SPX trading across venues and periods and the characteristics of each strategy. We classify each complex package following Li (2022) based on the configuration of its option legs. Packages are formed from the Cboe complex execution identifiers and floor agreement timestamps described in Section IA.IV. We group strategies by package structure. Combo-based packages comprise Combos, Index Combos, Combo Rolls, Box Spreads, and Multiple Combos. A Combo is a synthetic future that pairs a long call with a short put (or vice versa) at the same strike and expiration, and the other combo-based packages combine a Combo with additional legs or with other Combos. Table IA.IX defines each strategy, Table IA.X reports dollar volume shares by strategy group, and Table IA.XI reports average trade characteristics by strategy.

TABLE IA.IX HERE

TABLE IA.X HERE

Table IA.X reports the share of total complex SPX dollar volume by strategy group. Panel A pools floor and electronic trades within each period. Combo-based packages account for 58.12% of all complex SPX dollar volume before the closure and 62.38% after the opening, with a similar share of 57.05% during the closure. Within this group, the Index Combo share falls from 43.03% to 26.36% during the closure, while Box Spreads and Combo Rolls gain share. At the other end of the spectrum, Calendars and Diagonals, which average 12 to 47 contracts per trade (Table IA.XI) and trade predominantly electronically, gain share during and after the closure as electronic execution trades a broader range of order flow.

Panel B disaggregates by venue. Before the closure, combo-based packages account for 76.71% of floor dollar volume but only 1.93% of electronic volume. During the closure, when all order flow executes electronically, combo-based packages account for 57.05% of electronic volume. After the opening, the combo-based share of electronic volume falls back to 11.04%, while combo-based packages make up 85.01% of floor volume. The migration largely reverses once the floor becomes available again. Index Combos show this pattern most clearly, with an electronic share of 0.27% before the closure, 26.36% during the closure, and 0.13% after the opening.

TABLE IA.XI HERE

Table IA.XI reports average delta exposure, hedge cost, and trade size for each complex SPX strategy outside the closure period, with floor and electronic trades pooled. Strategies are grouped by package structure and sorted by descending hedge cost within each group. At the top, Index Combos, Combo Rolls, and Box Spreads trade in by far the largest dollar

sizes. Index Combos average 881 contracts ($5.77 million), Combo Rolls 838 contracts ($7.30 million), and Box Spreads 249 contracts ($13.93 million). Simple Combos carry the highest absolute delta (0.50 per underlying) and the highest hedge cost ($118 per trade). Simple Combos are much smaller than the other combo-based packages. They average 63 contracts ($0.92 million) and account for 0.26% of floor dollar volume before the closure and 0.08% after the opening (Table IA.X, Panel B). Index Combos pair a Combo with additional legs and carry an absolute delta of only 0.01 per underlying, but their size still produces the third-highest hedge cost ($60 per trade). Moving down the table, strategies become progressively smaller and more balanced. Ratio Spreads carry hedge costs between $10 and $32 per trade. Structures such as Butterflies, Calendars, and Vertical Spreads are near-delta-neutral with hedge costs roughly $10 or less per trade and trade sizes between 12 and 89 contracts on average, about one-tenth or less of the Index Combo average.

Table IA.IX. Complex Strategy Definitions

This table defines the complex strategies identified by the classification of SPX packages. Packages are classified following Li (2022) from the option type, strike, expiration, and contract ratio of each leg, and from the side of each leg in the Cboe trade-by-trade data. Strategies are grouped by package structure as in Tables IA.X, IA.XI, and IA.III. Names with (Put) or (Puts) in parentheses denote the put version of the call strategy. Leg sizes are in a 1:1 ratio unless the definition states otherwise. "Bought" and "sold" describe one side of the package; the counterparty holds the opposite position. Strategies marked with † have fewer than 100 packages outside the closure period and are not reported separately in Table IA.XI.

Strategy	Legs	Definition
Combos		
Combo	2	A call and a put with the same strike and expiration, one bought and one sold, in equal sizes (a synthetic future).
Index Combo	3+	One or more Combos executed together with one or more additional option legs that do not form further Combos, as a single package; the sizes of the additional legs need not equal those of the Combos.
Combo Roll	4	Two Combos with different expirations, in opposite directions, with the same or different strikes.
Box Spread	4	Two Combos with the same expiration and different strikes, in opposite directions.
2, 3, or 4 Combos†	4–8	Two to four Combos executed as one package that do not form a Box Spread or a Combo Roll (Multiple Combos in Table IA.X).
Ratio Spreads		
Vertical Call (Put) Ratio	2	Two calls (puts) with the same expiration and different strikes, one bought and one sold, in unequal sizes.
Ratio Calendar Call (Put)	2	Two calls (puts) with the same strike and different expirations, one bought and one sold, in unequal sizes.
Ratio Diagonal Call (Put) Spread	2	Two calls (puts) with different strikes and different expirations, one bought and one sold, in unequal sizes.
Straddle Ratio	2	A call and a put with the same strike and expiration, both bought or both sold, in unequal sizes.
Strangle Ratio	2	A call and a put with different strikes and the same expiration, both bought or both sold, in unequal sizes.
Straddles and Strangles		
Straddle	2	A call and a put with the same strike and expiration, both bought or both sold.
Strangle	2	A call and a put with different strikes and the same expiration, both bought or both sold.
Calendars and Diagonals		
Calendar Call (Put)	2	Two calls (puts) with the same strike and different expirations, one bought and one sold.
Diagonal Call (Put) Spread	2	Two calls (puts) with different strikes and different expirations, one bought and one sold.
Double Calendar Call (Put)	4	Two call (put) Calendars at different strikes over the same two expirations, both bought or both sold.
Double Calendar	4	A call Calendar and a put Calendar at different strikes over the same two expirations, both bought or both sold.
Vertical Spreads		

continued on next page

Table IA.IX continued

Strategy	Legs	Definition
Vertical Call (Put) Spread	2	Two calls (puts) with the same expiration and different strikes, one bought and one sold.
Vertical Rolls		
Vertical Call (Put) Roll	4	Two call (put) Vertical Spreads with different expirations, in opposite directions.
Butterflies and Condors		
Call (Put) Butterfly	3	Three calls (puts) with the same expiration and equally spaced strikes in a 1:2:1 ratio; the middle strike is traded in the opposite direction of the two outer strikes.
Call (Put) Broken Wing Butterfly	3	Three calls (puts) as in a Call (Put) Butterfly, but with unequally spaced strikes.
Call (Put) Ratio Butterfly†	3	Three calls (puts) as in a Call (Put) Butterfly, with two outer legs of equal size, but in an integer ratio other than 1:2:1, with leg sizes that are not all equal.
Call (Put) Butterfly (Custom)	3	Three calls (puts) with the same expiration and three strikes, the middle strike traded in the opposite direction of the two outer strikes, and either (i) outer legs of equal size, unequally spaced strikes, and an integer ratio other than 1:2:1, with leg sizes that are not all equal; or (ii) outer legs of different sizes, with the middle leg equal to the two outer legs combined (e.g., 2:3:1), and any strike spacing.
Iron Butterfly	4	A put and a call at a common middle strike, traded in one direction, and a lower-strike put and a higher-strike call traded in the other direction, all with the same expiration.
Condor Spread with Calls (Puts)	4	Four calls (puts) with the same expiration and four strikes; the two outer strikes are traded in one direction and the two inner strikes in the other.
Iron Condor	4	Two puts at the two lowest strikes and two calls at the two highest strikes, all with the same expiration; the two inner legs are traded in one direction and the two outer legs in the other.
Other Strategies		
Risk Reversal	2	A call and a put with different strikes and the same expiration, one bought and one sold; Li (2022) include this structure in the Combo.
Call (Put) Ladder	3	Three calls (puts) with the same expiration and three strikes in a 1:1:1 ratio; the lowest-strike call (highest-strike put) is traded in one direction and the other two legs in the other.
Non Standard	–	Packages that match none of the configurations above. They account for 4.9% of complex trades and enter only the volume shares, never the spread or other analyses.

Table IA.X. Complex Strategy Composition

This table reports the percentage of total SPX complex dollar volume attributable to each strategy group across the pre-closure (January 2 to March 13, 2020), closure (March 16 to June 14, 2020), and opening (June 15 to October 21, 2020) periods. Strategies are classified following Li (2022) based on the configuration of option legs within each package and grouped by package structure; Table IA.IX defines the strategies in each group. Panel A pools floor and electronic trades within each period; the closure period contains electronic trades only by construction. Panel B disaggregates by venue. For the pre-closure and opening periods, the Floor and Electronic columns report separate volume shares summing to 100% within each venue-period. The closure column reports electronic volume shares only.

Panel A: Volume by Strategy and Period

Strategy	Pre-Closure	Closure	Opening
Combo	0.27	0.24	1.00
Index Combo	43.03	26.36	37.42
Combo Roll	6.87	11.32	6.20
Box Spread	7.36	17.05	17.11
Multiple Combos	0.59	2.08	0.65
Ratio Spreads	5.19	4.39	3.07
Straddles and Strangles	3.20	3.71	3.17
Calendars and Diagonals	7.14	10.24	10.87
Vertical Spreads	13.29	12.75	10.87
Vertical Rolls	4.53	4.68	2.55
Other Strategies	0.44	0.25	0.47

Panel B: Volume by Strategy, Venue, and Period

Strategy	Electronic			Floor	
	Pre-Closure	Closure	Opening	Pre-Closure	Opening
Combo	0.30	0.24	3.09	0.26	0.08
Index Combo	0.27	26.36	0.13	57.18	53.86
Combo Roll	0.60	11.32	1.05	8.95	8.47
Box Spread	0.76	17.05	6.77	9.54	21.67
Multiple Combos	0.00	2.08	0.00	0.78	0.93
Ratio Spreads	7.17	4.39	4.60	4.54	2.40
Straddles and Strangles	6.05	3.71	6.48	2.26	1.71
Calendars and Diagonals	17.04	10.24	25.80	3.86	4.29
Vertical Spreads	30.78	12.75	26.08	7.51	4.17
Vertical Rolls	12.64	4.68	7.22	1.85	0.49
Other Strategies	1.03	0.25	0.43	0.24	0.49

Table IA.XI. Complex Strategy Characteristics

This table reports average trade characteristics for each complex SPX strategy type, excluding the floor closure period (March 16 to June 14, 2020). Floor and electronic trades are pooled. Strategies are defined in Table IA.IX, grouped by package structure, and sorted by descending hedge cost within each group. |Delta| is the absolute net delta of the package per unit of the underlying index. Hedge Cost is defined as the absolute net delta exposure of the trade multiplied by the half quoted spread in E-mini S&P 500 futures prevailing at the time of the option trade, reported in hundreds of dollars. Trade Size is reported in contracts and millions of dollars. |Net Delta| Per Trade is the absolute net delta exposure of the trade in units of 10,000 delta shares. Non Standard groups all packages that the classifier cannot assign to a named strategy; its hedge cost is not reported. Strategies with fewer than 100 packages outside the closure period (marked with † in Table IA.IX; 194 packages in total) are not reported.

			Trade Size		
Strategy	|Delta| Per Underlying	Hedge Cost $ Hundreds Per Trade	Contracts	Million $	|Net Delta| Per Trade 10,000 Shares
Combos					
Combo	0.50	1.18	63.05	0.92	0.32
Index Combo	0.01	0.60	881.13	5.77	0.07
Combo Roll	0.00	0.17	838.26	7.30	0.02
Box Spread	0.01	0.06	248.86	13.93	0.01
Ratio Spreads					
Strangle Ratio	0.04	0.32	292.55	0.62	0.04
Ratio Calendar Call	0.07	0.24	71.68	0.28	0.02
Ratio Calendar Put	0.04	0.22	228.20	0.37	0.03
Ratio Diagonal Put Spread	0.03	0.21	209.12	0.54	0.04
Ratio Diagonal Call Spread	0.04	0.19	105.60	0.45	0.05
Straddle Ratio	0.03	0.15	47.80	0.38	0.01
Vertical Call Ratio	0.03	0.11	60.13	0.14	0.01
Vertical Put Ratio	0.02	0.10	85.57	0.19	0.01
Straddles and Strangles					
Straddle	0.05	0.15	36.40	0.34	0.01
Strangle	0.04	0.10	30.49	0.11	0.01
Calendars and Diagonals					
Diagonal Put Spread	0.05	0.13	46.55	0.18	0.02
Diagonal Call Spread	0.05	0.12	30.29	0.17	0.02
Double Calendar Call	0.01	0.03	24.11	0.08	0.00
Calendar Put	0.01	0.03	16.08	0.10	0.00
Calendar Call	0.01	0.02	11.74	0.08	0.00
Double Calendar Put	0.01	0.02	21.50	0.11	0.00
Double Calendar	0.01	0.02	30.09	0.10	0.00
Vertical Spreads					
Vertical Call Spread	0.04	0.08	19.76	0.03	0.01
Vertical Put Spread	0.03	0.07	25.44	0.04	0.01
Vertical Rolls					
Vertical Call Roll	0.02	0.07	36.68	0.21	0.01
Vertical Put Roll	0.01	0.06	63.59	0.42	0.01
Butterflies and Condors					
Put Butterfly (Custom)	0.01	0.09	89.32	0.37	0.01
Put Butterfly	0.01	0.07	65.40	0.17	0.01
Iron Butterfly	0.03	0.06	17.35	0.04	0.01
Call Butterfly (Custom)	0.02	0.05	26.65	0.04	0.00
Condor Spread with Puts	0.01	0.04	48.99	0.14	0.00
Call Butterfly	0.02	0.04	23.19	0.04	0.00
Iron Condor	0.01	0.03	25.40	0.02	0.00
Condor Spread with Calls	0.01	0.03	36.47	0.06	0.00
Call Broken Wing Butterfly	0.02	0.03	23.59	0.04	0.00
Put Broken Wing Butterfly	0.01	0.02	25.13	0.11	0.00
Other Strategies					
Risk Reversal	0.27	0.66	53.19	0.21	0.11
Put Ladder	0.02	0.16	98.82	0.56	0.02
Call Ladder	0.03	0.14	54.83	0.15	0.01
Non Standard	0.04	–	109.41	0.46	0.04

IA.IV: Complex Order Reconstruction and Strategy Classification

We construct SPX complex packages from the Cboe Trade-by-Trade (TBT) data. Each TBT record reports one side of an execution in a single option series, with that side's capacity (customer, professional customer, firm, broker-dealer or market maker), direction and trade type. The two sides of an execution share an execution identifier (*exec_id*). We merge the two records of each execution into one fill and identify the initiating side from the pair of trade types.[1] The legs of a complex trade are separate executions; we link them into packages using the complex execution identifier (*complex_exec_id*) for electronic trades and the floor agreement time (*floor_action_ts*) for floor trades, as described below.

A. Electronic Complex

Each leg of an electronic complex trade is a separate execution, and the legs of the same trade share a *complex_exec_id*. Table IA.XII shows an example: a vertical call spread on SPXW options expiring the same day, executed on the complex limit order book (CLOB) on September 4, 2020. The trade has two legs, the 3480 and 3490 calls, and each leg is one execution, identified by its *exec_id*. Each execution appears twice, once for each side. In each execution, the customer side with trade type *Complex Remove* is the initiator: its incoming order executes against the resting order on the other side, which has trade type *Complex Add*. The initiating customer sells the 3480 call at 0.90 and buys the 3490 call at 0.65 for 10 contracts each, that is, it sells 10 call spreads for a net credit of 0.25 per spread. Because both executions carry the same *complex_exec_id*, we link them into one package, and we classify the package as a vertical call spread from its legs: two calls with the same expiration and different strikes, one bought and one sold, in equal sizes (a 1:1 ratio).

An electronic order can execute against several counterparties, for example when it trades against more than one resting order on the CLOB or is split among several responders in an auction. Cboe records each of these fills as a separate complex execution with its own *complex_exec_id* and reports no order identifier. We therefore reconstruct orders by combining packages that have (i) the same legs (the same option series, each traded on the same side), (ii) the same ratio of leg sizes, (iii) the same initiator capacity, (iv) the same execution mechanism, and (v) the same timestamp, which Cboe reports to the microsecond. These are identical complex packages, initiated by the same type of trader (e.g., a retail investor), executing in the same microsecond.

TABLE IA.XII HERE

[1] The trade types also identify the venue and mechanism of each execution: the trading floor or, electronically, the complex limit order book (CLOB), the Complex Order Auction (COA) or the Automated Improvement Mechanism (AIM). For instance, if one side has trade type AIM Agency - Complex and capacity customer, and the other side has AIM Contra - Complex and capacity market maker, the customer's agency order is the initiator and the market maker is the contra party paired with it in the AIM auction.

B. Floor Complex

Unlike electronic trades, floor complex trades cannot be grouped using *complex_exec_id*, which can differ across fills. Instead, all records of a floor trade carry the time at which the trade was agreed upon on the trading floor (*floor_action_ts*), recorded to the millisecond. Because floor trades are agreed upon by open outcry, two separate trades are unlikely to share the same timestamp to the millisecond.[2] We therefore group floor records into packages on *floor_action_ts*. Table IA.XIII shows an example: a customer buys 375 vertical put spreads, buying the 2625 put and selling the 1925 put. The order is filled in two executions per leg, of 275 and 100 contracts, each with a market maker. The two fills have different execution identifiers and are reported at different times (*transact_time*), but all eight records share the same *floor_action_ts*, which groups them into one package.

We identify the initiator of each floor execution in three steps. First, where the trade types (*trade_type*) distinguish the two sides, we use them. For instance, a market maker's record carries *MM In-crowd - Complex*, which Cboe defines as a market maker order executed in the crowd, while the record of the order represented in the crowd carries *Floor Legged In - Complex*. Second, where the trade types do not distinguish the two sides, we treat the side that is not a market maker as the initiator, since floor brokers typically bring retail and institutional orders to the trading crowd, where in-crowd market makers fill them by open outcry. In Table IA.XIII, both sides carry the same trade type (*Floor Order on PAR - Complex*), so the customer is the initiator as the side that is not a market maker. Third, where the trade types do not distinguish the two sides and both or neither are market makers, we assign the direction of the other signed executions on the same option in the same package. If there are none, or they disagree, we cannot sign the execution and drop the whole package; this applies to 14.7% of floor packages before exclusions.

The NBBO in the TBT data is recorded at *transact_time*, when the trade is reported. For floor trades, however, reporting may occur several minutes after the trade is agreed, so the corresponding TBT quotes can reflect market conditions after the agreement.[3] We therefore use LSEG quote data for floor trades, matching each leg to the most recent quote prevailing before *floor_action_ts*, the timestamp at which the trade was agreed, and construct the package-level synthetic quote from these leg-level quotes.[4] For 30.4% of floor trades, the LSEG quote differs from the TBT quote. For electronic trades, which are reported contemporaneously with execution, we use the NBBO recorded in the TBT data.

TABLE IA.XIII HERE

[2] A single order has one initiator, so all its legs have the same initiator capacity. Only 21 of the 55,664 floor packages in our sample have legs with initiators of different capacities. We drop these 21 packages from our analysis.

[3] The median time between *floor_action_ts* and *transact_time* is 78 seconds.

[4] The median age of the matched quote is 2 seconds.

Table IA.XII. An Electronic Complex Trade in the Cboe TBT Data
This table shows the Cboe Trade-by-Trade (TBT) records of one SPX complex trade executed electronically on the complex limit order book (CLOB) on September 4, 2020: a vertical call spread on SPXW options expiring the same day. A customer sells 10 spreads, selling the 3480 call at 0.90 and buying the 3490 call at 0.65. Each leg is one execution (*exec_id*) with two records, one for each side; the side with trade type Complex Remove initiates against the resting Complex Add order. Both legs share one complex execution identifier (*complex_exec_id*), which links them into one package. Execution identifiers have been randomized.

Time	Strike	Type	Side	Size	Price	Capacity	Type	exec_id	complex_exec_id
10:41:10.113946	3480	C	Buy	10	0.90	Customer	Complex Add	Y1D0180LM	YF801L01M
10:41:10.113946	3480	C	Sell	10	0.90	Customer	Complex Remove	Y1D0180LM	YF801L01M
10:41:10.113946	3490	C	Sell	10	0.65	Customer	Complex Add	81YML100E	YF801L01M
10:41:10.113946	3490	C	Buy	10	0.65	Customer	Complex Remove	81YML100E	YF801L01M

Table IA.XIII. A Floor Complex Trade in the Cboe TBT Data

This table shows the Cboe Trade-by-Trade (TBT) records of one SPX complex trade executed on the trading floor on September 28, 2020: a vertical put spread on SPX options expiring October 16, 2020. A customer buys 375 spreads, buying the 2625 put at 1.50 and selling the 1925 put at 0.15. Each leg is filled in two executions (*exec_id*) of 275 and 100 contracts with market makers, and each execution has two records, one for each side. All records share the time at which the trade was agreed on the floor (*floor_action_ts*), which links them into one package; *transact_time* is the time at which each execution is reported (shown to the millisecond). Trade type: PAR-C = Floor Order on PAR – Complex. Both sides of each execution have the same trade type; the customer is the initiator, as the side that is not a market maker. Times and execution identifiers have been randomized.

floor_action_ts	transact_time	Strike	Type	Side	Size	Price	Capacity	Trade Type	Exec ID
14:01:02.822	14:02:57.762	2625	Put	Buy	275	1.50	Customer	PAR-C	51X4L000E
14:01:02.822	14:02:57.762	2625	Put	Sell	275	1.50	Market Maker	PAR-C	51X4L000E
14:01:02.822	14:02:57.930	2625	Put	Buy	100	1.50	Customer	PAR-C	X0L1H0054
14:01:02.822	14:02:57.930	2625	Put	Sell	100	1.50	Market Maker	PAR-C	X0L1H0054
14:01:02.822	14:02:57.762	1925	Put	Sell	275	0.15	Customer	PAR-C	XF501L004
14:01:02.822	14:02:57.762	1925	Put	Buy	275	0.15	Market Maker	PAR-C	XF501L004
14:01:02.822	14:02:57.930	1925	Put	Sell	100	0.15	Customer	PAR-C	X1I0050L4
14:01:02.822	14:02:57.930	1925	Put	Buy	100	0.15	Market Maker	PAR-C	X1I0050L4

IA.V: Pre-Trade Midpoint Drift

We measure effective spreads against the synthetic midpoint at the time the package price is fixed, which is the agreement timestamp for floor trades and the execution timestamp for electronic trades (Section I). If quotes move systematically before trades, and if these moves differ across venues, the choice of benchmark time could affect the comparison between floor and electronic execution. Floor orders are represented in the crowd before the price is agreed, so timing effects of this kind would most likely show up on the floor.

To check this, we recompute the effective spread of each package against the synthetic midpoint 15, 10, and 5 minutes before the price is fixed and subtract the effective spread against the midpoint 1 minute before. The difference is the change in the midpoint over the interval, signed by the direction of the trade. A positive value means that the midpoint moved in the direction of the trade before execution.

Table IA.XIV reports the results. The midpoint drifts in the direction of the trade in every venue, and the drift grows with the length of the window. Between 15 minutes and 1 minute before execution, the midpoint moves by 4.81 cents for floor packages, 4.26 cents for AIM, 3.81 cents for COA, and 1.61 cents for the CLOB. The drift on the floor is close to the drift in the two electronic auctions, whose execution timestamps are set by the exchange. It is therefore not specific to the floor or to the way floor trades are timestamped. The pattern is consistent with traders submitting orders after the midpoint has moved in their direction.

Because the drift is positive in every venue and largest on the floor, a benchmark taken before the price is fixed would raise effective spreads everywhere, and more on the floor than in any electronic venue. Measuring spreads when the price is fixed therefore does not tilt the comparison toward higher floor costs. An arrival-price benchmark would, if anything, widen the gap between floor and electronic execution.

TABLE IA.XIV HERE

Table IA.XIV. Pre-Trade Midpoint Drift by Execution Venue
This table reports the average change in the quote midpoint before complex SPX trades, signed by the direction of the trade, by execution venue. For each package, we compute the effective spread against the synthetic midpoint 15, 10, and 5 minutes before the time the package price is fixed and subtract the effective spread computed against the synthetic midpoint 1 minute before. The difference equals the change in the synthetic midpoint over the interval multiplied by the trade direction. A positive value means that the midpoint moved in the direction of the trade before execution. The price is fixed at the agreement timestamp for floor trades and at the execution timestamp for electronic trades. Electronic fills are grouped into orders. Packages is the number of complex packages in each venue. Drift is measured in cents per unit of the underlying.

		Midpoint Drift (Cents)		
Venue	Packages	15 min → 1 min	10 min → 1 min	5 min → 1 min
Floor	48,946	4.81	4.08	2.63
AIM	292,942	4.26	3.35	2.01
COA	2,294,101	3.81	3.03	1.86
CLOB	2,300,181	1.61	1.15	0.58

IA.VI: Trade Initiators by Venue

Table IA.XV reports the composition of complex SPX packages by the capacity of the initiating party on the trading floor, the CLOB, AIM, and the COA. On the floor, retail investors initiate 42.0% of packages and institutional investors 49.9%. Market makers initiate only 1.2%, consistent with their role on the floor of responding to the orders that brokers bring to the crowd. Market makers initiate 94.5% of CLOB packages, and retail investors 4.8%. The two electronic auctions serve retail investors, who initiate 89.4% of AIM packages and 89.1% of COA packages.

Table IA.XV. Trade Initiators by Venue

This table reports the composition of complex SPX packages by the capacity of the initiating party on each venue: the trading floor, the complex limit order book (CLOB), the Automated Improvement Mechanism (AIM), and the Complex Order Auction (COA). The sample covers the pre-closure, closure, and opening periods (January 2 to October 21, 2020). The table reports the percentage of packages initiated by market makers, customers, firms, broker-dealers, and professional customers, together with the number of packages; no packages are initiated by other capacities.

	Packages by Initiator Capacity (%)					
Venue	Market Maker	Customer	Firm	Broker-Dealer	Pro Customer	Packages
Floor	1.2	42.0	49.9	4.6	2.3	51,577
CLOB	94.5	4.8	0.1	0.0	0.5	2,272,717
AIM	2.6	89.4	6.6	1.0	0.3	161,536
COA	0.0	89.1	0.0	0.0	10.9	1,294,196

IA.VII: Institutional Details of Cboe Floor Trading and Hybrid Execution

A. Scope and sources

This Appendix documents the institutional details of open outcry floor trading at Cboe Options Exchange ("C1") and its interaction with electronic execution mechanisms in the hybrid market design. Our primary institutional sources are (i) the C1 Exchange Rule Book effective October 7, 2019 and contemporaneous rule filings, (ii) Cboe circulars and operational updates surrounding the March 2020 COVID-19 floor closure and subsequent opening, and (iii) Cboe's published technical and market-structure documents describing complex-order handling and the boundary between electronic and floor-eligible trading.[5]

B. Hybrid market architecture and trading sessions

Hybrid model. Cboe operates a hybrid options market in which the same option classes may be executed electronically (through an automated matching engine and displayed order books) and, for designated classes, through in-person open outcry on the Chicago trading floor. The hybrid design dates back to Cboe's "Hybrid Trading System" transition in the early 2000s, which integrated electronic executions and quoting with an open outcry floor auction.[6]

Trading sessions. C1 distinguishes between Regular Trading Hours (RTH) and Global Trading Hours (GTH). Open outcry is an RTH functionality: during GTH, C1 operates in an electronic-only mode for the relevant products, and orders that would otherwise route to the floor (via PAR) are not accepted for floor handling.[7] Within RTH, open outcry trading in a given series begins only after the series has opened for electronic trading.[8]

The trading floor as a price-discovery venue for selected products. In modern U.S. listed options, most volume is electronic, but Cboe continues to operate a floor that is

[5]For readability, we refer to the exchange as "Cboe" and the venue as "C1" when quoting specific rulebook mechanisms that apply to Cboe Options Exchange. Rules and functionalities change over time; the institutional description here is anchored to the rule set applicable to the 2019–2020 period and to the temporary amendments adopted during the COVID-related inoperability of the physical floor.

[6]See, for example, SEC approval orders and related rule filings for Cboe's Hybrid system evolution, including the original Hybrid approval and later enhancements. In the current C1 rulebook structure, open outcry trading is governed primarily by Chapter 5, Section G, and routing/eligibility for floor handling is governed by the PAR rules and complex-order rules (discussed below).

[7]Operationally, the floor-based PAR routing and open outcry mechanisms are inactive in GTH; by rule, C1 may return orders that route to PAR during GTH.

[8]This sequencing matters empirically: the opening process is electronic, and floor trading begins only after the electronic opening rotation for the series/class has completed.

materially relevant in index products. Cboe market-structure publications emphasize that open outcry remains commonly used for (i) certain products, (ii) certain complex strategies, and (iii) large orders, particularly during volatile periods.[9]

C. Participants, roles, and floor governance

Open outcry trading is organized around a *trading crowd* for each class (often described colloquially as a "pit" or "post"). The main participant types are:

- **Floor Brokers (agency intermediaries).** Floor brokers are representatives of member firms who handle customer agency orders on the floor. They are obligated to use due diligence to execute orders at the best available prices under the rules, including requesting quotes from the crowd and continuously representing certain marketable orders.[10]

- **Market Makers (liquidity suppliers).** Market makers are exchange-registered liquidity providers that may trade for their own accounts and are subject to quoting and market-quality obligations in their appointed classes.[11]

- **Designated liquidity providers (DPM/LMM).** In some classes, C1 appoints a Designated Primary Market-Maker (DPM) and/or a Lead Market-Maker (LMM) and may grant these participants a *participation entitlement* in allocations at a given price when they are quoting at the best price.[12]

- **PAR Officials (exchange-appointed order handlers).** PAR Officials are exchange personnel (or contractors) designated to operate a PAR workstation and effect proper executions of orders in their custody. They have explicit display and due-diligence obligations, are not affiliated with market-making firms, and are compensated exclusively by the Exchange.[13]

[9] Cboe and industry market-structure publications also report that open outcry constitutes a meaningful share of SPX and VIX volume in recent data, illustrating that the floor is not purely residual even after broad electronification.

[10] See Rule 5.91 for the detailed due-diligence obligations, representation requirements, and constraints on discretion.

[11] See, for example, Rules 5.51–5.52 for market-maker obligations and quoting.

[12] These entitlements exist both in electronic allocation and in open outcry allocation, subject to class-by-class application. In open outcry, the entitlement is applied after Priority Customer orders in the Book at the execution price are satisfied, as described below.

[13] See Rule 5.90.

Floor access, clerks, and supervision. The rulebook restricts authority to transact on the trading floor to approved Trading Permit Holders performing floor functions (e.g., Floor Brokers and Market Makers). Clerks have badge and booth requirements and are limited in how and where they may stand relative to trading crowds. The floor is overseen by Floor Officials, who can interpret certain rules in real time and may nullify or adjust transactions that violate key trading rules.[14]

D. Order capacity, agency and proprietary orders, and trade initiators

Every order on C1 carries a capacity code for the account behind it. This subsection sets out who may trade on the floor and in which role, separates agency from proprietary orders, and explains what the initiator labels in the paper reveal. Rule numbers refer to the rule book in force in 2020 (Cboe Global Markets, 2019). Table IA.XVI summarizes.

Capacity codes and account definitions. Trading on the Exchange requires a Trading Permit, and its holder is a Trading Permit Holder (TPH) (Rule 3.1(a)). Rule 1.1 defines the eight capacity codes listed in Table IA.XVI. A Public Customer is any person that is not a broker-dealer, and every TPH counts as a broker-dealer. A Professional is a person that is not a broker-dealer and places more than 390 listed options orders per day, on average over a calendar month, for its own beneficial accounts. A Priority Customer is a Public Customer that is not a Professional, and Priority Customer orders resting in the Book have first priority at a price (Rule 5.85(a)(2)(A)).

Capacity in the data and the labels in the paper. The Cboe trade-by-trade file reports five buckets: Customer (C), ProCustomer (U), BrokerDealer (B), Firm (F, J, and L), and MarketMaker (M and N). The eight codes cannot be recovered from it. "Agency" in the paper means capacity C, the Priority Customers. The label is legal and can include institutions, such as asset managers or pension funds, that are not broker-dealers and stay below the Professional threshold. "Proprietary" means the Firm bucket: proprietary accounts of clearing members, joint back office accounts, and accounts of affiliates that are not TPHs. A TPH that does not know the capacity of an order may report B and correct it later (Rule 6.1(h)).

Who trades on the floor and in which role. Only TPHs approved for a floor function trade on the floor (Section IA.VII.C). The in-crowd market participants are in-crowd market makers, on-floor DPMs or LMMs, and Floor Brokers or PAR Officials representing orders in the crowd (Rule 1.1).

[14] See Rule 5.80 for floor admission and conduct; and the Floor Officials' authority to adjust/nullify trades that violate, among others, the open outcry allocation rules.

- **Floor Brokers act as agents.** A Floor Broker accepts orders from TPHs and registered broker-dealers, and from public customers only as the nominee of a TPH organization approved to do business with the public (Rules 3.35 and 3.50(a)). It can represent an order of any capacity, including a market maker's own order, whose owner it must name on request (Rule 5.91(b)).
- **PAR Officials act as agents.** They are Exchange employees or contractors who execute the orders placed with them (Rule 5.90).
- **In-crowd market makers trade as principals.** They must be physically present in the crowd (Rule 5.50(e)) and must give a two-sided quote when a Floor Broker or PAR Official asks for one (Rule 5.52(e)). An on-floor DPM or LMM quoting the best price can also receive the participation entitlement of Section IA.VII.G (Rule 5.85(a)(2)(B)).

No TPH may initiate a trade on the floor for an account in which it has an interest unless it is a registered market maker, error positions excepted (Rule 8.22). Market makers are therefore the only principals in the crowd. Any other order, including a firm's proprietary order, reaches the crowd only through a Floor Broker or a PAR Official. Priority Customer and broker-dealer orders resting in the Book can also trade with a represented order (Rule 5.85(a)(2)).

Agency and proprietary orders. Agency and proprietary describe the account behind an order, not the person who represents it. An order is proprietary when it is for a TPH's own account, the account of an associated person, or an account over which either exercises investment discretion (Rule 5.87(b)(3)). The rule book calls a TPH that represents customer orders as agent, or that trades for its own account without being a market maker, an Order Entry Firm (Rule 3.51). One firm can act in both ways on the same trade: a clearing firm can bring a customer order (C) and take the other side for its own account (F). Rule 5.9(a) allows it to trade as principal against an order it represents as agent only after the order, or its own quote, has been exposed on the System for one second, or by crossing the two orders under Rule 5.87 on the floor or in an AIM auction. Proprietary orders on the floor also need an exemption from Section 11(a)(1) of the Exchange Act. Under the "G" exemption, they yield priority to Priority Customer and broker-dealer orders resting in the Book at the same price (Rules 5.85(a)(2)(E) and 5.87(b)).

Original orders and contra orders in floor crosses. The crossing mechanisms of Section IA.VII.H assign each order a role. In a facilitation cross, a Floor Broker holds a Public Customer order of a TPH organization and a facilitation order of the originating firm. The Floor Broker requests a market, bids or offers on the customer's behalf at a better

price, and crosses the remainder with the facilitation order only after the other in-crowd participants have had the chance to trade (Rule 5.87(d)). The customer order, of capacity C or U, is the original order, and the firm's order is the contra. In a solicited transaction, the TPH representing an original order, which the rule book calls the TPH "initiating" it (Rule 5.86(d)), solicits a TPH, a public customer, a broker-dealer, or a market maker to trade against it, and the solicited order is the contra (Rule 5.86 and Interpretation .01).

The crossing entitlement applies to original orders of at least the eligible size, which the Exchange sets at no less than 50 contracts. The Floor Broker may cross part of such an order with another order it holds, with the originating firm's facilitation order, or, where the Exchange allows, with a solicited order (Rule 5.87(f)). A DPM or LMM that is also the originating firm receives no participation entitlement on that cross (Rule 5.87(f)(3)). When a Floor Broker crosses two orders it holds under Rule 5.87(c), the rule treats them alike and names no original order. In classes the Exchange designates, a TPH holding an option order of at least the designated size, never below 500 contracts, may also hedge it in stock or futures before announcing it, if it then offers the whole hedge to the in-crowd participants who trade the order (Rule 5.87, Interpretation .07).

Initiators in the paper. On the floor, the paper's initiator is the order represented in the crowd, and the in-crowd market makers who respond to it are its counterparties. Section IA.IV describes the three signing steps: the trade types, then the side that is not a market maker, then the direction of other fills on the same leg. The fills that remain unsigned are crosses of two represented orders, mostly between two market makers and in some cases between two firms, and the paper excludes the orders that contain them. A market maker initiates when its record is that of an order represented in the crowd, such as an order that a Floor Broker holds for it (Rule 5.91(b)). This is the case in 1.2% of floor packages. Execution costs are end-user measures, taken from the initiator, or from its counterparty when a market maker initiates (Section II.C).

An initiator capacity gives the account type of the original order. A customer-initiated order is an agency order for a Priority Customer. A firm-initiated order is an order for a clearing member's proprietary account, a joint back office account, or an affiliate's account, which a Floor Broker or PAR Official represents as agent. The capacity names neither the customer nor the firm. A firm counterparty in a customer-initiated order can be the originating firm's facilitation order, a solicited order of another firm, or a firm order that another Floor Broker represents in the crowd. A market maker counterparty can be an in-crowd responder or a market maker solicited as a contra. The closure filing describes large customer orders on the floor as filled primarily by appointed market makers solicited as contras (Section IV.D). The file has no participant identifiers, so these cases cannot be told apart.

Electronic counterparts. Market makers have no quoting obligation for complex strategies (Rule 5.33, Interpretation .01). The Exchange can also bar market makers' complex orders from the COB, the paper's CLOB, and admit them only in narrow cases, such as opposite resting Priority Customer complex orders (Rule 5.33(b)(2)). Market makers initiate 94.5% of CLOB packages by taking resting orders, and the end user is the resting side. In AIM, the Initiating TPH pairs an order it represents as agent, the Agency Order, with an Initiating Order that is its own principal interest or a solicited order (Rules 5.37 and 5.38). In single-leg AIM, the Initiating Order may not be for a market maker appointed in the class, and complex AIM has no such restriction. The rule book's Initiating Order is thus the contra, and the paper's initiator is the Agency Order, a retail order in 89.4% of AIM packages. AIM is the electronic route for facilitation and solicitation, with a timed auction in place of the request for a market (Rule 5.9(a)). In the COA, the incoming complex order starts the auction and is exposed to responses (Rule 5.33(d)). Retail orders initiate 89.1% of COA packages.

Table IA.XVI. Order Capacity and Roles on the Cboe Floor
This table lists the capacity codes of Rule 1.1 of the Cboe rule book in force in 2020 (Cboe Global Markets, 2019), the account each code designates, the bucket that the Cboe trade-by-trade file reports for the code, with the label used in the paper in parentheses, and how an order of that capacity can trade in open outcry. The file reports the five buckets, not the eight codes. Only registered market makers trade in person for their own account (Rule 8.22). All other orders reach the crowd through a Floor Broker or a PAR Official, who act as agents.

Code	Account	Data bucket (paper label)	How it trades on the floor
C	Public Customer; Professionals are coded U, so C is a Priority Customer	Customer (agency)	Agency order represented by a Floor Broker. Original order in a facilitation cross (Rule 5.87(d)). First priority when resting in the Book (Rule 5.85(a)(2)(A)).
U	Professional: not a broker-dealer, more than 390 orders per day on average	ProCustomer (professional customer)	Agency order represented by a Floor Broker. A Public Customer for facilitation crosses, handled like a broker-dealer order in the Book (Rule 1.1).
B	Broker or dealer, including a foreign broker-dealer	BrokerDealer (broker-dealer)	Order represented by a Floor Broker. Can be an original or a solicited order. Fourth priority when resting in the Book (Rule 5.85(a)(2)(D)). Default code when the capacity is unknown (Rule 6.1(h)).
F	Proprietary account of a clearing member of the Options Clearing Corporation	Firm (proprietary)	Proprietary order represented by a Floor Broker. Cannot trade in person (Rule 8.22). Original order when the firm initiates, or the facilitation or solicited contra to a customer order. Needs an exemption from Section 11(a)(1) (Rule 5.87(b)).

continued on next page

Table IA.XVI continued

Code	Account	Data bucket (paper label)	How it trades on the floor
J	Joint back office account; participants are registered broker-dealers (Rule 11.4)	Firm (institutional)	Order represented by a Floor Broker. Cannot trade in person (Rule 8.22). Can be an original or a solicited order.
L	Account of an affiliate that is not a TPH	Firm (institutional)	Order represented by a Floor Broker. Can be an original or a solicited order.
M	Account of a Cboe market maker	MarketMaker (market maker)	Trades in person in the crowd and must quote on request (Rules 5.50(e) and 5.52(e)). DPM or LMM entitlement if appointed. Can be solicited as a contra (Rule 5.86). Its own order can be represented by a Floor Broker (Rule 5.91(b)).
N	Account of a market maker on another options exchange	MarketMaker (market maker)	Order represented by a Floor Broker. Not an in-crowd participant.

E. Order entry, systematization, and the PAR routing channel

Systematization and audit trail ("enter it before you trade it"). A central institutional feature of floor trading on C1 is that orders handled on the floor must be captured in a systematized format to satisfy audit-trail requirements. The rulebook requires that orders transmitted to the floor be recorded on approved order forms (or electronic media), including time-of-receipt information, and that cancellations/changes be similarly recorded. If the Exchange's systems are disrupted, participants must maintain written records and subsequently input the information into Exchange systems by prescribed deadlines.[15]

PAR as the operational bridge between electronic and floor handling. C1's *Public Automated Routing system (PAR)* is the operational channel by which eligible orders are routed to the trading floor for manual handling and potential open outcry execution. A PAR workstation is an exchange-provided order management tool used on the trading floor by Trading Permit Holders and PAR Officials.[16] Orders may route to PAR in two primary ways: (i) the user explicitly designates the order as *Direct to PAR*, or (ii) the order is designated as *Default* and cannot be electronically processed under the system rules, in which case it routes to PAR subject to user instructions.[17]

[15] See Rule 5.7 for systematization requirements, contingency procedures during malfunctions, and audit-trail/COATS-related interpretations. The rulebook explicitly recognizes that hand-signal communications may initiate or modify orders on the floor, but such actions must still be systematized.

[16] See Rule 5.82.

[17] See Rule 5.82(b).

What can be done with an order on PAR. Once an order is on PAR, the floor broker/PAR Official can, subject to rules and user instructions: (i) submit the order (or a portion) back into the System for electronic processing if eligible, including participation in a complex order auction (COA); (ii) execute the order (or a portion) in open outcry on the floor (including executions against Priority Customer orders and other interest resting in the electronic Book at the best price); (iii) route to an away exchange (if eligible for routing); (iv) route to another PAR workstation; or (v) cancel the order.[18] PAR is an order-handling interface that permits hybrid "work" of an order across electronic and floor mechanisms.

Direct-routing and order marking. After Cboe's migration of C1 onto a Bats-technology platform, Cboe issued operational guidance that orders intended for manual open outcry handling must be properly marked and routed with exchange-specified directives and PAR workstation identifiers.[19]

F. What is floor-only (or effectively floor-only) in the C1 rule set?

The C1 rulebook and technical specifications create explicit *eligibility boundaries* between electronic processing and floor/manual handling. These boundaries are important for understanding why the floor closure in March 2020 required rule adjustments.

Simple orders. On a class-by-class basis, C1 may permit certain simple order types, instructions, and times-in-force to be routed to PAR for manual handling and open outcry.[20]

Complex orders: the ratio boundary as a central segmentation device. For complex orders, C1 imposes a ratio-based boundary that (in normal operation) determines whether a multi-leg order may be processed electronically or must be handled manually on the floor. Specifically, *other than Index Combo orders*, a complex order with a ratio less than one-to-three (0.333) or greater than three-to-one (3.00) may only be submitted for manual handling and open outcry trading.[21] C1 may make additional complex-order instructions available for PAR routing, including (among others) Complex Only, Compression/PCC, Multi-Class Spread, RFC, and SPX Combo.[22]

[18] See Rule 5.82(c) for the menu of permitted PAR actions.

[19] See Cboe Exchange Notice on direct routing to PAR and related order-marking requirements; such notices specify how default vs. direct-to-PAR directives determine whether an order reaches the floor or is processed electronically.

[20] Rule 5.83(a) enumerates available simple order types and instructions that the Exchange may make available for PAR routing.

[21] See Rule 5.83(b). This ratio boundary is a key institutional feature for empirical work that compares electronic and floor execution.

[22] See Rule 5.83(b)(2) for the list of complex-order instructions made available for PAR routing.

Additional technical/operational constraints that create floor concentration. Beyond the 3:1 ratio boundary, Cboe technical specifications for the U.S. options complex book process document several restrictions that are directly relevant for substitution and "strategy splitting":

- **Maximum number of legs differs by electronic vs floor routing.** Electronic complex orders have a maximum number of legs (set by the exchange and generally far below "everything you could write down"), while floor-routed orders may support creation/handling of complex instruments with substantially more legs (including up to very high leg counts for floor-routed non-FLEX complex orders).[23]
- **Cash-priced complex orders and other special price types.** Cash-spread (PriceType=C) complex orders may be rejected by the electronic complex book and instead require floor routing to PAR.[24]
- **Certain trade-condition codes require floor routing.** For example, "CMBO" (combo) trade-condition handling for SPX/SPXW may require the order to be directed to PAR and is not accepted for electronic processing.[25]

These restrictions imply that a trader seeking to execute an economically meaningful but "electronically non-eligible" package must either (i) route the package to the floor (when the floor is operational), or (ii) decompose the desired package into multiple electronically eligible orders (or into a sequence of electronically eligible complex and simple orders), which may introduce legging risk, exposure to interim price changes, and additional information leakage.

G. Open outcry price formation and allocation

Open outcry as a competitive auction with explicit priority rules. Open outcry is a verbal/visible auction mechanism in which floor brokers (and PAR officials) represent orders to the trading crowd and in-crowd liquidity providers respond with bids/offers. Cboe regulatory guidance emphasizes that trades executed on the floor must be executed in open outcry and may not be executed as private "cross-only" transactions; bids and offers must be requested and exposed in the crowd (with narrowly defined exceptions such as certain electronic crossing mechanisms).[26]

[23] See Cboe's US Options Complex Book Process documentation for the maximum-leg rules and the distinction between electronic and floor-routed complex instrument creation.

[24] See the Complex Book Process specification section on Cash Spreads and the routing requirements for such orders.

[25] See the Complex Book Process specification on Combo trade-condition handling and floor-routing requirements.

[26] See Cboe Regulatory Circulars on open outcry trading requirements and prohibited practices. Similar requirements appear in older Cboe circulars from the 1990s.

Execution price priority. For an order represented in open outcry, bids/offers at the best price have priority.[27]

Priority at the same price: interaction with the electronic book. A distinctive feature of C1's hybrid design is that open outcry executions are not insulated from the electronic Book. At a given execution price, Priority Customer orders resting in the electronic Book have first priority. If multiple Priority Customer orders are at the same price, the System allocates them in time priority (order of receipt by the System).[28] After Priority Customer orders, the rulebook specifies additional priority layers, including potential DPM/LMM participation entitlements and then in-crowd responses.

Participation entitlement (DPM/LMM). If the Exchange applies a participation entitlement to a class, and the DPM/LMM is quoting at the best price, the DPM/LMM receives the greater of: (i) the amount it would receive under the general in-crowd allocation rules, and (ii) a percentage entitlement of the remaining non-Priority Customer contracts after Priority Customer orders in the Book are satisfied. The entitlement percentage depends on the number of other in-crowd quotes at that price (e.g., 50% if one other ICMP quote, 40% if two, and 30% if three or more), and it is capped by the size of the DPM/LMM quote.[29]

In-crowd market participant responses. In-crowd market participant (ICMP) bids/offers made at the time the market is established have priority after Priority Customer orders (and any applied entitlement). The floor broker/PAR official (or DPM/LMM/Market-Maker, depending on who is controlling the execution) determines which participants responded at the time the market was established and the sequence in which responses were made. If two or more bids/offers are simultaneously made and the sequence cannot be determined, the remaining contracts are allocated equally among the ICMPs at that price.[30]

Then, non-customer book interest. After Priority Customer orders and in-crowd responses, remaining contracts may execute against non-customer order interest and quotes resting in the electronic Book, with allocation governed by the electronic allocation rules applicable to that class.[31]

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[27] Rule 5.85(a)(1).

[28] Rule 5.85(a)(2)(A).

[29] Rule 5.85(a)(2)(B). The rulebook also distinguishes "on-floor" vs "off-floor" DPM/LMM entitlements when both are appointed, with on-floor entitlements applying to orders represented in open outcry.

[30] Rule 5.85(a)(2)(C).

[31] Rule 5.85(a)(2)(D) references application of the allocation algorithm in Rule 5.32 for Book interest at the execution price.

Floor broker and PAR official obligations: prompt representation and time-stamping. The rulebook codifies detailed obligations intended to preserve auction integrity and auditability:

- **Floor Brokers:** must use due diligence in handling orders (including requesting quotes and ensuring compliance with firm-quote obligations), must continuously represent market and marketable limit orders at the trading station, and must electronically record the time of initial representation via PAR (subject to systematization requirements).[32]
- **PAR Officials:** must display promptly certain customer limit orders that improve the best disseminated quote (subject to exemptions), must keep "Autobook" active, and must electronically record representation times.[33]

Trade reporting and "late" flags. Open outcry executions must be reported within a short time window; transactions not reported within the required window are marked late, and patterns of late reporting can be treated as rule violations.[34]

H. Crossing, facilitation, and solicitation on the floor

A second distinctive institutional feature of floor trading is that open outcry supports brokered crossing activity (facilitation and solicitation) subject to exposure rules, and that the rulebook provides explicit *crossing entitlements* that can be interpreted as incentives to bring order flow to the floor.

Facilitated and solicited transactions. A facilitated transaction is an agency order crossed with the originating firm's proprietary interest (i.e., the broker/firm "facilitates" the customer order). A solicited transaction is an agency order crossed with pre-arranged contra interest ("solicited" liquidity). These transactions are permitted only if executed under open outcry exposure and allocation rules.[35]

Exposure and competition. Even when a broker intends to cross or facilitate, the order must be exposed to the crowd so that other in-crowd liquidity providers can compete. In practice, the broker requests quotes, and the cross may only occur at a price consistent with

[32]Rule 5.91.

[33]Rule 5.90, including the "30-second standard" for display under normal conditions and the enumerated exemptions.

[34]See Rule 6.1 on trade reporting (including the 90-second requirement and outage procedures).

[35]See Rule 5.86 for the detailed mechanics and conditions.

the crowd's market and the protection of priority interest (including Priority Customer orders in the Book).[36]

Crossing entitlement. If the relevant conditions are met, the broker (facilitator/solicitor) may receive 20% or 40% of the contracts remaining after Public Customer orders in the Book and then in the crowd are satisfied. In the C1 framework, the entitlement depends on the circumstances and the type of cross (and may also be limited by combined entitlements when a DPM/LMM also has a participation entitlement).[37]

Why this matters empirically. Crossing entitlements and the ability to solicit contra interest provide a clear microstructure channel through which floor execution can differ from purely anonymous, time-priority electronic execution. These features may affect execution costs and the distribution of who supplies liquidity, especially for large and/or complex packages.

I. Complex orders: electronic mechanisms, constraints, and the floor boundary

Definition and representation. A complex order is a multi-leg package (two or more legs) executed at a net debit/credit price (and, in some cases, may include an underlying stock or security future leg). C1 maintains a complex order book (COB in Cboe's rules, the CLOB in this paper) where eligible complex orders may rest and interact with other complex orders and, subject to restrictions, with liquidity implied from the individual-leg (simple) books.[38]

Electronic complex execution pathways. For eligible complex orders, electronic execution may occur through: (i) the continuous complex order book (resting complex liquidity interacting under price-time rules); (ii) complex order auctions (COA), which expose a complex order to solicited/competitive responses; and (iii) other auction/crossing mechanisms (e.g., complex variants of the Automated Improvement Mechanism (AIM) or SAM, where available).[39]

[36] See Rule 5.86 and Rule 5.87 (Crossing Orders) for the request-for-market and price constraints; see also floor-trading circulars on the open outcry and exposure requirements.

[37] Rule 5.87(f)(2) sets out the 20% or 40% crossing entitlement, and Rule 5.87(b)(2) caps the combined DPM/LMM participation and crossing entitlements at 40%.

[38] See Rule 5.33 for the definition of complex orders, COB handling, and the synthetic best bid/offer concepts used for complex order evaluation and price protections.

[39] Rule 5.33 describes complex order auction handling and interaction with COB; Cboe also publishes technical specifications describing these mechanisms and their eligibility constraints by class and session.

Key eligibility constraints that differentiate electronic vs floor execution. Cboe's rulebook and technical specifications impose multiple constraints that define the boundary between electronic and floor execution for complex strategies. Three are particularly relevant:

1. **Contract ratio constraint (the "3:1 boundary").** Under normal (pre-COVID) rules, other than Index Combo orders, complex orders with contract ratios outside the range [1/3, 3] are not eligible for electronic processing and must be submitted for manual handling and open outcry.[40]

2. **Leg-count and legging restrictions.** Even for electronically eligible ratios, complex orders face restrictions on (i) the maximum number of legs permitted for a given electronic pathway and (ii) whether the complex order may "leg" into the simple book. The system may permit legging only for strategies with a small number of legs (e.g., two, three, or four), and may prohibit legging for certain sign patterns (e.g., all-buy or all-sell three/four leg strategies) and for stock-option orders.[41]

3. **Special price types and strategy codes.** Cash-priced complex orders (cash spreads) and certain strategy/trade-condition codes (e.g., specific combo handling for SPX/SPXW) may be rejected in the electronic workflow and require floor routing.[42]

Complex order priority protections in open outcry. C1 also specifies when a complex order executed in open outcry may execute at a net price without "giving priority" to equivalent bids/offers in the individual-leg markets. The rulebook distinguishes between (i) complex orders with ratios within the standard electronic range and (ii) high-ratio complex orders that are floor-only in normal times. For example, for certain standard-ratio complex orders, open outcry execution without giving priority to resting leg interest is permitted only if at least one leg improves the corresponding best bid/offer of a Priority Customer order in the Book by at least one minimum increment. For high-ratio complex orders (outside [1/3, 3]), the rulebook requires more stringent improvement conditions across legs.[43]

[40] Rule 5.83(b).

[41] See Rule 5.33(g) on legging restrictions and the exchange-determined maximum number of legs for legging eligibility.

[42] See the Cboe US Options Complex Book Process specification for Cash Spreads and trade-condition-code handling, including PAR routing requirements.

[43] See Rule 5.85(b) for the detailed conditions for open outcry complex execution without giving priority to equivalent bids/offers in the individual series. These conditions are central to how the hybrid market protects Priority Customer interest when floor-negotiated net prices do not map one-for-one onto the displayed leg markets.

Substitution and "strategy splitting" when electronic constraints bind. These eligibility constraints generate a natural substitution margin between floor and electronic execution. When the desired strategy falls outside the electronically supported set (e.g., a ratio backspread with ratio > 3:1, a package with many legs beyond the electronic maximum, a cash-price spread, or a strategy requiring a trade-condition code not accepted electronically), a trader must either (i) execute the package via PAR/open outcry (when available), or (ii) approximate the same economic exposure through multiple electronically eligible orders (for example, splitting a high-ratio strategy into two or more lower-ratio complex orders, or combining one complex order with one or more single-leg orders). The second approach can introduce *legging risk* (interim exposure to market moves between component executions), can worsen information leakage (multiple prints and/or multiple exposures), and can alter priority/exposure mechanics relative to a single negotiated net-price package.

J. COVID-19 floor closure: operational shift and confirmatory rule changes

On March 16, 2020, Cboe temporarily closed its Chicago options trading floor due to the COVID-19 pandemic. During the closure period, open outcry trading ceased, and C1 operated in a screen-based only configuration.[44]

Operational consequences. Cboe announcements emphasized that, with the floor closed: (i) orders with instructions to route to the trading floor would be rejected/returned, and (ii) floor brokers and market makers could continue to participate electronically from remote locations using approved order-entry tools (e.g., institutional front ends and exchange-connected interfaces).[45]

Temporary expansion of electronic complex-order eligibility. Because the floor previously served as the execution channel for high-ratio complex strategies, Cboe implemented confirmatory/temporary rule changes to accommodate that order flow in the electronic environment. In particular, for exclusively listed index option classes, Cboe temporarily expanded the definition/eligibility of complex orders for electronic processing to include ratios up to 25-to-1 (and down to 1-to-25), and set corresponding minimum increment rules for these complex orders.[46] This is the institutional channel through which the 2020 floor closure changes the feasible set of complex strategy executions.

[44] See Cboe operational updates and circulars regarding the March 2020 closure and the opening date.

[45] See Cboe floor-closure updates describing how floor participants could continue to trade electronically and how floor-directed orders were handled.

[46] See the screen-based-only environment provisions regarding loss/inoperability of the floor and Cboe's COVID-related rule change filings and circulars that expanded complex-order ratio eligibility.

Temporary activation of AIM for SPX. During the floor closure, Cboe also activated AIM for SPX/SPXW complex orders during RTH, a function AIM had not previously performed for that class, and deactivated it when floor trading resumed on June 15, 2020 (Section I.C). The CLOB and COA remained available after the opening, with the 3:1 ratio limit on electronic complex orders reinstated. AIM was the only electronic execution mechanism withdrawn.[47]

K. Why does the floor still exist?

Although the listed options market is predominantly electronic, the Cboe floor persists in part because it supports *package-level negotiation and execution* for certain large and/or complex orders whose economics are not always well served by standardized electronic constraints. Three institutional reasons stand out:

1. **Feasible-set differences (what the floor can do that the electronic workflow restricts).** Ratio constraints (e.g., the 3:1 boundary in normal times), leg-count limits, cash-spread handling, and special trade-condition codes can make some strategies effectively floor-only. Even when an economic exposure is replicable electronically, it may require splitting into multiple orders, increasing execution risk and complexity.

2. **Liquidity formation for certain complex instruments.** Cboe has acknowledged that some complex instruments (e.g., box spreads, box swaps, and jelly rolls) have historically traded predominantly in open outcry and that electronic liquidity formation for these packages can be constrained in important index products.[48]

3. **Hybrid incentives and governance.** Floor-specific participation entitlements and crossing mechanics (facilitation/solicitation), coupled with exposure rules and customer-priority protections, create a distinct trading protocol that can be valuable for executing large packages and for market makers supplying liquidity in stressed conditions.[49]

Implication for empirical interpretation. These institutional features imply that the floor is a complementary execution channel that (i) expands the feasible set of package executions, (ii) changes the exposure/information environment of execution, and (iii)

[47] See Cboe Global Markets (2020d) and Cboe Global Markets (2020c). The March 12, 2020 notice states that the AIM activation applied only during all-electronic trading and would be reverted when open outcry trading resumed.

[48] See Cboe documentation on the Quoted Spread Book (QSB) initiative and related statements about why certain spreads historically concentrated on the floor.

[49] See Rules 5.85–5.87 and related Regulatory Circulars for the incentive/protection design embedded in open outcry allocations and crossing.

can concentrate liquidity for large multi-leg index packages. The closure of the floor on March 16, 2020, its opening on June 15, 2020, and the temporary changes to electronic complex-order eligibility and AIM between those dates provide variation for studying execution outcomes across mechanisms.